kasalis



PIXID 500 PRO

USER MANUAL | 2022 REV20

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Declaration of Conformity

We,

Manufacturer:	Kasalis Incorporated
Address:	11 North Avenue Burlington, MA 01803 United States of America

in accordance with the following Directives:

2004/108/EC	EMC Directive
2006/95/EC	Low Voltage Directive

hereby declare that:

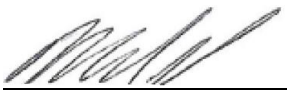
Type of Equipment:	Automated Alignment System for Camera Modules
Model #:	Pixid 500 Pro/Test

is in conformity with the applicable requirements of the following documents:

EN61000-6-4	Radiated and Conducted Emissions for Industrial Environments
EN61000-3-2	Harmonics
EN61000-3-3	Flicker
EN61000-6-2 - 2005	Electromagnetic Compatibility – Generic Standards. Immunity for Industrial Environments
EN60204-1	Safety of Machinery. Electrical Equipment of Machines. General Requirements.

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CE

Signature: 

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The information may be incorporated into the user's documentation.

Revision History

Date	Rev	Comments	Initials
06/07/2015	14	Initial release for dispense module operations in details	A.F
06/22/2016	15	Editing system start up section due to software changes Editing MasterProcessLogData Editing setup a new device setup section –dispense teach	A.F
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5/1/2018	20	Added specifications to section 2, updated drawing views in section 4.2 to more recent model	SJW
8/14/23	21	Added IT Recommendations section to the Installation chapter	MRK

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1 Introduction

1.1 Overview

This introduction provides an overview of the Pixid 500 Active Alignment System for 5 or 6 degree of freedom alignment, assembly, and test of camera modules.

The Pixid 500 consists of seven major modules:

- Rotary Module, which can rotate the fixtures from station to station
- Load Station, which lifts the fixtures off the work plate for easy and safe loading and unloading
- Dispense Module, for the adhesive dispense. This module may also include electrical, particle and machine vision tests
- Alignment module, for 5 DOF (Degree of Freedom) or 6 DOF alignment and test
- Target Module. This is either with intermediate diopter with narrow field of view target or without diopter but with corner targets for wide field of view cameras
- Base Module including Industrial computer & touch screen

1.2 Documentation

The documentation provided with this Pixid 500 Camera Module Active Alignment System provides a complete document package for selection, installation, operation, maintenance and repair.

Part 1: Introduction: This section provides an overview of the Pixid 500 and its subsystems.

Part 2: Safety: This part provides an overview of the safety concerns and requirements for the Pixid 500.

Part 3: Installation: This part provides site preparation, unpacking, and installation information for the Pixid 500. This part includes all set-up procedures, including initial checks and alignment.

Part 4: Subsystems: This part provides detailed information on the various subsystems of the Pixid 500.

Part 5: Operational Interfaces: This part provides detailed information on the interfaces to the Pixid 500.

Part 6: Operation: This part describes all operating procedures for the Pixid 500 including an overview of all controls and indicators.

Part 7: Adjustments and Calibration: This part provides all standard adjustments and calibrations required for proper operation of the Pixid 500.

Part 8: Software Reference: this part provides an overview of the software control features for the Pixid 500 and provides a complete error reference.

Part 9: Maintenance: This part provides maintenance schedules and procedures for standard maintenance of the Pixid 500.

Part 10: Troubleshooting: This part provides troubleshooting information for the Pixid 500.

Part 11: Drawings: This part provides all drawings and BOM supplies with the Pixid 500.

1.3 Manual Usage

This manual is designed to be used as an engineering, maintenance and operator's reference.

General information relating to all functions is provided in:

Part 2: Safety

Part 4: Subsystems

Engineering information is provided in:

Part 3: Installation

Part 5 Operational Interfaces

Part 8: Software Reference

Part 11: Appendices

Part 12: Attached Drawings

Maintenance information is provided in:

Part 3: Installation

Part 5: Operational Interfaces

Part 7: Adjustments and Calibration

Part 9: Maintenance

Part 10: Troubleshooting

Part 11: Drawings

Operational information is provided in:

Part 5: Operational interfaces

Part 6: Operation

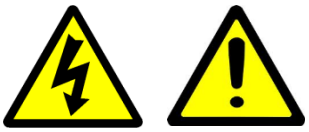
1.4 Manual Notation

This part describes the notation used in this manual for notes, cautions and warnings.



CAUTION

A caution notes actions or situations where equipment damage could result if the proper precautions are not taken.



DANGER

A danger notice emphasizes actions or situations where severe personal injury or death could result if the proper precautions are not taken. The type of danger symbol used indicates the type of hazard: electrical (lightning shaped arrow) or general (exclamation point).



WARNING

A warning describes actions or situations where personal injury could result if the proper precautions are not taken. The type of warning symbol indicates the type of hazard: electrical, radiation (radiating ball), or general.

2 Specifications

Specifications are subject to change without notice.

Dimensions and Weight:

- 1.4m W x 1.5m D x 2.5m H (see section 4.2 for dimensioned drawing)
- Machine Weight: approx. 1300 lbs (590 Kg)
- Shipping Weight: approx. 1800 lbs (820 Kg)

Electrical Requirements:

The Pixid 500 is configured for one of the following power configurations. Check the machine nameplate for the correct voltage for your machine:

- Input Voltage:
  - 115 VAC (North America)
  - 230 VAC (Europe and Asia)
- Current Draw:
  - 10A / 115 VAC
  - 5A / 230 VAC
- Recommended Outlet Service: 15A
- Input Frequency: 50/60 Hz

Compressed Air:

- Pressure: 80 psi (0.55 MPa) minimum
- Consumption: < 0.2 scfm (5.7 L/min)
  - On-board vacuum option: add 3.5 scfm (100 L/min)
  - Ionized blowoff option: add .1 scfm (3L/min)
- Input connection: 8mm SMC One-Touch fitting

Vacuum (optional):

- Input connection: 8mm SMC One-Touch fitting

Machine Throughput:

- Up to 240 UPH for camera alignments

Controls PC

- Operating System: Windows 7 Embedded, 32-bit
- Real Time Operating System: VxWorks
- CPU: Intel Core i7 2.3 GHz
- Memory: 8 GB
- Expansion slots: 2 PCIe x4, 1 PCIe x1
- Hard drives:
  - OS: 32GB CFAST Drive
  - Data: 500 GB SATA drive
- Display: 15” touchscreen

DUT Camera Interfaces

- Native camera interfaces:
  - MIPI (1-4 lanes)
  - HiSPi
  - Parallel
- Other Supported Interfaces (requires conversion at fixture:
  - Serialized LVDS (Automotive)
  - sub-LVDS (Sony)
  - SLVS-EC (Sony)
  - Ethernet
- Camera control:
  - I2C
  - SPI

Environmental Requirements:

- Temperature: 15-30°C
- Humidity: recommended 35-65%

Laminar Flow Exhaust (optional):

- Outgoing connection: 4" trade size ducting port (either top or side exit)

3 Safety

This section is provided to supply necessary safety guidelines for the Pixid 500.

Please note: These safety recommendations are basic guidelines. If the facility where the Pixid 500 is installed has additional safety guidelines, they should be followed as well, along with the applicable national and international safety codes.

3.1 Personnel Safety Guidelines



The Pixid 500 may impart several direct safety hazards to personnel if not properly installed or operated. Please adhere to the following safety precautions:

- Personnel operating the Pixid 500 should be properly trained.
- Possible injury can result from the operation of the Pixid 500.
- Know the location of the following:
  - Fire extinguisher
  - First Aid station
  - Emergency eyewash and/or shower
  - Emergency exit
- Be aware of sharp edges while working around the Pixid 500.
- Eye protection and safety shoes should be worn prior to servicing the Pixid 500.
- Observe the facility guidelines pertaining to loose clothing while working around the Pixid 500.
- Perform a complete review of the **Material Safety Data Sheets** (MSDS) for each adhesive material used to operate service or clean the Pixid 500. These sheets are provided by the supplier. Basic safety information sheets are supplied at the end of this section, but it is recommended to refer to the supplier MSDS for up-to-date information.
- Ergonomic hazards may exist with certain repair procedures.

3.2 Equipment Safety Guidelines

The user is accountable for the following safety concepts:

- If hazardous materials are to be present, users must take responsibility to observe the proper safety precautions.
- The Pixid 500 is provided with an Emergency Machine Off (EMO) device.
- Do not place cables where they could cause a safety hazard.
- Do not place the Pixid 500 where it may be subject to physical damage.
- Ensure that all power connections to the Pixid 500 are properly grounded.



CAUTION

Use of the Pixid 500 for any purpose other than camera module assembly or test is not recommended and may cause damage to the Pixid 500.

3.3 Disconnect Devices

The Pixid 500 incorporated a main disconnect device and an Emergency Machine Off (EMO) device.

3.4 Lockout/Tagout

The Pixid 500 is provided with a lockout/tagout provision. It is recommended to use the lockout/tagout when servicing the machine.

3.5 Mechanical Hazards

The Pixid 500 is a complex device and operators must be trained to stay clear of parts operating automatically. Proper precautions such as wearing protective glasses and shoes, operating with caution, and using protocols recommended in Kasalis training, should be taken.



WARNING

The Pixid 500 is heavy and moving it or operating it without taking proper precautions could result in personal injury. The machine should be uncrated and moved into position by properly trained personnel only.

3.6 Electrical Hazards



DANGER

All power to the Pixid 500 must be disconnected before servicing to prevent the risk of electrical shock. The main power disconnect on the side of the machine should be locked and tagged using the LOTO procedure.

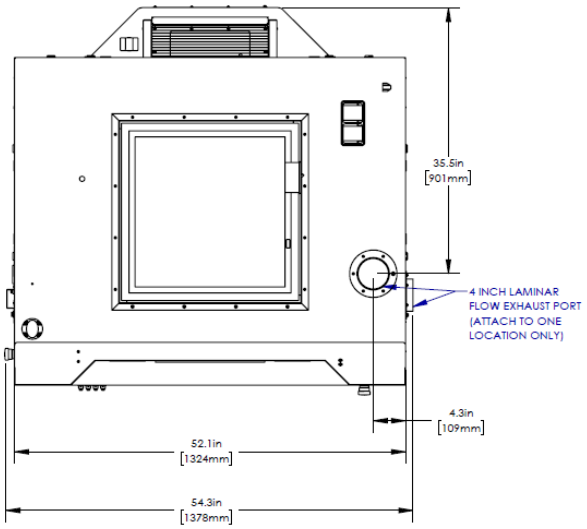
4 Installation

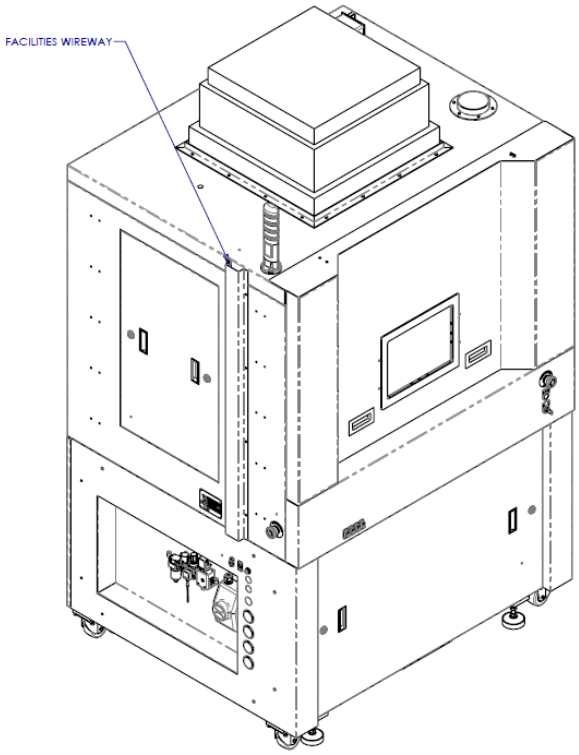
4.1 Site Requirements

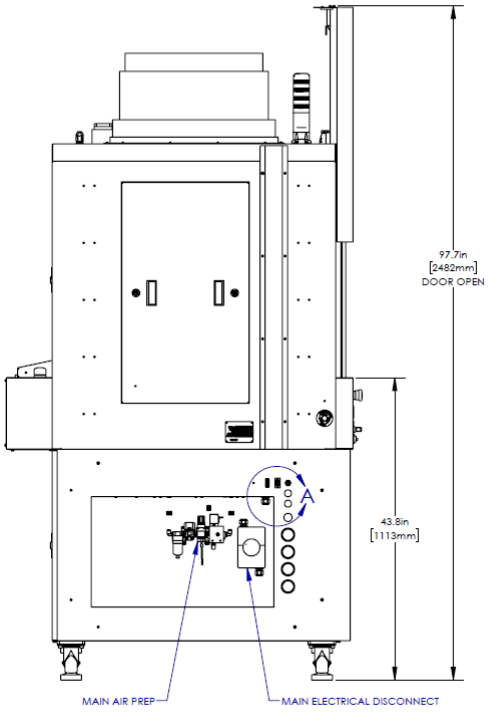
Before installing the Pixid 500, the site where it will be located must be properly prepared and checked for the appropriate electrical and communications connections. The site should be big enough to ensure proper clearance for the operation and servicing of the Pixid 500. These space requirements can be determined by the user, with the understanding that access to the sides and back of the machine are required for power, but the machine is on wheels to facilitate ease of arrangement. Environmental requirements include an operating temperature of 15 to 30 degrees C and a controlled humidity. Large variations in temperature and/or humidity can result in inconsistent alignments.

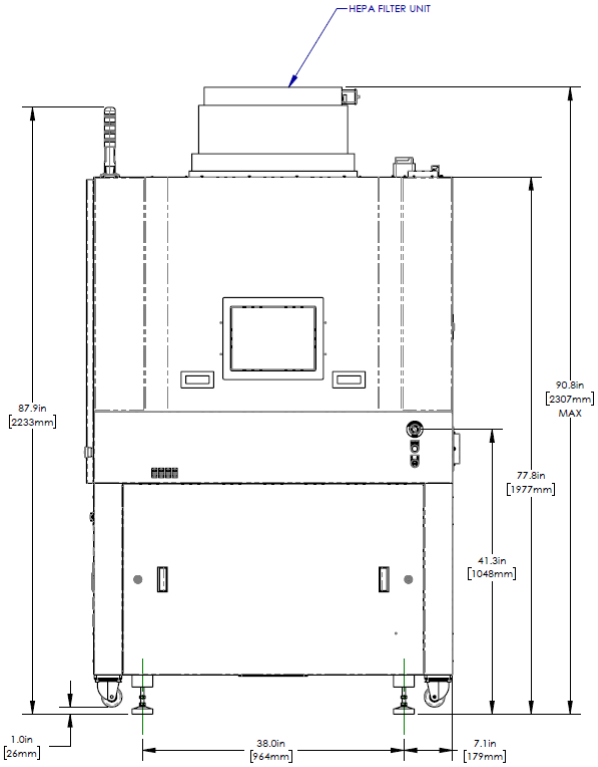
4.2 Pixid 500 Dimensions

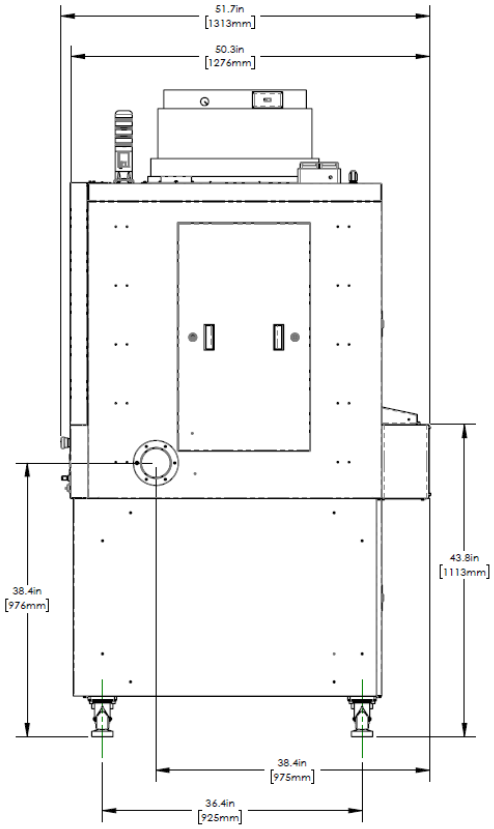
The following images show the external dimensions of the Pixid 500.

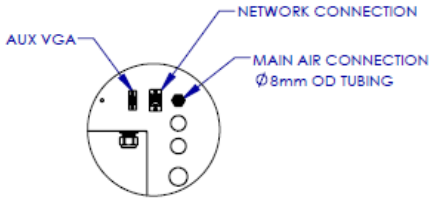












DETAIL A
SCALE 1 : 8

4.3 Unpacking and Inspection

Unpack the Pixid 500 with care. Inspect and verify its contents against the checklist provided below. Should anything seem missing, please report this to Kasalis.

Table 3.2: Packing Checklist

Contents	Checkbox
Pixid 500 Base (stand with platform)	❒
Rotary Module (x-shaped)	❒
Target Module (with image)	❒
Dispense Module	❒
Alignment Module	❒
Loading Station	❒
Pixid 500 Enclosure (five-sided top with doors and touchscreen panel)	❒
User's Manual (this one)	❒

4.4 IT Warnings and Recommendations

This section describes how to install am Pixid machine on the factory domain.

4.5 Overview of Pixid Industrial Computer

The Pixid industrial computer runs two different operating systems:

- Windows 10 Enterprise
- VXWorks (a customized version created by B&R Automation and branded “ARemb”)

These two operating systems are managed by a bare-metal Type 1 Hypervisor created by B&R Automation. The machine control software that is executed on the computer runs under both operating systems, and there is a local virtual network adapter “Real-Time Hypervisor PCI Network Adapter” created by B&R Automation that is used to carry a large amount of continuous communication between the control software executing on both operating systems using multiple TCP/IP sockets.

4.6 Use of Digital Guardian

It has been determined empirically with Pixid machines deployed at a Jabil facility that full deployment of the computer on a factory domain with all domain policies applied including Digital Guardian causes issues indicated by severe degradation in system performance of the machine control software. For this reason, it is strongly recommended not to deploy DG on Pixid industrial computers.

4.7 Use of SCCM

Previous versions of Pixid industrial computers ran Windows Standard Embedded 7 and an earlier version of the VXWorks OS customized by B&R Automation and branded “ARWin”. These earlier systems implemented a Type 2 Hypervisor. B&R Automation had informed Kasalis that Windows Updates could not be deployed on these older industrial computers as doing so had been shown to damage the ability of the Type 2 Hypervisor to manage the ARWin OS running as a guest in the WSE7 host operating system. However, B&R Automation has also informed Kasalis that Windows Updates can be deployed on the new industrial computers running Windows 10 and the Type1 Hypervisor. Because of this new guidance it is possible that use of SCCM to control Windows Updates may be acceptable on the new Pixid industrial computers running Window 10, but this has not been tested on Pixid systems deployed in the factory.

4.8 Use of Windows Defender and other Anti-Virus Solutions

The Pixid industrial computers ship from Kasalis with Windows Defender enabled but configured with several folder exclusions as shown below. This Windows Defender configuration has proven to work in a robust manner in providing protection against viruses and malware on the computer without jeopardizing system performance, and it is strongly recommended to recreate these local folder exclusions with any anti-virus solution installed on the industrial computer as part of joining the computer to a Jabil domain.

- C:\BRAutomation
- C:\Pixid
- C:\Program Files\BrAutomation

- C:\Program Files\Python37
- D:\Pixid

4.9 Use of Bitlocker Drive Encryption

It is recommended to not enable Bitlocker driver encryption on the boot or data drives of the industrial computer. This will mitigate reduction in control system performance during writing of data files generated by the control system software during processing of devices by the Pixid machine.

4.10 Windows Firewall

It is recommended to disable Windows Firewall on the Pixid industrial computer. If this is not done it may be possible to run the control system software by adding exclusions for these executable programs:

- "C:\Pixid\Bin\KSvisionServer.exe"
- "C:\BRAutomation\VisualComponents\BR.VC.ProjectLoader.exe"

Additional exclusions may also be required to allow operation of other control system software components however this has not been tested on Pixid systems deployed in the factory.

4.11 Local Admin Rights

It is recommended that the domain account used by the Pixid industrial computer is granted local admin rights. If this is not done it may be possible to run the control system software by changing the Pixid software executable "C:\Pixid\Bin\KSvisionServer.exe" shortcuts on the Windows Start Menu and Windows Desktop to be run "As Administrator". Additional changes may also be required to allow operation of other control system software components however this has not been tested on Pixid systems deployed in the factory.

4.12 Installation Procedure

4.12.1 Machine Leveling

Once the machine is in place, the 4 leveling feet on the bottom of the machine need to be lowered to a point where the machine is no longer resting on the transportation casters on the machine. This is a critical step that is required in order to adequately isolate the machine from external vibrations.

Once the machine is raised up on the four feet, the machine must be leveled. Adjust the feet accordingly in order to level the machine. Place a level on the index table at the align station for feedback during the leveling process.

4.12.2 Connecting Power to the Machine

The Pixid 500 ships with a power cord and international locking style power connector. If the user site requires that a cord is hard-wired to the machine, follow the procedure below to connect power to the machine directly.

Step 1: Remove the cover on the main disconnect switch:



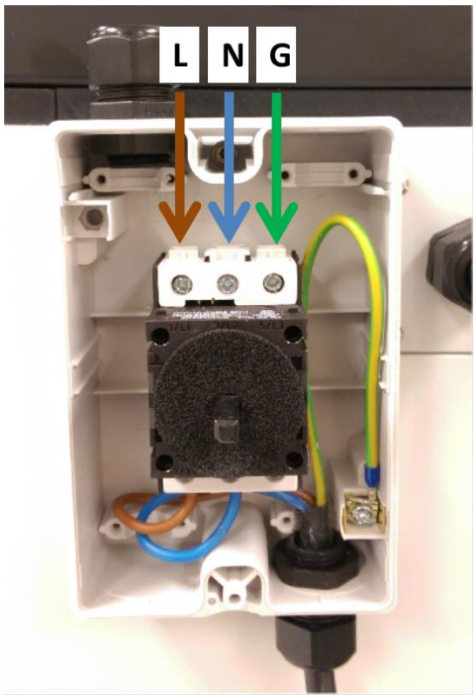
Turn handle to expose screw, then remove the screw



Remove handle

Remove screw

Step 2: Connect power cable to switch as shown below:



4.13 Initial Power-up Sequence

Please see Program Startup Initialization, for the list of defaults and commands associated with the program startup.

5 Subsystems

5.1 Pixid 500 Internal Subsystems

The Pixid 500 Pro is comprised of several functional modules that are integrated into one complete system used for volume manufacturing of camera modules. The modular design of the Pixid systems allows for streamlined assembly and delivery of systems as well as simplified maintenance and operation. The individual modules that are included in with Kasalis' standard Pixid 500 Pro systems are described below:

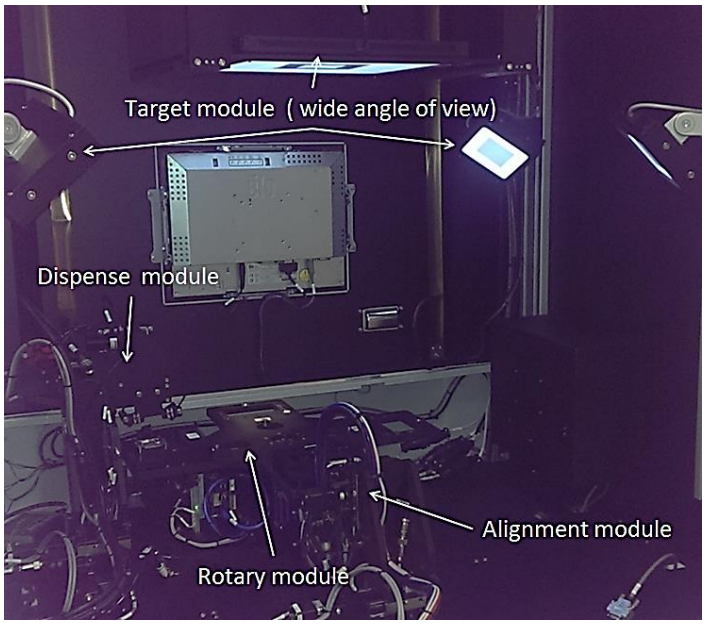


Figure 1: Pixid 500 Pro Subsystem

5.2 Alignment Module

The alignment module comprises a 5/6-axis (X, Y, Z, θx, and θy) motion stack for lens positioning during the active alignment process. Customized lens grippers are attached to the alignment module to hold the lens so that it may be moved relative to the image sensor in order to find the position yielding best focus and optical centration. UV delivery wands are attached to the lens grippers to allow for the lens housing to be attached to the sensor housing via UV cure of adhesive after the optimal alignment position is determined through the active alignment process.

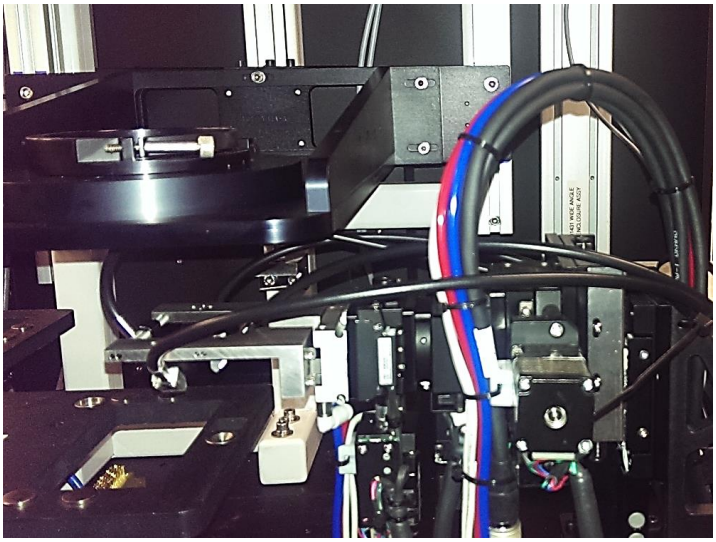


Figure 2: Alignment module

Alignment Module Specifications:

Degrees of Freedom	5 (X, Y, Z, θx, θy) – Optional 6th DOF Available
Linear (X,Y,Z)Resolution	< 0.2 µm
Angular (θx, θy, θz) Resolution	< 0.01 deg
Linear Travel Range (X,Y,Z)	50 mm
Rotation Travel Range (θx, θy, θz)	+/- 5 deg
UV Cure Source	Dymax BLueWave LED Prime UVA

5.3 Dispense Module

The dispense module offers automated, precision dispense of UV-curable adhesive. DELO DUALBOND AD 345 is chosen to assemble modules. The dispense module is equipped with a machine vision camera and illumination in order to automatically determine the dispense site for each camera module presented to the machine to accommodate part to part variation. Additionally, the dispense module's vision system can be used to inspect the dispensed adhesive to verify the size and continuity of the adhesive bead. The module also includes an XYZ calibration mechanism to determine the position of the dispense tip relative to the camera whenever the dispense tip is replaced or the dispense head is removed from the system for cleaning/maintenance

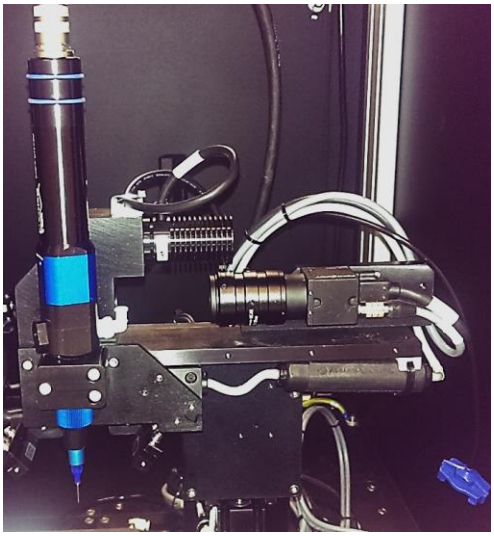


Figure 3: Dispense Module

Dispense Module Specifications:

Dispense Technology	Positive displacement pump
Site Recognition and Dispense Verification	Machine vision camera with illumination
Dispense Pattern Options	Lines, Arcs, Dots (or any combination)
Minimum Dispense Bead Width	300-500um (adhesive dependent)
Linear Positional Resolution	< 1 µm
Dispense Positioning Accuracy (X, Y, Z)	25 µm

5.4 Rotary Module

The indexing module is a motorized rotation table that moves the camera module from one process module to another. Attached to the indexing module are custom designed camera fixtures that are used to mechanically and electrically engage camera modules loaded into the machine.

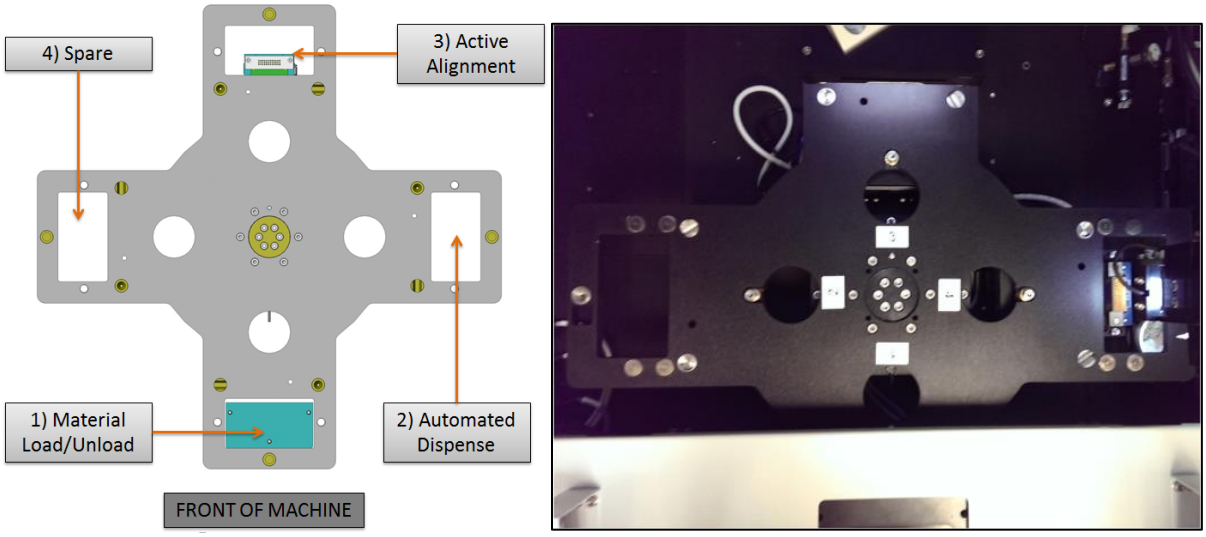


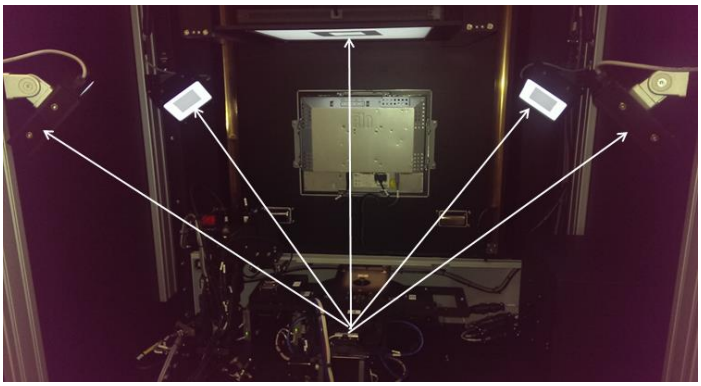
Figure 4: Rotary module (right image) & rotary motion (left image)

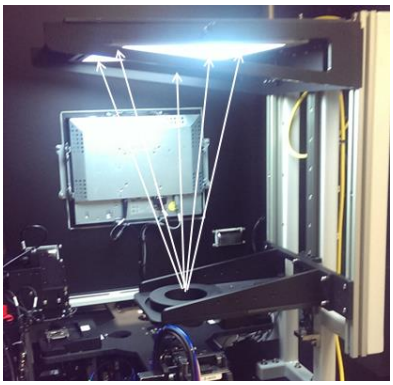
5.5 Target Module

The target module includes a backlit targets for feedback during the alignment process as well as for functional testing of the camera module. The backlit targets provide uniform illumination across the entire target area for optimal alignment and testing measurements. The alignment/test target is placed at desired height from lens surface to the target. The target contains features for performing standard optical tests. The features are five slanted edge targets; 1 on-axis (center) and 4 off-axis (corner) target features. Slanted square target feature is used to calculate MTF based on the ISO 12233 standard. MTF feedback is used for optimizing focus and for verifying that the camera module's focus satisfies pass/fail criteria after assembly.

The target modules can also be configured for:

1. Visible spectrum
2. Near IR spectrum
3. Far IR spectrum





fffff

Target Module Specifications:

Diagonal FOV Between Corner Targets	40deg (full angle)
Nominal Target Object Distance	8.0mm
Target Positioning Resolution (Mechanical)	<0.1mm (accuracy of setup will depend on camera module focus sensitivity)
Backlight Uniformity Max Variation	10%
Backlight LED Wavelength	NIR
Feedback for Active Alignment	MTF based on ISO 12233



5.6 Base Module and Safety Enclosure

The base station incorporates the AC Distribution and DC distribution panels and the industrial control PC and the touchscreen monitor.

The safety enclosure protects the operator from moving parts and UV light exposure and also prevents external light from affecting the alignment and testing.

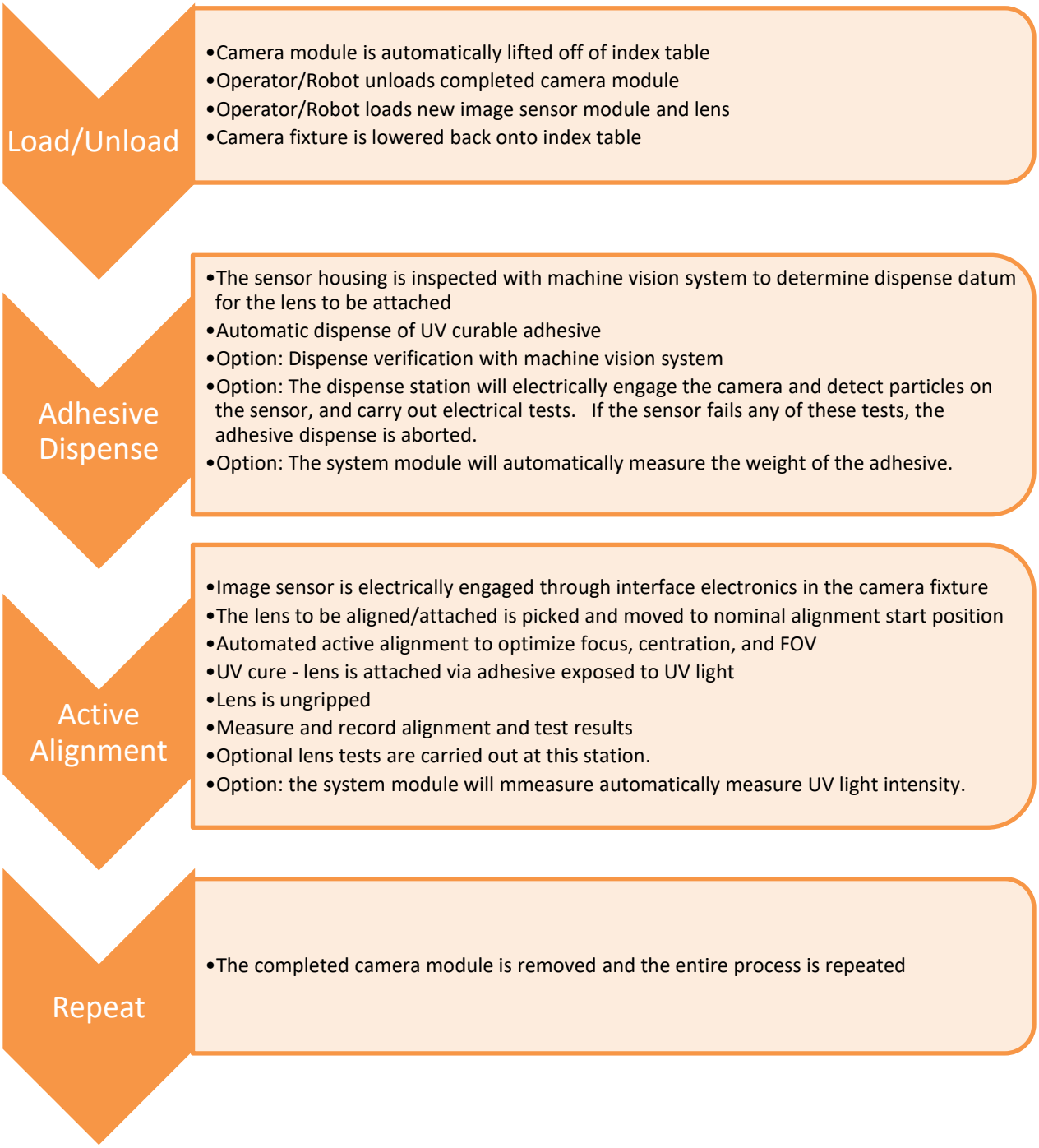
5.7 Spare Station

The Pixid 500 has a 4th station, which is unused. The spare stations can be used for custom user tests as secondary UV cure, fixture barcode scan…etc

6 Process integration

6.1 Top level process flow

The below process flow chart shows how material will move through the Pixid 500 system in order to obtain the optimal TAKT time.



6.2 System Sub-Process Definitions

The Pixid 500 Pro system is comprised of three main process stations: the load/unload station, the dispense station, and the align station.

6.2.1 Load/unload station

1) The camera fixture is automatically lifted off of the indexing module such that it is brought to the loading area of the machine.
2) The operator is prompted to unload the completed camera module and to load a new sensor module and lenses for assembly. The operator prompt will indicate if the completed module has passed or failed the final testing.
3) The operator unloads the completed module and loads a new lens and a sensor module into the camera fixture. The operator acknowledges that the parts have been loaded using a finger switch.
4) The camera fixture is automatically lowered back onto the indexing module.

6.2.2 Dispense station

1) Optional: If the particle test is enabled, the system will execute the test prior to adhesive is dispense. If the sensor fails due to particle test, the sensor will not be dispensed on.

2) The dispense module's vision camera with illumination is moved over the image sensor and using machine vision feedback, the datum for the dispense patterns is determined.

3) Based on the dispense datum feature's position, the UV curable adhesive is dispensed onto the image sensor housing. The dispense datum feature's position depends the calibration between the dispense tip and the camera system, and the configurable parameters for the dispense pattern (lines, dots, and/or arcs).

4) The dispense module's camera with illumination is moved over the image sensor and using machine vision feedback, the dispense patterns are verified for proper thickness and continuity of dispense.

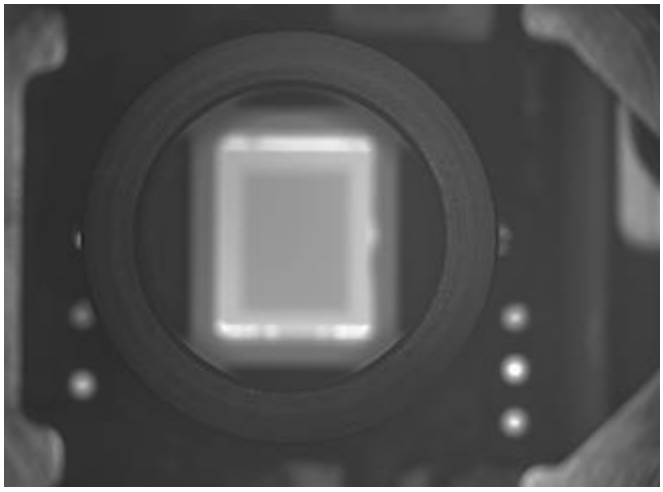


Figure 5: Pre Dispense Inspection to Determine Dispense Pattern

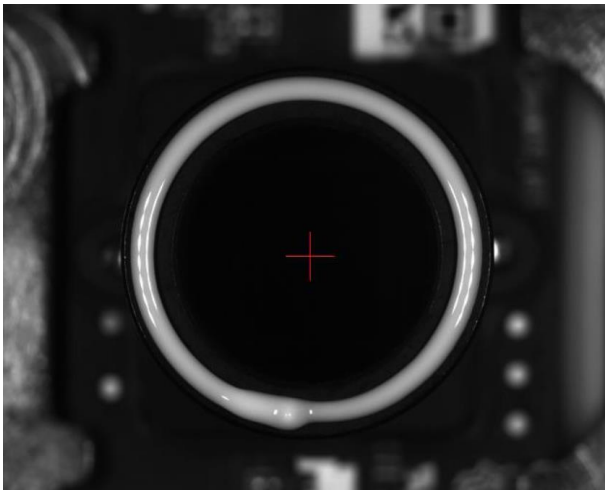


Figure 6: Post Dispense Inspection using machine vision camera

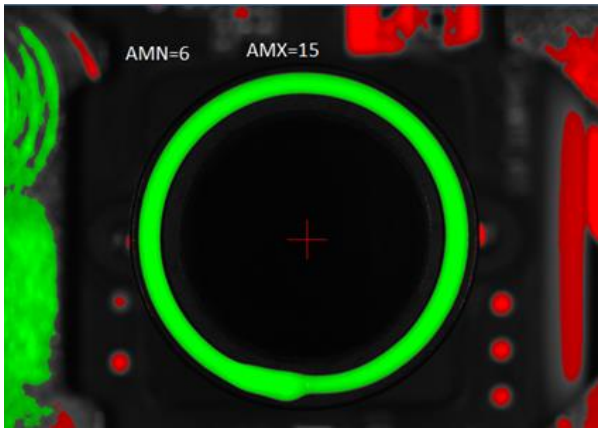


Figure 7: Figure 13–Post Dispense Inspection results

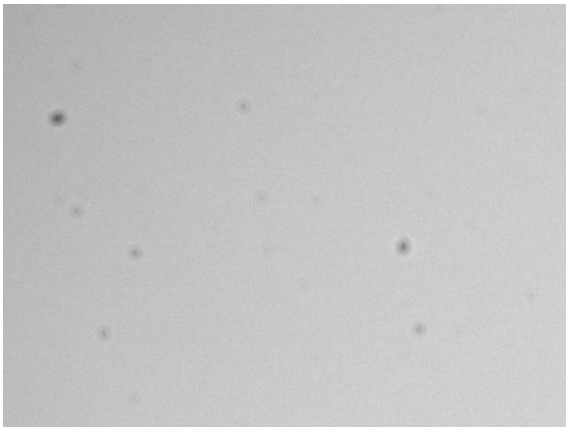


Figure 8: Particles Test before the image is processed

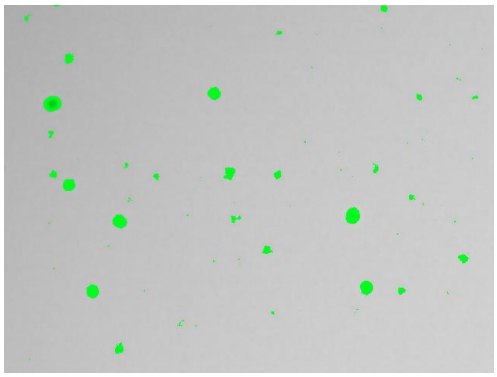


Figure 9: Particles test after the image is processed

6.2.3 Align station

1) When the camera fixture reaches the align station, the camera fixture is automatically engaged electrically from below in order to power on the image sensor and stream images.

2) Using the customized target module's motion stage, the target is moved upward to allow room for the lens to be picked and placed.

3) The lens is picked from the camera fixture by the alignment grippers and moved over the image sensor to the nominal alignment start position.

4) The target module is lowered to a configurable position to set the object distance for active alignment.

5) The lens is actively aligned to the image sensor. During active alignment, the tip/tilt of the lens is adjusted to be within a configurable threshold, the lens is optically centered within a configurable threshold, and the lens is focused to a position optimized for on and off-axis performance based on a configurable weighting factor.

6) On and off-axis MTF is measured and compared to configurable PASS/FAIL limits.

7) The lens is moved upward by a configurable distance to compensate for the shrinkage of the UV curable adhesive.

8) UV cure is executed to attach the lens housing to the image sensor housing.

9) The lens is un-gripped.

10) On and off-axis MTF is measured and compared to configurable PASS/FAIL limits.

11) The target module is moved upward to allow for clearance for the grippers to retract away from the assembled camera module.

12) The electrical engagement to the camera fixture is automatically disconnected so that the camera fixture may be indexed.

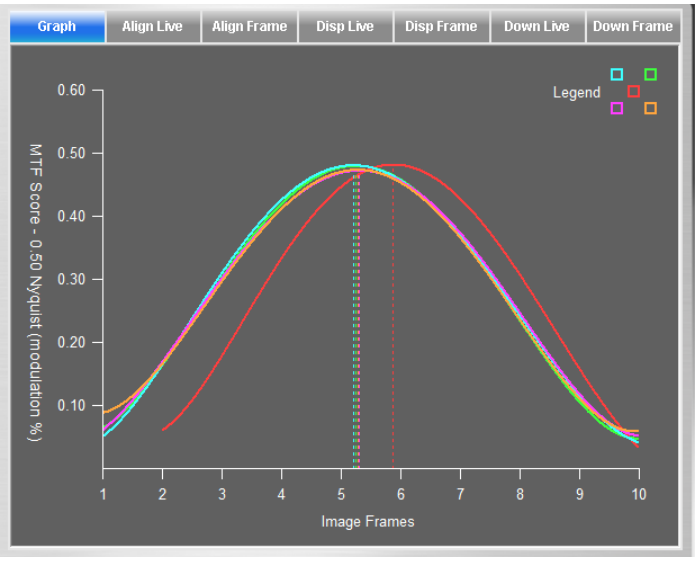


Figure 10: Thru focus curve during alignment process & tip/tilt

6.2.4 Spare station

Pixid 500 pro is designed with extra station (spare station). The standard system, spare station is not used for any extras test. However spare station can be used for custom advanced optical tests by customer request. Spare station can be used for secondary UV cure, fixture barcode scan…etc

6.2.5 Process Tracking and Data Reporting

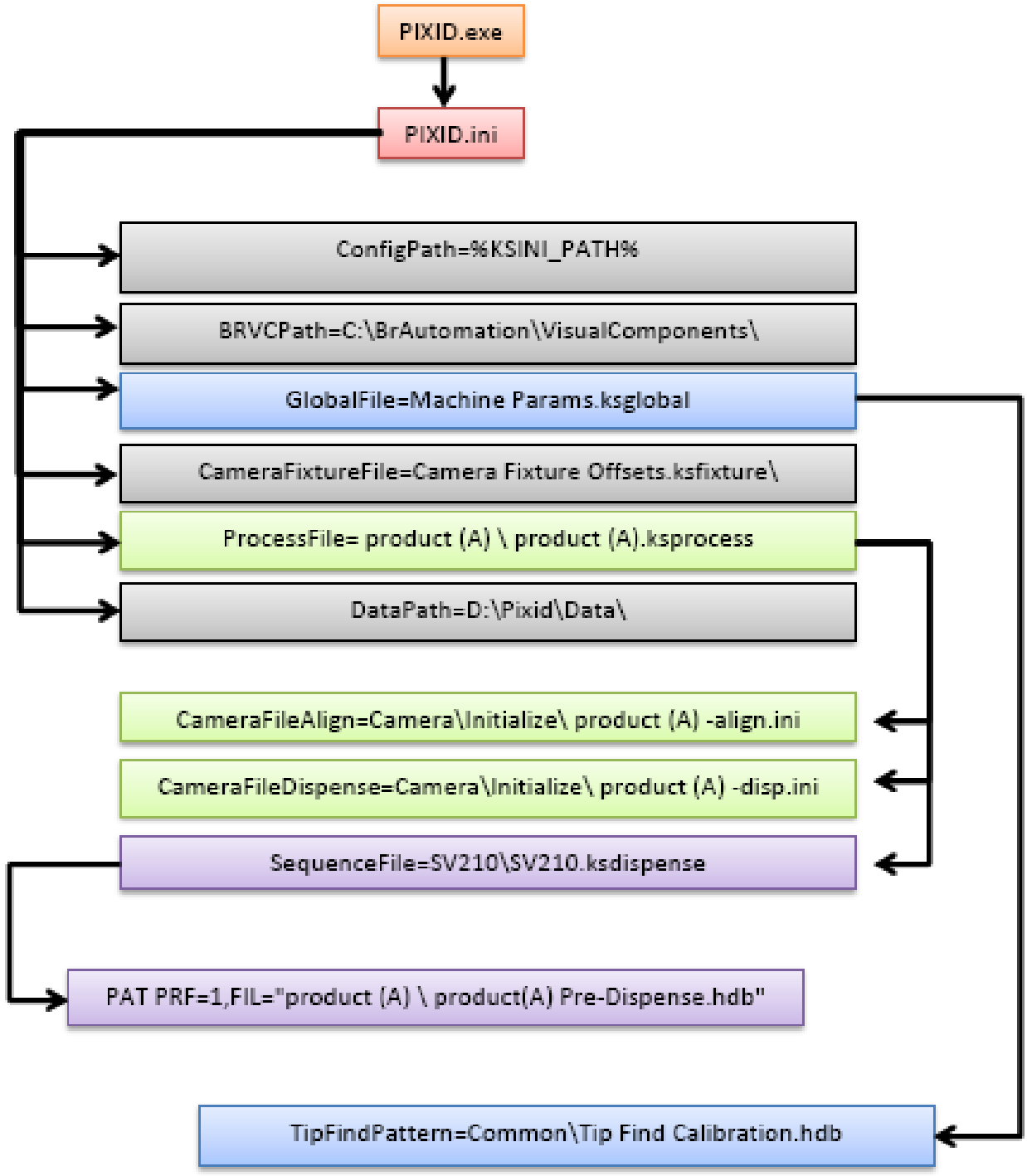
All alignment results and test results will be logged into a database for process tracking. All known process data will be stored to csv files for off-line analysis. In addition to process tracking, each image captured for automatic MTF verification and other tests shall be optionally stored to file. Further, the Pixid software includes an MES interface option. As part of the MES data interface option Kasalis provides a basic C++ example software solution including full source code that demonstrates how to receive device data output generated by the Kasalis Pixid software for each device that is completed on the machine. This software solution is created in the form of two Windows console applications: One that acts as an emulator of the Pixid software to generate the data, and the other that acts as the recipient of the data. The data is sent to the recipient via a single TCP/IP socket, the creation and management of which is developed by the customer. Sample code demonstrating this functionality is provided to assist with the customer integration of the MES interface. The software solution is presented as a MS VS2010 un-managed C++ solution.

Data reports for uptime, downtime, idle time, yield, failures, and UPH will be automatically generated. Optionally, these reports can be set to be emailed to specific personnel should a critical downtime condition occur, if yield falls below a certain value, failures per hour is above a certain level, or if UPH falls below a specified rate

6.2.6 Control System

The control system software is designed to allow for process setup/modifications, process sequencing, process tracking, status reporting, operator prompts, maintenance operations, system calibration,

display of alignment/testing results, and data logging to standard database formats. The interface allows the customer to easily change process parameters to optimize or modify the entire process. The control system software includes multiple password protected access levels which grants access to different machine functions for qualified personnel. All login attempts are logged to file and emailed reports can be sent if there are excessive failed logins



7 Basic machine functions definitions

Pixid software provides a variety of function and test. The table below summarizes the basic machine functions

Operation	Definition
Build	Automated camera build sequence for production
Retest	The automatic re-test sequence is used to measure the performance of a built camera as well as determine its pointing error measured as X and Y error in the 2 dimensional image output by the camera. It performs the following steps: Opens the camera at the align station and measures its performance by collecting MTF scores
Stop Cycle	Stops the build or re-test sequence after processing of the current alignment and dispense operations are complete. Does not index the rotary table
Run out	Stops the build or re-test when there is no more cameras or lenses for production
Park	Returns all motion axes and I/O solenoids to their park positions. This operation un-grips the lens at the alignment station if necessary. If there is a lens gripped at the alignment station it will return it to the fixture lens pocket if the lens has not yet been in contact with adhesive or sealant. If the lens has been in contact with adhesive or sealant activates the UV cure system to bond the lens to the device PCB before parking
Dispense	starts the dispense operation running for the part in the fixture at the dispense station. This allows the testing of the current dispense operation. If the Dispense Continuously check box is checked, it repeats the dispense operation until unchecked or Dispense button is pressed to stop operation
Tip Find	The Tip Find button starts the location of the dispense tip in X, Y, and Z axis location from its home location to accurately position tip during the dispense process. Any time a new tip is replaced or new adhesive is replaced the Engineer should run tip find
Calibration	starts with a tip find then proceeds to locating the calibration fiducial cross hear markings on the top of the tip find assembly
Teach(under dispense tab)	teach XYZ positions for dispense process
Head 1 Reset	allows the Engineer to update the adhesive information, type, batch ID, expiration

	date, volume left in tube, and hours left before returning to freezer
Prime Head 1	The Prime Head 1 button will prime adhesive into cup and return to dispense home position
Reset Prime	If dispense prime cup full, this option allows the engineer to reset the volume
Align	starts the align operation running for the part in the fixture at the alignment station. This allows the testing of the current alignment operation. If the Cure and Ungrip after Align check box is checked, it will activate the UV source to cure and run the ungrip tests. CAUTION: make sure that the front door is closed or proper safety glasses are worn before using this option
Characterize	Execute manual thru focus curve for the current camera at align station
Final Test	starts the final test (MTF scores, Centration, Focus Symmetry, Field Curvature, and User Test) with the current lens alignment or finished cured lens
Teach (under align tab)	Teach lean pick and place positions for alignment process
Return Lens	allows the user to return the lens from the placed alignment position to pick position (lens holder)
Repeatability	allows The user to repeat alignment process for one particular camera from specific number of time
Stability	The Stability test acquires image frames every interval in seconds, based on value in the Intv (sec) field, for a total of what is in the Reads field
UV Measurement	This is optional function: The UV Measurement button starts the automated measurement process of measuring the light intensity of each of the 4 UV light source positions
Initialization	Before operating the machine it must be initialized. This procedure homes all of the motion axes and places the machine in a state where it is ready to be operated. Machine initialization is performed by pressing the Initialize button in the program control section of the user interface.

8 System startup

To start the Pixid 500 control software click either the Pixid 500 icon on the Windows Start Menu or the Pixid 500 icon on the Windows desktop.

The startup screen will be presented as the software performs its initialization procedure.



Figure 11: Pixid 500 startup screen

Once the software completes its initialization the main user interface will be shown.

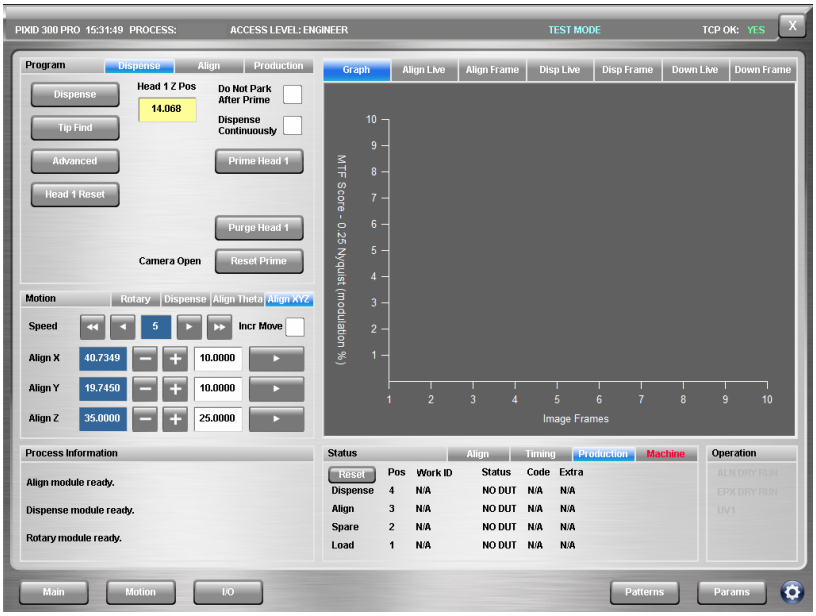
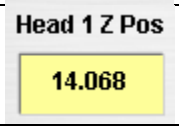
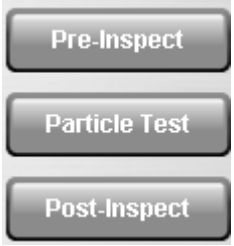
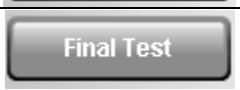
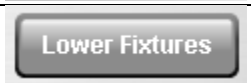
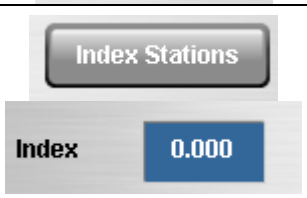
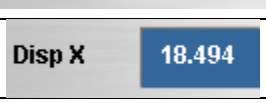
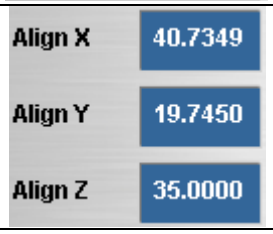
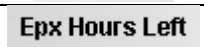


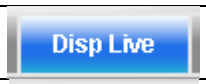
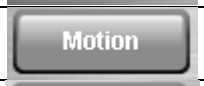
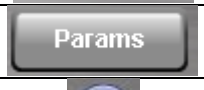
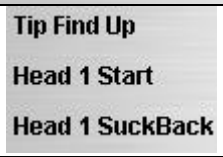
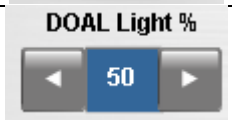
Figure 12: Pixid 500 Pro software GUI

8.1 PIXD Software Function Definitions

	Execute automated dispense operation only as defined in Ksdispense file
	Execute tip find routine to define dispense tip position
	Promote for advanced dispense operations
	Allows the Engineer to update the adhesive information, type, batch ID…etc
	Check the box to move XYZ axis to home position after prime head1
	Check the box to run dispense operation continuously
	Function to prime adhesive flow into cup and return to dispense home position
	Function to purge adhesive into cup to ensure that the adhesive is flowing through dispense tip
	Function to zero prime volume used
	Stream live image from image sensor at dispense station
	The current Z-height of dispense tip during dispense operation
	Function to that starts with a tip find then proceeds to locating the calibration fiducial cross hear markings on the top of the tip find assembly (Automated dispense calibration
	Teach Z height foe dispense tip with respect to dispense surface of image sensor
	Move XYZ axis to pre inspection position
	Move XYZ axis to particle test position
	Move XYZ axis to post dispense inspection position
	Execute active alignment operation
	Execute thru focus curve according to parameter ins Ksprocess file
	Run final test with the current lens alignment (MTF score, centration, focus symmetry)

	Setup pick and place positions
	Return lens from place position back to pick position (this operation is done after alignment process is completed)
	Execute repeatability test, setup number of alignments as required and dispense number of alignments runs during repeatability test
	Execute stability test, specify total # of runs as required at specific intevral (sec) and dispense total # of run during stability test
	Check the box to turn on UV cure after alignment is completed (not production mode)
	Stream live image from image sensor at alignment station
	Open and close the grippers at alignment station
	Build modules in production mode Execute final test in production mode for assembled modules Stop the current cycle on the machine (dispense and alignment operations) Stop the machine when there is no more modules for production Move index table to park position at load station System initialization before production mode
	Lower down air cylinders at load station, dispense station and alignment station all at once
	Switch from engineering mode to operator mode Switch from engineer mode to technician mode
	Move index table to 90 degrees Dispense the current position of index table in degrees (0deg-load station, 90deg dispense, 180deg alignment , 270deg spare, 360deg load)
	Display the current position of Dispense X axis

	Display the current position of Dispense Y axis
	Display the current position of Dispense Z axis
	Move axis in positive of negative direction.
	Move axis to absolute position as defined in the box. The axis will move to 21.126 mm
	Check the box to move increments moves instead of absolute position
	Move axis increments moves in positive of negative direction as defined in the box (0.010= 10 microns) units are mm
	Display the current position of TX Display the current position of TY Display the current position of TZ
	Display the current position of Align X axis Display the current position of Align Y axis Display the current position of Align Z axis
	Display the current position of index table (Position 1 at alignment station)
	Display unique WorkID generated for each camera
	Display pass and fail modules
	Display error codes for failed modules
	Display extra error code for failed modules
	Display adhesive volume information for production purposes
	Display the current setup for On/off axis %
	Display expiration date for adhesive
	Display align status if alignment is dry run Display dispense status if there is no adhesive is dispensed (dry run) Indicates if UV cure is on/off after alignment process is completed
	Display thru focus curves during active alignment process
	Display live image from image sensor at Align station
	Display last image saved by image sensor at Align station

	Display live image from image sensor at dispense station
	Display last image saved by image sensor at dispense station
	Display a live image from machine vision camera at dispense station
	Display last image saved by machine vision camera at dispense station
	Switch from other screen to Main PIXID GUI screen
	Switch to motion screen for XYZ axis
	Switch to input/output controls screen
	Switch to pattern design screen for pre dispense inspection
	Access to process parameters
	Switch to general configuration parameters
	In motion screen; Turn on/off axis
	In motion screen; move the axis to home position
	In motion screen; Acknowledge error if the axis is stopped/crashed
	In I/O screen; Open/Close grippers In I/O screen; Turn on/off UV light
	In I/O screen; Increase/decrease target light intensity
	In I/O screen; move Tip Find air cylinder up/down In I/O screen; Turn on/off dispense motor In I/O screen; Turn on/off dispense motor in reverse direction
	In I/O screen; Increase / decrease dispense motor speed In I/O screen; Increase/decrease particle light intensity
	In I/O screen; Increase/decrease DOAL light intensity

	In I/O screen; Increase/decrease Line light intensity
	In I/O screen; Move Align pogo pins up/down
	In I/O screen; Move dispense pogo pins up/down
	In I/O screen; Move load station up
	In I/O screen; Move load station down

9 Dispense module

9.1 Dispense module overview

Dispense module is assembled using XYZ stage, DOAL light (Diffuse on Axis Light), LED Line Light, Particles detection light (PD light), machine vision camera and dispense dosing unit.

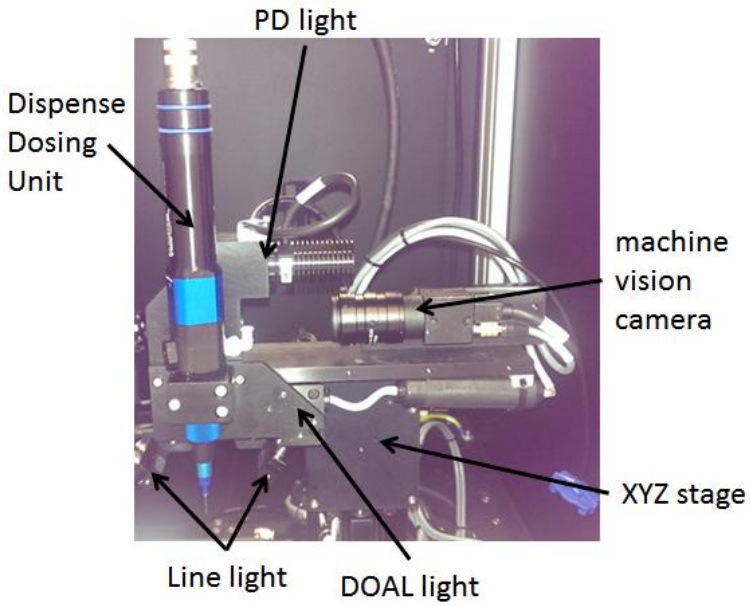


Figure 13: Dispense module components

There are various number of tests that can be executed at dispense station such as

- Particle detection test (optional)
- Pre-dispense inspection (standard)
- Post dispense inspection (standard)
- Dark signal test(optional)
- Hot/dead/defective pixel test pixel test(optional)
- Temporal noise test (optional)
- Automated glue weight (optional)

9.2 Machine Vision Pattern

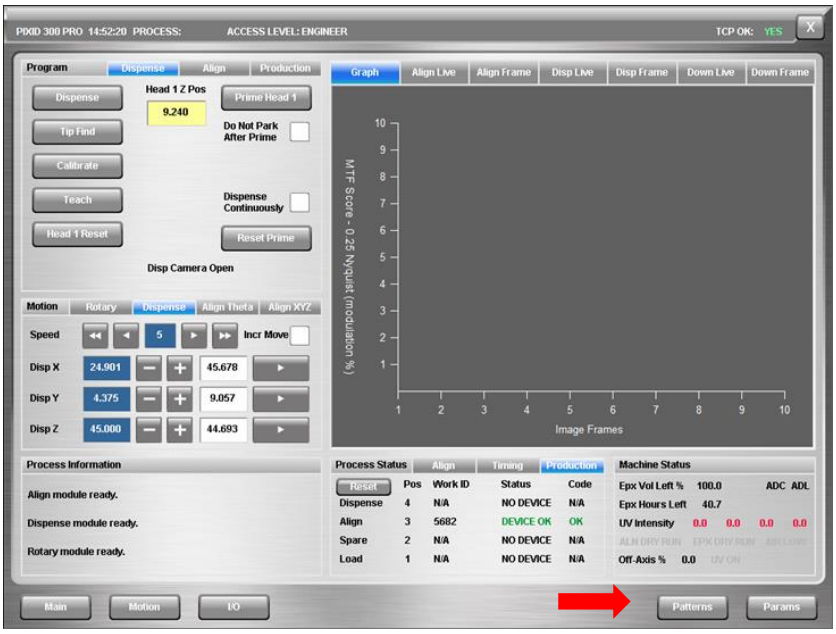
The Pixid system provides a down looking vision camera system to help with the Dispense operation such as

- Pre-dispense inspection

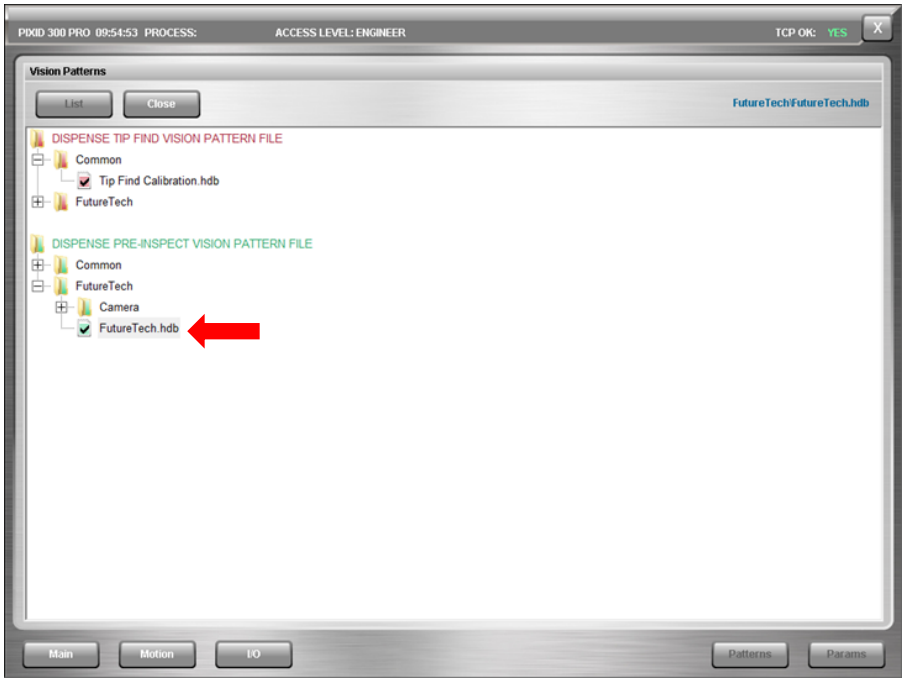
- Post-dispense inspection
- Particles test
- Tip find & Dispense calibration

Device vision patterns are stored in model files and editing is controlled from Pixid500 GUI screen. Patterns models represent the geometric contours of the object to help it accurately and effectively recognize object instances. An image in good focus, increase the accuracy of locating part featuers

The user can access vision patterns list from PIXID500 GUI screen by clicking on patterns button at the bottom right of GUI screen.



Next screen show the list of vision pattern that are stored in model files. The user can select the desired file by check the green box or edit the file by double click.

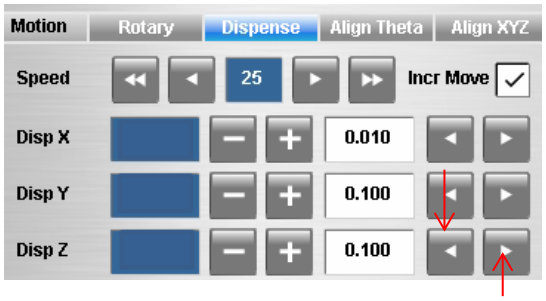


The list has two sections i) DISPENSE TIP FINP DVISON PATTERN FILE ii) DISPENSE PRE INSPECT VISION PATTERNS FILE.

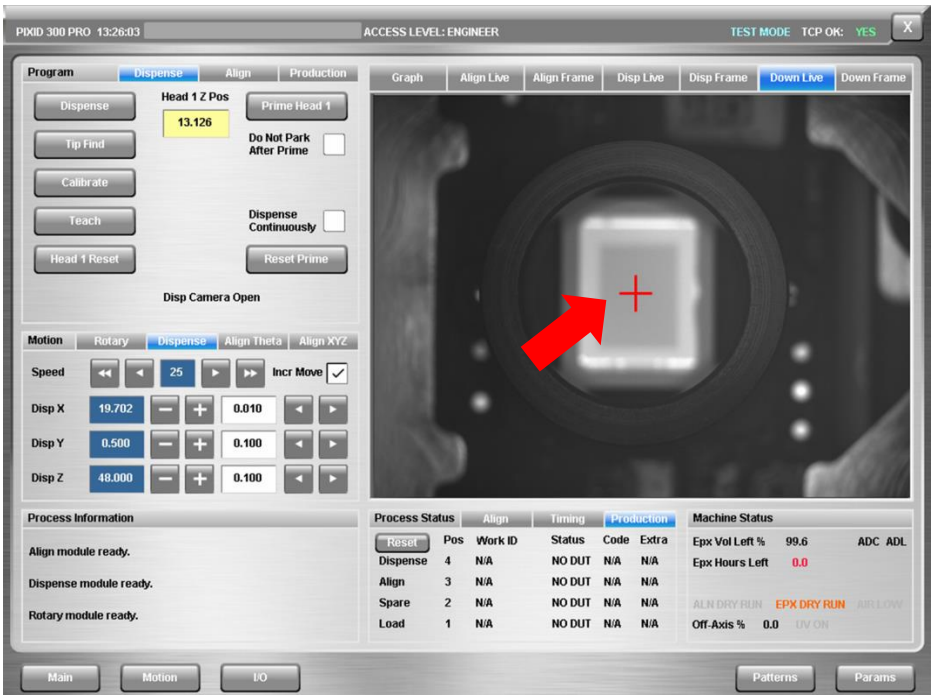
Under DISPENSE PRE INSPECT VISION PATTERNS FILE section the user can open the desired file depends on the product in production.

9.3 Design a new pattern

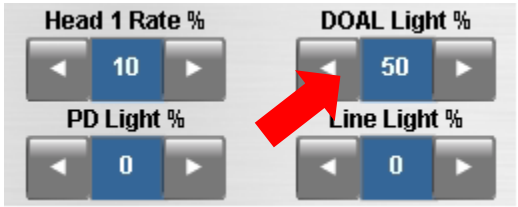
1. Load the new camera into fixture one.
2. Index to dispense station.
3. Manually move “dispense Z-axis” up and down until the image is in good focus.



4. Manually move dispense X & Y axis until it is center with the lens holder (image)



5. Set up DUAL light level to 50 as starting point.



6. Click on pattern file at the bottom right of GUI screen

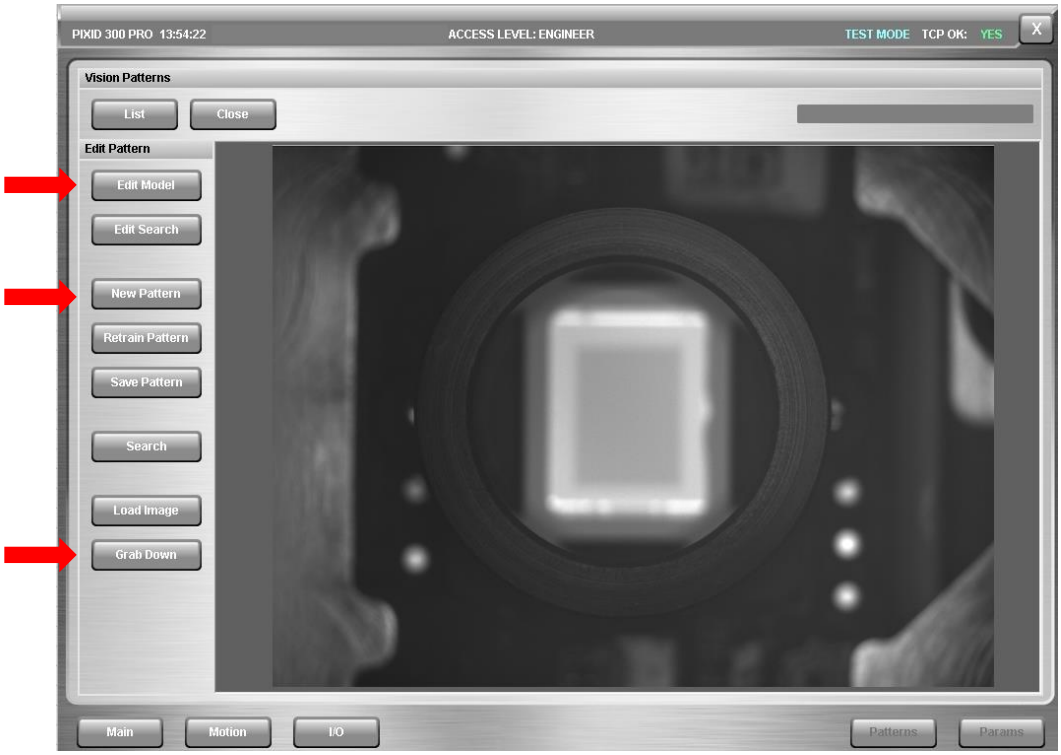


7. Under DISPENSE PRE INSPECT VISION PATTERNS FILE, double click on one old pattern model image



8. Click on below buttons in order to create a new pattern

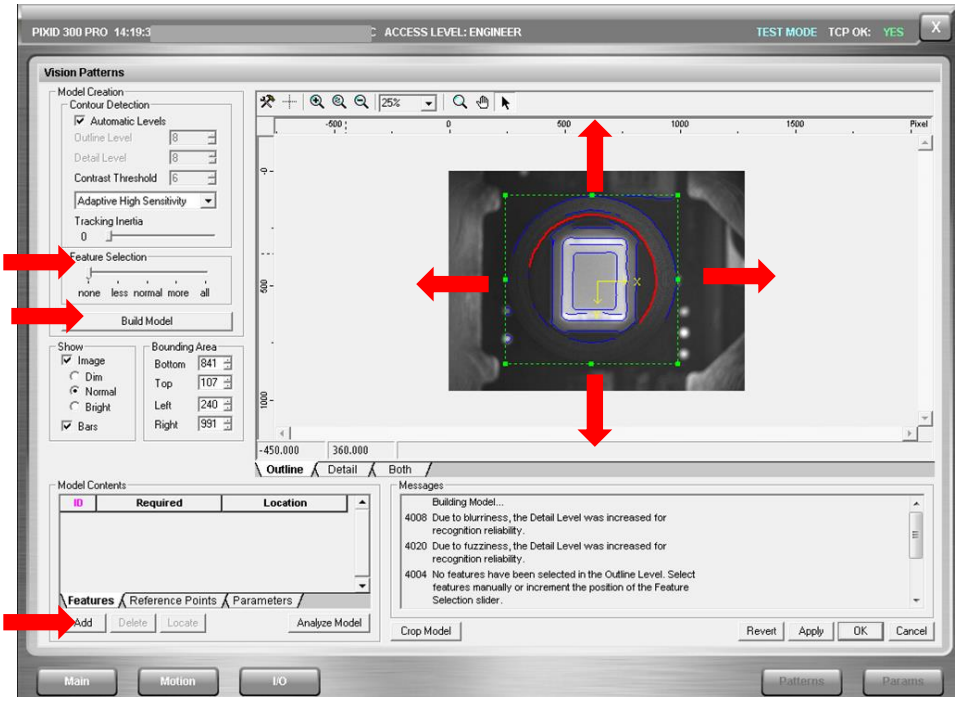
    a. Grab Down
    b. New Pattern
    c. Edit Model



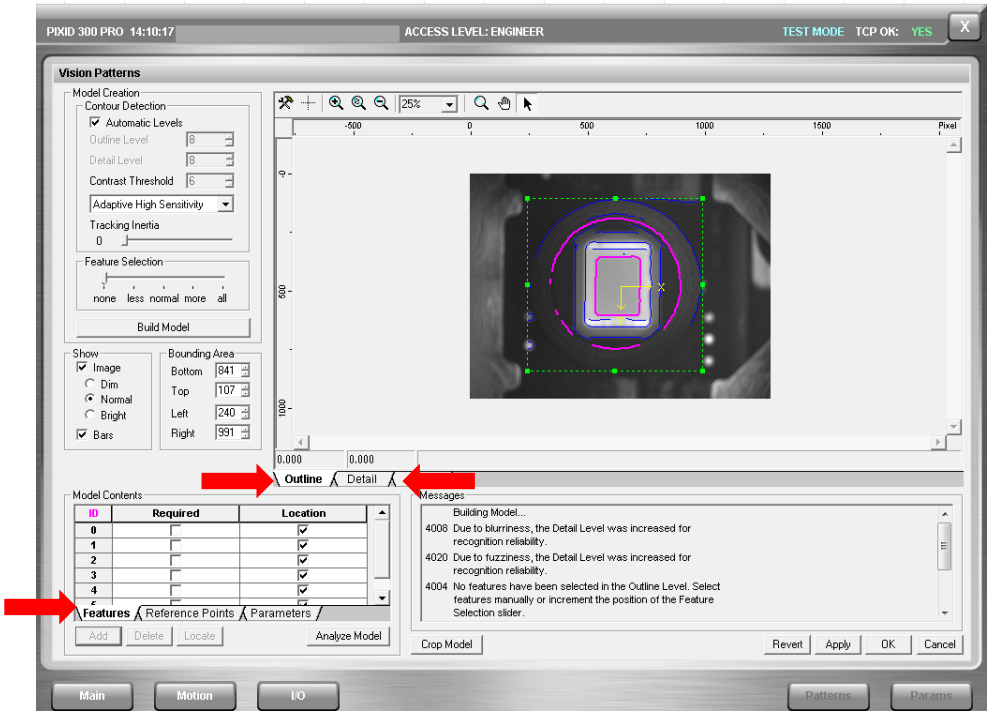
9. Adjust the green box size (bounding box or search box) to outline the area of the image to select contours that will be used to build the model to help detect and locate devices during the Dispense process.

    a. Adjust the size of bounding box
    b. Under feature selection section, move the bar from normal to none
    c. Click on build model button

d. Some features in the image of the right side of the screen will turn blue
    i. Blue: the software recognize those image features
    ii. Red: The features is selected but it is not added to the model contents
    iii. Purple : the features is selected and added to model contents
e. Double click on the blue features that will be used for dispense process. The blue features then turn Red.
f. Click on the add button at the bottom left side of the screen to add the feature to model contents.



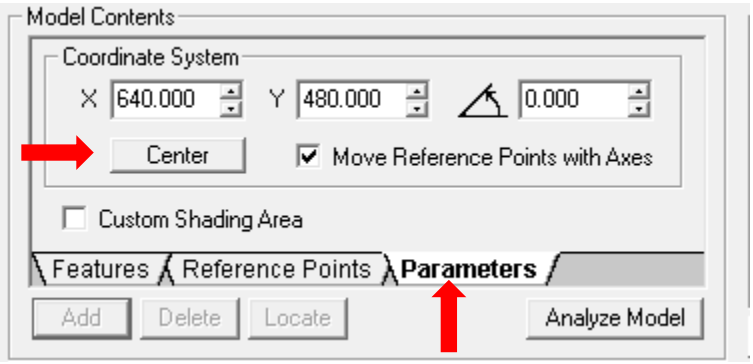
10. That features that has been added to model contents should turn into purple color.



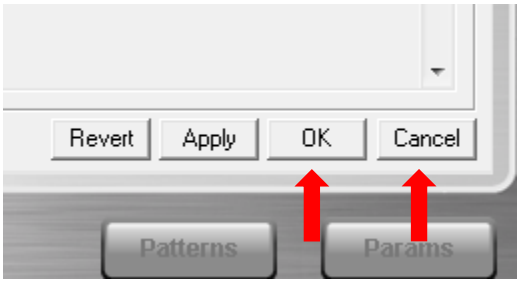
11. Follow previous steps to do detail section pattern design and select same features.

12. Click on parameters button under model contents section

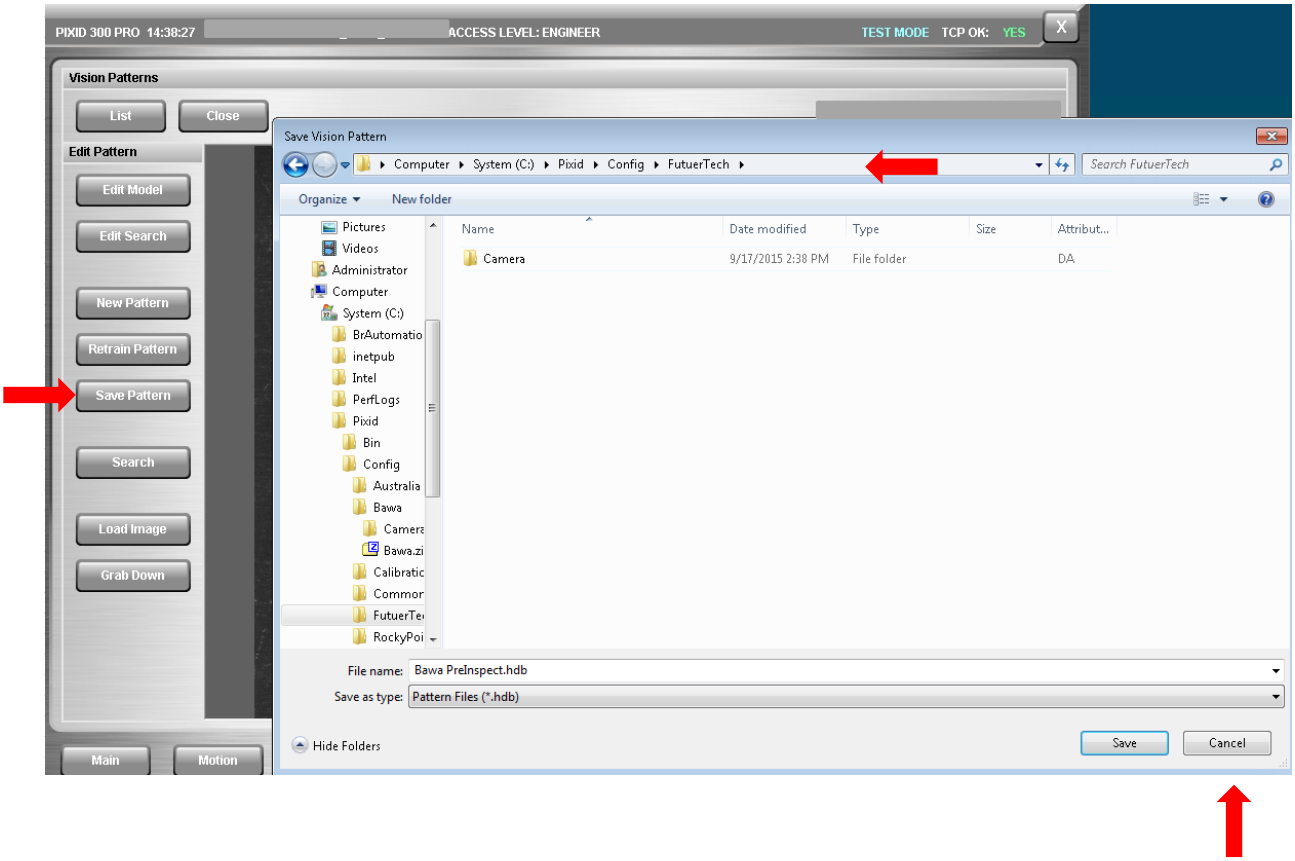
13. Click on center (this function will move the yellow X & Y axis into the center of the bounding box therefore the edges of the bounding box must overlaid the outside diameter of the circle in order to use the center point as the center of the circle) .



14. Click on apply button and OK button at the bottom right of the screen.



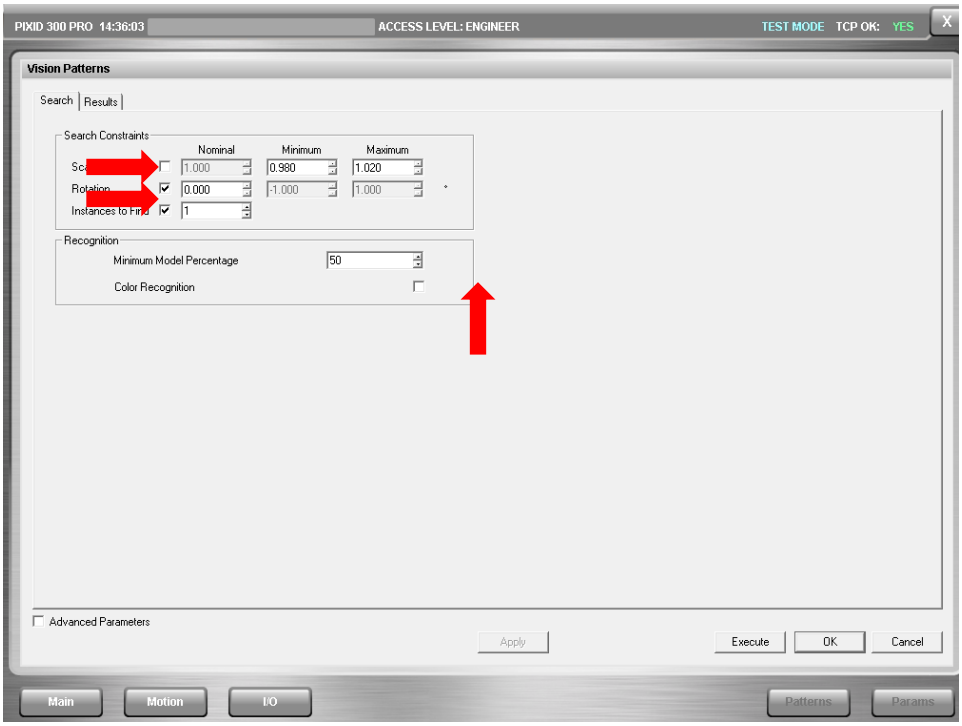
15. Click on save pattern to save the new pattern. Save the new pattern under C:\Pixid\Config\Futuer Tech folder (any folder under config has the camera process and dispense recipe)



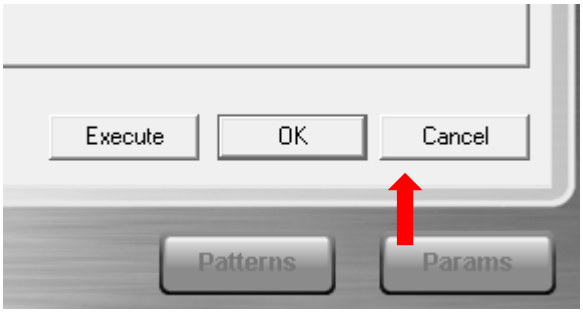
Then click on Edit search button

a. Note: Scale and rotation setting should change according to lens holder profile

b. Recognition must be at least 50% for all camera types.



16. Click on OK button at the bottom right side of the screen



17. Repeat step number 15

18. Click on close button at the top left of the screen to back to PIXID500 GUI Screen.



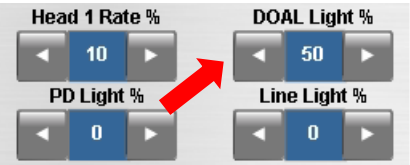
9.4 Pre-dispense inspection operation

PIXID systems uses machine vision-guided camera to determine the location, position and orientation of object features in order to accurately dispense adhesive on desired features of image sensors. Image sensor features are pre-defined in patterns file where each camera has a unique pattern design (see section above). Machine vision guided software ensure that the correct camera is loaded and dispense pattern is consistent.

Setup

Pre-dispense inspection parameter are located in .ksdispense file. Steps below need to be followed in order to execute pre-inspection operations:

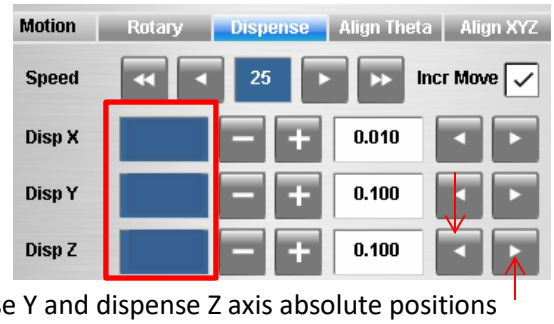
1) Design a new dispense pattern (see section “design a new pattern”)
2) Reference Pre-inspection dispnse fie in .ksdispense file as following example:
    i) PAT PRF=1,FIL="605-4687\605-4687 PreInspect Production.hdb";
    ii) where PAT: Defines a vision pattern recognition file
    iii) PRF: An integer to be used to refer to this pattern by one or more PRE operations
    iv) FIL : Full or relative path to vision pattern file (*.hdb) used for the pre-inspection. Must exist in the root of the configuration path or somewhere in a sub-folder or nested sub-folder of the configuration path. See *.ksini file ConfigPath parameter.
3) Load the new camera into fixture one.
4) Index to dispense station.
5) Setup DUAL light level to 50 as starting point.



6) Manually move dispense X & Y axis until it is center with the lens holder (image)



7) Manually move dispense Z-axis up and down until the image is in good focus.



8) Record dispense X, Dispense Y and dispense Z axis absolute positions
9) Setup pre-dispense parameter as the following example
    i) PRE IN=1,PRF=1,LW=0,LD=50,LL=90,EXP=80.0,XA=19.494,YA=1.588,ZA=44.671
        (a) Ensure that, XA,YA are at the center of the pattern.
        (b) Where PRE : Performs a pre-dispense inspection
        (c) HD : [optional, default=1] head number (1 or 2) where 1 is primary dispense head, 2 is secondary dispense head in two head configurations
        (d) ID : [optional, default=1] index number for array camera applications with non-monolithic lens attach where separate multiple dispense/align operations are performed on the same device
        (e) IN : reference index for this pre-inspection, used by other operations to specify reference coordinates relative to this operation
        (f) PRF : reference id of the pattern to be used in this pre-inspection (see PAT dispense operation)
        (g) LW : [optional, default=0] vision light delay (0 to 1000 ms) - typically set to zero unless performing consecutive inspections with different light intensities
        (h) LD : vision light intensity for DOAL light (0 to 100% where 0% disables the light)
        (i) LL : vision light intensity for line light (0 to 100% where 0% disables the light)
        (j) EXP : camera acquisition exposure time in milliseconds (0.05 to 1000 ms)
        (k) XA : x axis absolute position in mm to perform the vision inspection
        (l) YA : y axis absolute position in mm to perform the vision inspection
        (m) ZA : z axis absolute position in mm to perform the vision inspection
10) Replace XA,YA and ZA with recorded dispense X, Dispense Y and dispense Z axis absolute positions in step (8)

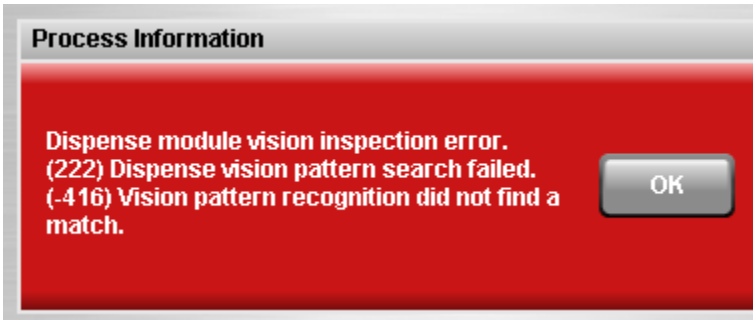
9.5 Dispense pattern verification

The purpose of verification test is to check that the pattern is accurate to detect the selected features.

1. Load a new camera into fixture number 1
2. Index the rotary table to dispense station.
3. In dry mode, click on dispense button
4. The dispense process should start in dry mode.

9.6 Vision Pattern troubleshooting

If the part failed dispense due pre inspection error.



Follow the instruction below for troubleshooting

1. Check if the right pattern is selected and try to dispense again

    a. Pattern file is located in .ksdispense file

```
PAT PRF=1,FIL="SV210\SV210 Pre-Dispense.hdb"
PRE HD=1,ID=1,IN=1,PRF=1,LW=0,LD=50,LL=0,EXP=20.000,XA=7.960,YA=32.174,ZA=38.000
```

2. If the correct file is check and the dispense fail again

    a. Check that the Doal light setting in the dispense file is the same setting that has been used when the pattern is designed (in this case it is 50). Make sure that the line light setting is set up to ZERO

```
PAT PRF=1,FIL="SV210\SV210 Pre-Dispense.hdb"
PRE HD=1,ID=1,IN=1,PRF=1,LW=0,LD=50,LL=0,EXP=20.000,XA=7.960,YA=32.174,ZA=38.000
```

3. If the dispense fail again, use search function to check for image selected image features.

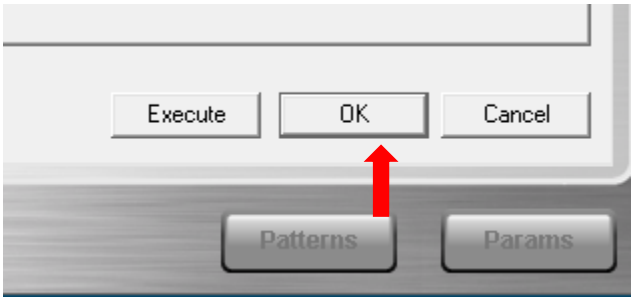
    a. Click on the pattern button and double click on pattern file

    b. Click on Grab Down button

    c. Click on Search button then click on Edit Search



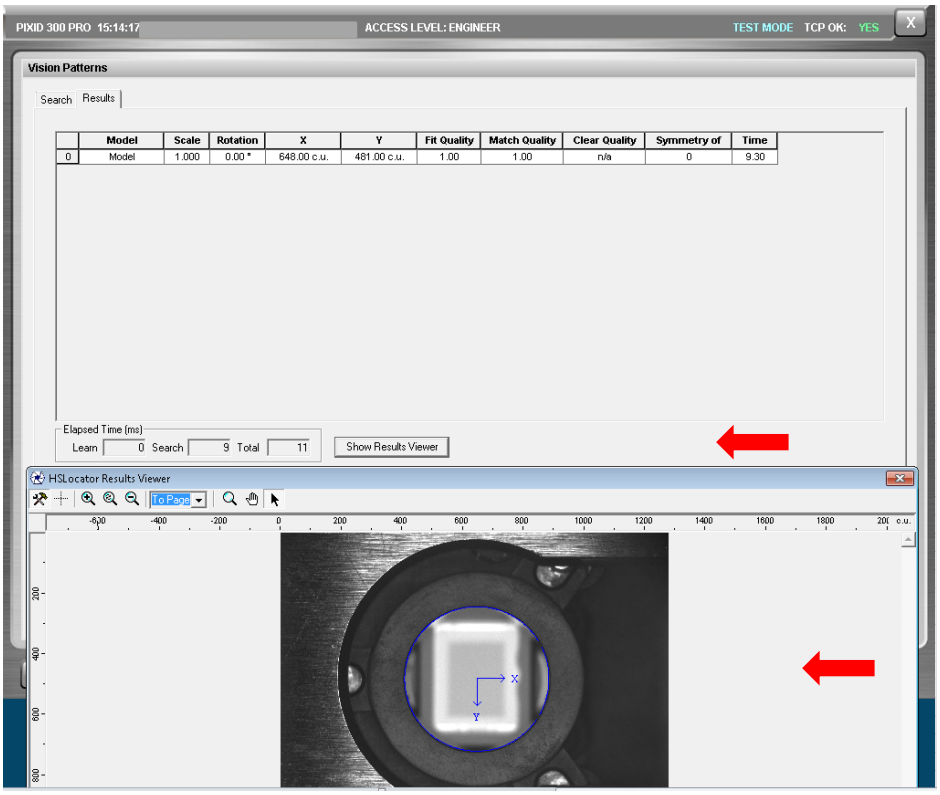
d. Click on Execute button at the bottom right of the screen



e. Click on the results button at the top left of the screen



f. Click on Show Results Viewer to check for the features that the machine vision camera recognize it.



g. If there is no features are selected,

   i. Repeat previous steps using different camera and check for pattern features

   ii. It the problem repeated with different cameras, re redesign dispense pattern using different camera.

9.7 Post dispense inspection

9.7.1 Theory of operation

Post dispense inspection algorithm is based on blob analysis; where the algorithm detect regions in the image that are differ in properties such as brightness, size or continuously compared to surrounding regions. The blob analysis is a region of an image where the pixels properties are somewhat consistent.

When dispense sequence is completed, machine vision camera at dispense station capture an image with user-specifiable illumination level. Posts dispense inspection algorithm process the image to compare the properties of specified image regions. The parameters can be set up to eliminate false positives while still detecting real dispense region.

9.7.2 Post dispense inspection operation

Post dispense inspection operations are specified by adding an operation called “VIS” to the [SEQUENCE] section of the .Ksdispsense file. An example of “VIS” function with all parameters specified in shown below:

VIS
HD=1,ID=1,LW=0,LD=0,LL=10.0,EXP=30,XO=0.000,YO=0.0,ZO=0,SM=1,ST=60,SH=10,RL=0,RT=0,RR=100,RB=100,AMN=10.0,AMX=15.0,BMN=60.0,BMX=70.0,PMN=1,PMX=1,XMN=0.25,XMX=0.25,YMN=0.25,YMX=0.25,CSA=1,CSR=3,CAW=1,CMN=50,CMX=100,CSO=50,CTW=200,CUT=3,COT=3

More than one “VIS” operation can be specified in the dispense sequence, allowing to execute post dispense inspection at multiple levels with different thresholds and filtering.

9.7.3 Post dispense inspection parameters

1. **HD :** [optional, default=1] head number (1 or 2) where 1 is primary dispense head, 2 is secondary dispense head in dual dispense head configurations

2. **ID :** [optional, default=1] index number for array camera applications with non-monolithic lens attach where separate multiple dispense/align operations are performed on the same device

    a. *These parameters do not affect the function of the post dispense inspection algorithm itself.*

    b. *They allow multiple VIS tests to be written for specific dispense heads and lenses on machines, which support these features.*

3. **LW :** [optional, default=100] Light Wait (0 to 1000 ms) - typically set to 100 ms to ensure the particle test light is updated prior to performing the test

4. **LD :** [not optional, no default] vision light intensity for DOAL light (0 to 100% where 0% disables the light)

5. **LL :** [not optional, no default] vision light intensity for line light (0 to 100% where 0% disables the light)

6. **EXP :** [not optional, no default] camera acquisition exposure time in milliseconds (0.05 to 1000 ms)

    a. *LD & LL & EXP are important to the function of the post dispense inspection algorithm, but it controls the environment in which the measurements are taken, and not the way in which the measurement is processed by the algorithm.*

    b. *It should be adjusted so that the adhesive have good contrast with the background image.*

    c. *It is the only non-optional parameter.*

7. **XO :** [optional, default=0] x axis offset in mm from configured pre-inspection position

8. **Y0 :** [optional, default=0] y axis offset in mm from configured pre-inspection position

9. **Z0 :** [optional, default=0] z axis offset in mm from configured pre-inspection position

    a. *These parameters affect the image captured by the machine, not the post dispenses inspection analysis itself.*

    b. *They adjust the center of the post dispense inspection region relative to the configured pre-inspection position.*

10. **SM :** [not optional, no default] segmentation mode (0 to detect dark objects, 1 to detect light objects, 2 to not apply inspection but acquire image only for manual dispense verification prompt)

11. **ST :** [not optional, no default] segmentation threshold (0 to 100%, or -1 to use automatic thresholding) for connected component analysis to locate dispensed material

12. **SH :** [optional, default=10.0] segmentation histogram tail percentage (0 to 100%) specifies the histogram tail percentage used when applying segmentation analysis with a partially adaptive segmentation threshold of >= 0

    a. *SM, ST, and SH are important parameters that define how the dispensed material is distinguished from its surrounding background in the machine vision camera.*

    b. *Set ST=-1 will apply fully adaptive blob segmentation and will ignore the SH parameter*

13. **RL :** [optional, default=0] Search ROI left (0 to 100% as a percentage of image width)

14. **RT :** [optional, default=0] Search ROI top (0 to 100% as a percentage of image height)

15. **RR :** [optional, default=100] Search ROI right (0 to 100% as a percentage of image width)

16. **RB :** [optional, default=100] Search ROI bottom (0 to 100% as a percentage of image height)

   a. *RL, RT, RR & RB are parameters to define the region of interest (ROI) where the software is going to look for the adhesive material.*

   b. *Top left corner of the image is the (0,0) position*

17. **AMN :** [optional, default=don't_care] min area of located dispensed material (0 to 100% as a percentage of image area)

18. **AMX :** [optional, default=don't_care] max area of located dispensed material (0 to 100% as a percentage of image area)

   a. *AMN & AMX are the threshold tests are applied after all the previous processing.*

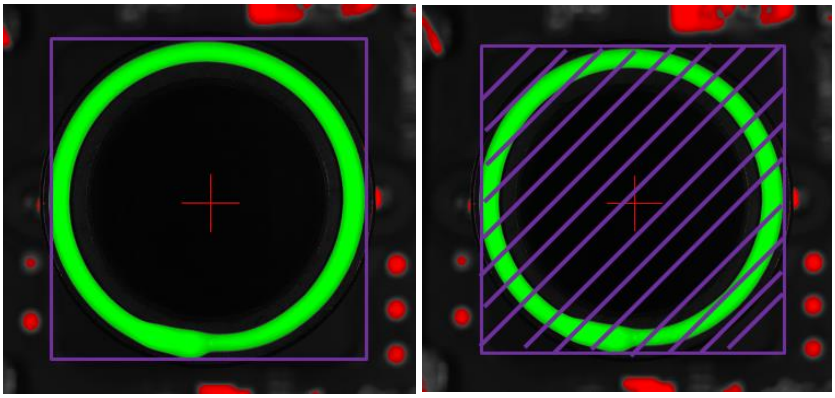
   b. *It is the maximum and minimum area of adhesive bead*

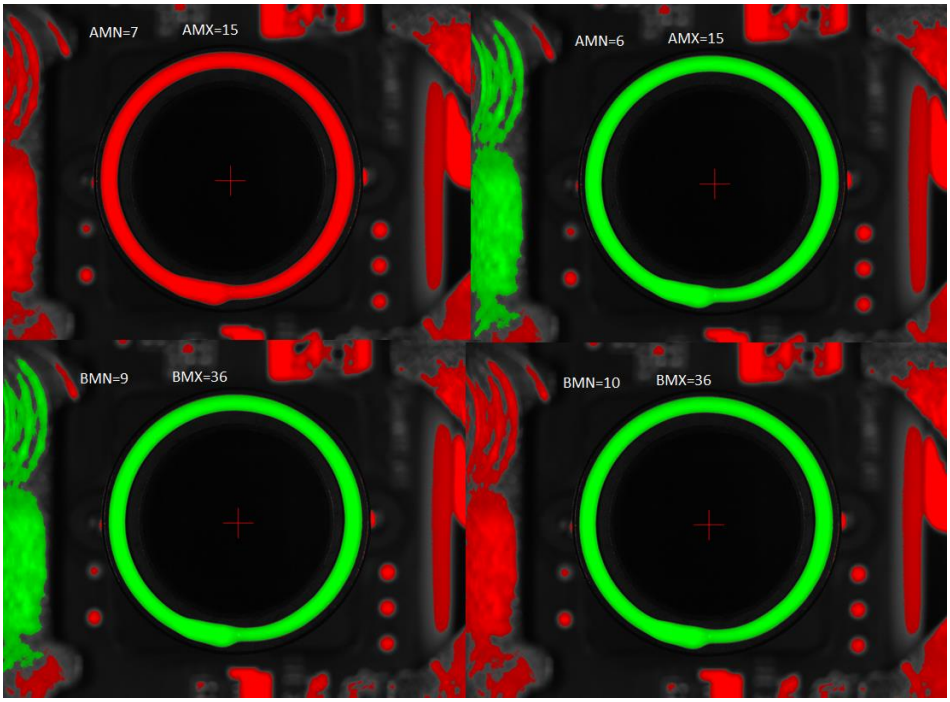
19. **BMN :** [optional, default=don't_care] min area of expected bounding box of dispensed material (0 to 100% as a percentage of image area)

20. **BMX :** [optional, default=don't_care] max area of expected bounding box of dispensed material (0 to 100% as a percentage of image area)

   a. *BMN & BMX are the threshold tests are applied after all the previous processing.*

   b. *After the software find the adhesive bead, it create a bonding box around the bead then it calculate the area of the adhesive and compare it to the area of the box*

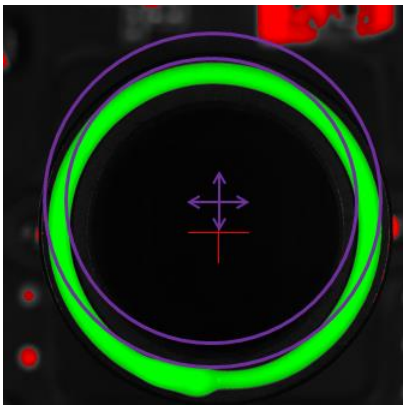




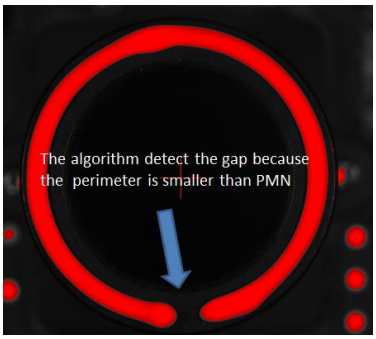
21. **XMN :** [optional, default=0=don't_care] min center x of expected bounding box of located dispensed material relative to the calibrated dispense tip position (mm)
22. **XMX :** [optional, default=0=don't_care] max center x of expected bounding box of located dispensed material relative to the calibrated dispense tip position (mm)
23. **YMN :** [optional, default=0=don't_care] min center y of expected bounding box of located dispensed material relative to the calibrated dispense tip position (mm)
24. **YMX :** [optional, default=0=don't_care] max center y of expected bounding box of located dispensed material relative to the calibrated dispense tip position (mm)

    a. *XMN, XMX, YMN & YMX define the center of bounding box with respect to the fiducial XY position in the image located by the dispense pre-inspection vision analysis. Note that this is not necessarily the center of the machine vision image.*

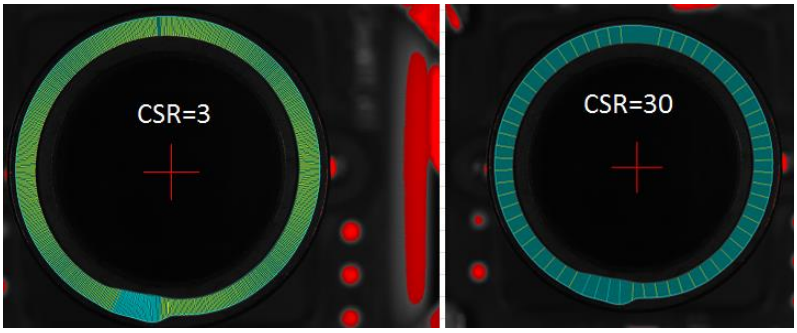
    b. *It is the center adhesive bead. For instance, If adhesive bead is shifted in +Y, the bounding box will be shifted in +Y and it will trigger and error because the adhesive is not dispensed on the desired area*



25. **PMN :** [optional, default=don't_care] min perimeter of located dispensed material (pixels)
26. **PMX :** [optional, default=don't_care] max perimeter of located dispensed material (pixels)
    a. PMN and PMX are important parameter to detect any gaps or bubbles in dispense material.



27. **CSA :** [optional, default=0] enable or disable the cross-sectional analysis step of post-dispense inspection
28. **CSR :** [optional, default=3] the denominator of the ratio of sampling points to contour points
    a. *Sampling ratio is the sample size/population size where sample size is 1 point and population size is CSR*
    b. *The algorithm does the analysis for 1 point every CSR point; for instance if CSR set up to three; the algorithm will analysis 1 point every 3 points. If CSR set up to 30, the algorithm will do the analysis for 1 point every 30 points*

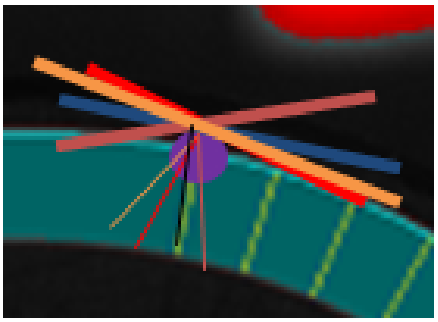


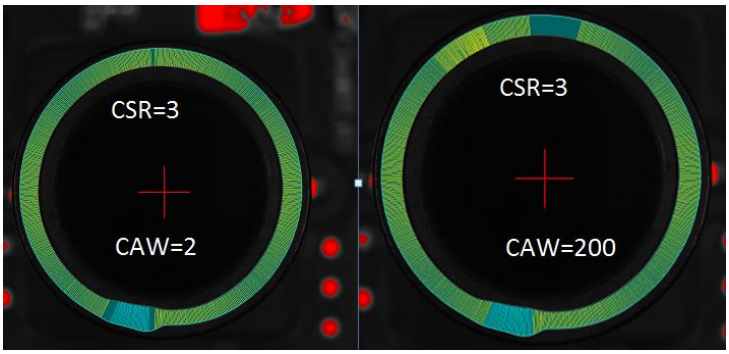
29. **CAW :** [optional, default=1] the number of measurements around the chosen point to average together

    a. *Define the number of measurement that would be taken around 1 point.*

    b. *The algorithm create multiple tangents lines around one point then it creates a perpendicular line at each tangent line. Then the algorithm takes the average of the length of the perpendicular lines.*

    c. *For instance if CAW is set up to two, the algorithm will take the average of two perpendicular lines at each point. If CAW is set up to 200, the algorithm will take the average of 200 perpendicular lines at each point*



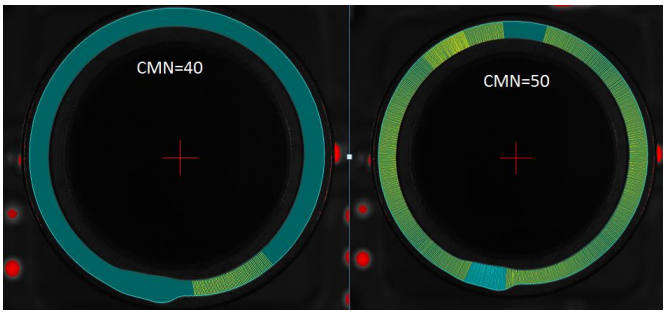


30. **CMN :** [optional, default=50] min acceptable dispense width (pixels)

    a. *Define Minimum length of the perpendicular line.*

b. *For instance: if CMN set up to 40, the algorithm will highlight the perpendicular lines that are less than 40 pixels in length (Yellow Lines)*

c. *For instance: if CMN set up to 50, the algorithm will highlight the perpendicular lines that are less than 50 pixels in length (Yellow Lines)*

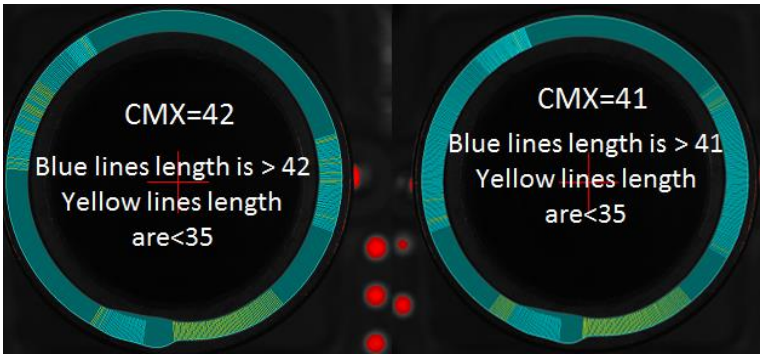


31. **CMX :** [optional, default=100] max acceptable dispense width (pixels)

a. *Define Maximum length of the perpendicular line.*

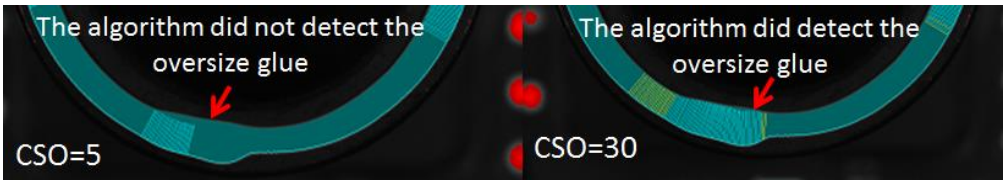
b. *For instance, if CMX set up to 42, the algorithm will highlight the perpendicular lines that are less than 40 pixels in length. (Blue Lines)*

c. *For instance, if CMX set up to 50, the algorithm will highlight the perpendicular lines that are less than 50 pixels in length. (Blue Lines)*



32. **CSO :** [optional, default=50] percentage beyond CMX to scan for oversize dispense

a. *See image below*



33. **CTW :** [optional, default=200] the size in contour points of the window used to approximate a tangent to the dispense contour

34. **CUT :** [optional, default=3] min number of consecutive undersize measurements that will trigger a failure

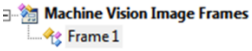
35. **COT :** [optional, default=3] min number of consecutive oversize measurements that will trigger a failure

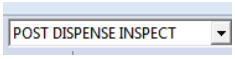
    a. *COT & CUT are important parameters that are used to trigger a failure. How many perpendicular lines are larger or smaller than the threshold (CMN & CMX).*

9.7.4 Setup post dispense inspection

Pixid KSvision utility should be used to set up post dispense inspection. Ksvision allows the user to load unprocessed image to be used as a sample to set up the threshold. The procedure to set up post dispense inspection as follow

1. Set up the dispense station to execute dispense sequence as production
2. Dispense adhesive on one image sensor as sample (ignore post dispense inspection error)
3. In I/O screen, turn off DUAL light and adjust Line Light intensity to gain good contrast of the adhesive compared to the background
4. Exit Pixid software and open Ksvision utility 

5. Open machine vision camera  then click on “acquire machine vision camera image to new frame”  the a new frame image will be added to machine vision image frames panel 

6. Select Post Dispnse Inspect from drop down menu 

7. Set up SM parameter as 1 to detect light objects and ST parameter as -1 to automaticlly locate dipsense material and CSA to 0 to disable cross section analysis

9.7.5 Set up the thresholds and filters

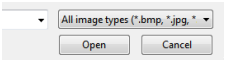
The proces of setting up the filter s and threshold start with

1. Zero out all the paramters except for the ones that are mentioned in step 7
2. Change the value for AMX (increase or decrease the number then press ENTER or ) untill only the adhesive is turned green (if there are other regions of image are green, it will be filtered out using the rest of the paramter)

3. Change the value for AMN (increase or decrease the number) untill only the adhesive is turned green (if there are other regions of image are green, it will be filtered out using the rest of the paramter)
4. Record AMX & AMN threshold levels (the numbers)
5. When AMX and AMN threshholds are setup, the user should zero out AMX & AMN in order to set up BMN and BMX
6. Repeat steps 2-4 in order to set up BMX & BMN.
7. When AMX, AMN , BMX & BMN threshholds are setup, the user should zero out AMX & AMN, BMN & BMX in order to set up PMX & PMN
8. Repeat steps 2-4 to set up PMN & PMX.
9. Once all parameter and threshold are identified; enter all the thresholds and press enter. Then verify that only located adhesive material is green. If it is not, re adjust the filters.
10. Only if needed use the cross section analysis (CSA) and follow same steps to set up CSA analysis and filttering. If the user don't need CSA; setup CSA=0
    a. The alogorithm does CSA analysis after blob analysis is done using the previous paramters.
    b. Therfore, CSA paramter need to be set up after identify located dispsne material.
11. Open .ksdispnse file and replace the old threshold and filter settings with the new settings
    a. VIS HD=1,ID=1,LW=0,LD=0,LL=10.0,EXP=30,XO=0.000,YO=0.0,ZO=0,SM=1,ST=60,SH=10,RL=0,RT=0,RR=100,RB=100,AMN=10.0,AMX=15.0,BMN=60.0,BMX=70.0,PMN=1,PMX=1,XMN=0.25,XMX=0.25,YMN=0.25,YMX=0.25,CSA=1,CSR=3,CAW=1,CMN=50,CMX=100,CSO=50,CTW=200,CUT=3,COT=3

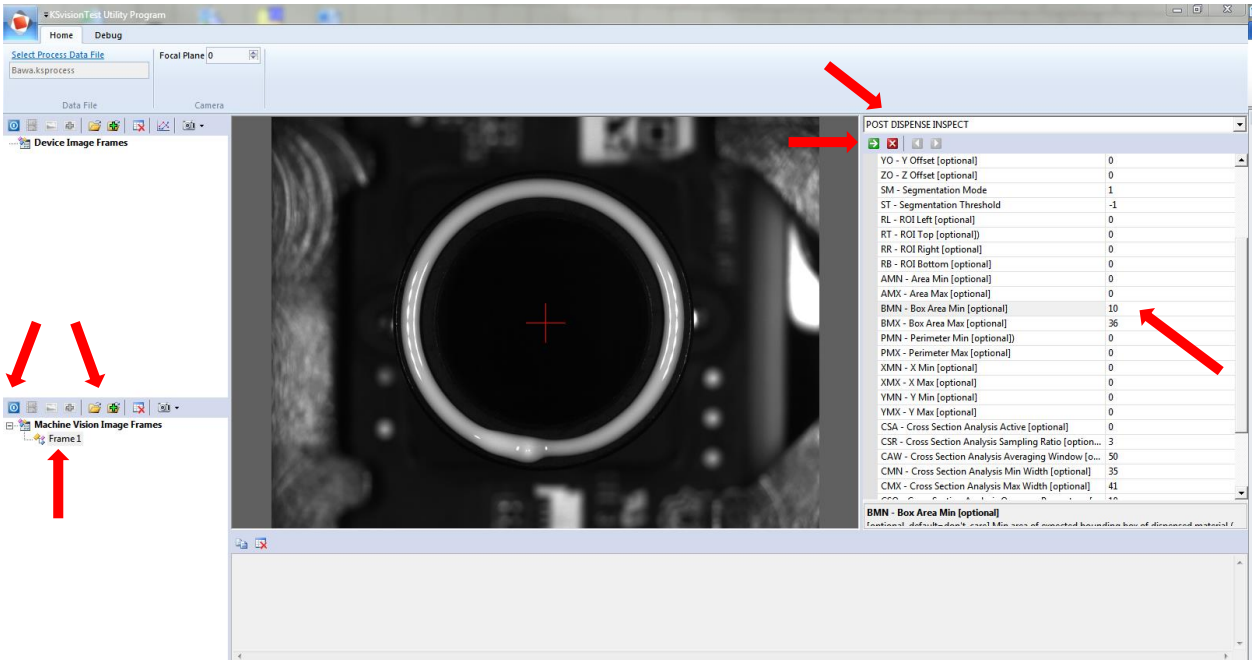
9.7.6 Working offline with .KSvision utility for post dispense inspection

Ksvision allows the user to load unprocessed image to do post dispense inspection offline. The user can capture multiple images at different light levels and do the analysis. The procedure to do offline analysis is as follow

1. Click on " load machine vision images to new frames" 
2. Select the desired images and click open 

3. All selected images will be displayed in machine vision image frames panel

4. Select one image from the frames panel and repeat steps in section 5.1 to set up the threshold



9.8 Particles test

When a device is being processed at the dispense station, the Pixid 500 allows the process designer to check for particulate contamination of the image sensor. The algorithm is based on filtering and connected component detection on the image from device sensor. In addition, an external light source is provided to highlight the particle contamination

As there may be multiple dispense operations in the dispense sequence, so there can be multiple particle tests as needed.

9.9 Particles test algorithm

An image is taken using device camera, at the dispense station, with user-specifiable illumination level. The image is filtered to remove noise. A connected component detection algorithm is used to identify particles on the sensor. Parameters may be specified to eliminate false positives while still detecting real particulate contamination.

Particles test is specified by adding an operation called “PRT” to the [SEQUENCE] section of the ksdispense file. More than one PRT operation can be specified in the dispense sequence, allowing scans for particles to be done at multiple scales with different thresholds and filtering.

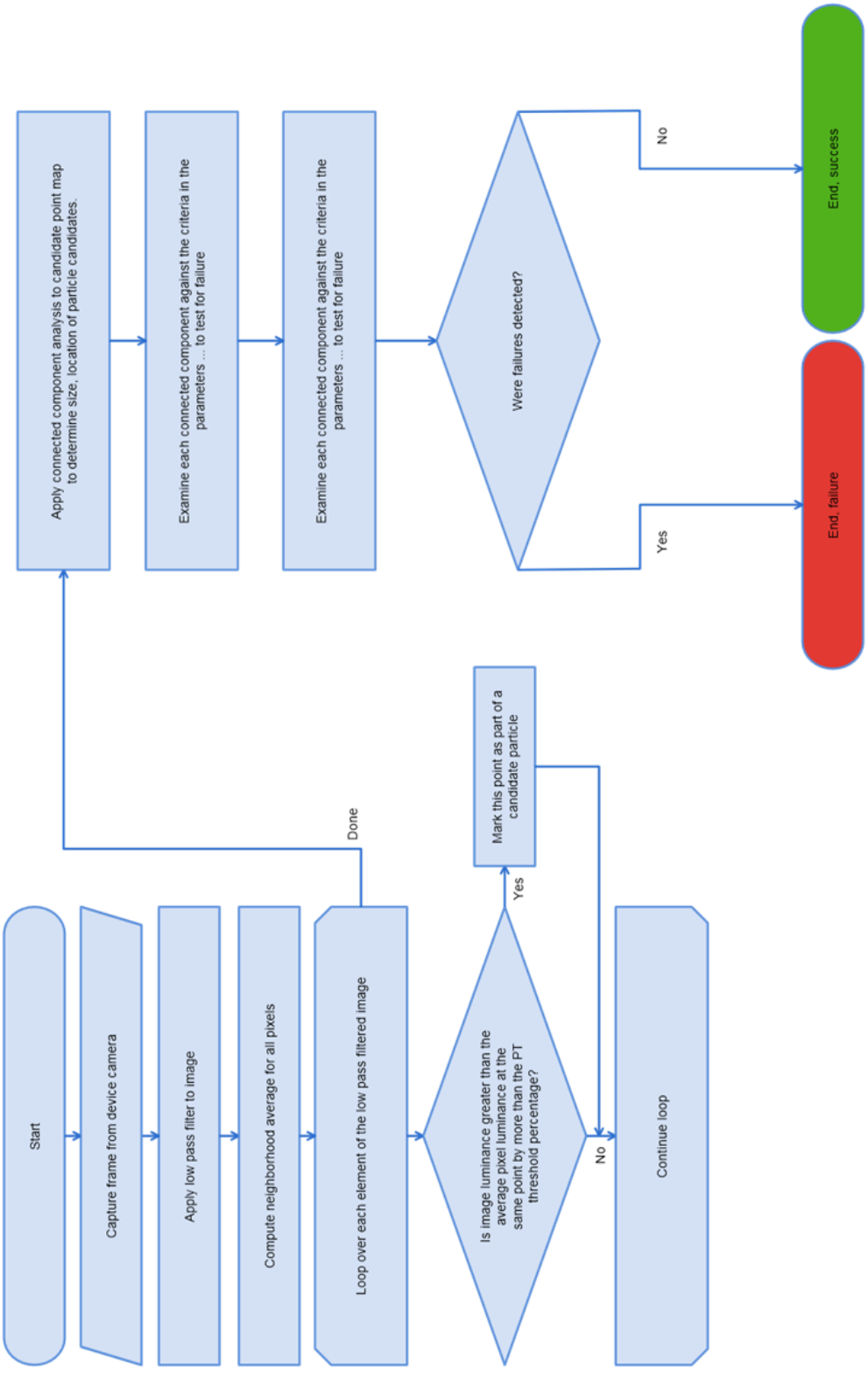


Figure 14: Particles test detection flowchart

9.10 Particles test parameters

Example of “PRT” function is shown below as

PRT
HD=1,ID=1,LW=30,LP=6.0,X0=0,Y0=0,Z0=0,BD=1,PT=11.0,IRH=100,IRW=10,ITC=30,ILS=20,ITS=120,OTC=5
,OLS=3,OTS=10,RE=1

1. HD : [optional, default=1] head number (1 or 2) where 1 is primary dispense head, 2 is secondary dispense head in dual dispense head configurations
2. ID : [optional, default=1] index number for array camera applications with non-monolithic lens attach where separate multiple dispense/align operations are performed on the same device
    a. These parameters do not affect the function of the particle detection algorithm itself.
    b. They allow multiple PRT tests to be written for specific dispense heads and lenses on machines which support these features.
3. LW : [optional, default=100] Light Wait (0 to 1000 ms) - typically set to 100 ms to ensure the particle test light is updated prior to performing the test
4. LP : [not optional, no default] Light Power – intensity setting for particle light (0 to 100% where 0% disables the light)
    a. LP is important to the function of the particle detection algorithm, but it controls the environment in which the measurements are taken, and not the way in which the measurement is processed by the algorithm.
    b. It should be adjusted so that particles have good contrast with the background image.
    c. It is the only non-optional parameter.
5. X0 : [optional, default=0] x axis offset in mm from configured pre-inspection position
6. Y0 : [optional, default=0] y axis offset in mm from configured pre-inspection position
7. Z0 : [optional, default=0] z axis offset in mm from configured pre-inspection position
    a. These parameters affect the image captured by the machine, not the particle analysis itself.
    b. They adjust the center of the particle inspection region relative to the configured pre-inspection position
8. BD : [optional, default=0] the size of the border (in pixels) to be excluded from the test image. Pixels within this distance from the edges of the image will not be examined by the particle detection algorithm.

9. FS : [optional, default=5] low pass filter size from 3 to 21 in pixels - must be an odd number The image inside the border will be smoothed using a low pass filter whose window size is given by this parameter.

10. NS : [optional, default=51] particle neighborhood size from 3 to 201 - must be an odd number When evaluating particles, each low-pass filtered pixel value is compared to the average value of the local neighboring pixels. If the value exceeds the average by a configurable threshold (see next slide, PT parameter), then that point is considered to be part of a contaminant particle.

    a. These parameters control the preprocessing of the image prior to particle detection

    b. They allow the system to be tuned to avoid false positives and control the rough scale of particles or other contamination which can be detected.

11. PT : [optional, default=0.8] Particle Threshold, a value from 0.0 to 100.0, gives the percentage by which a pixel value must exceed the average intensity of its neighboring pixels, in order for that pixel to be considered part of a particle. (neighborhood size given by NS parameter above).

    a. The particle detection algorithm examines each pixel in the low-pass filtered image, and compares the pixel value (luminance) of that pixel with the average luminance of the pixels in the neighborhood around that pixel, in order to determine if the pixel is part of a particle on the sensor. If the luminance of the pixel exceeds the average of those in its neighborhood by the percentage given in the PT parameter, that pixel is marked as part of a particle. Connected component analysis finds the groups of adjacent “particle” pixels and these are assessed for size and other characteristics which are detailed below.

12. PM : [optional, default=1] Particle Minimum size in pixels; candidate particles less than or equal to this size are not considered particles for the purposes of this test.

13. IRW and IRH specify the bounds of the “inner region” – as a percentage of the total width and height respectively of the full image. They default to 50 %.

    a. The remaining parameters are in two sets, prefixed “I” for the inner region and “O” for the outer.

14. “*TC” give the total count threshold for failure, and they default to 5 for “I” and “O”

15. “*LS” give the largest size for any given particle, in pixels, defaulting to 3.

16. “*TS” give the largest total size of all particles in the given region, defaulting to 10.

    a. The threshold tests are applied after all the previous processing, that is, after the final count of particles, sizes and positions is available.

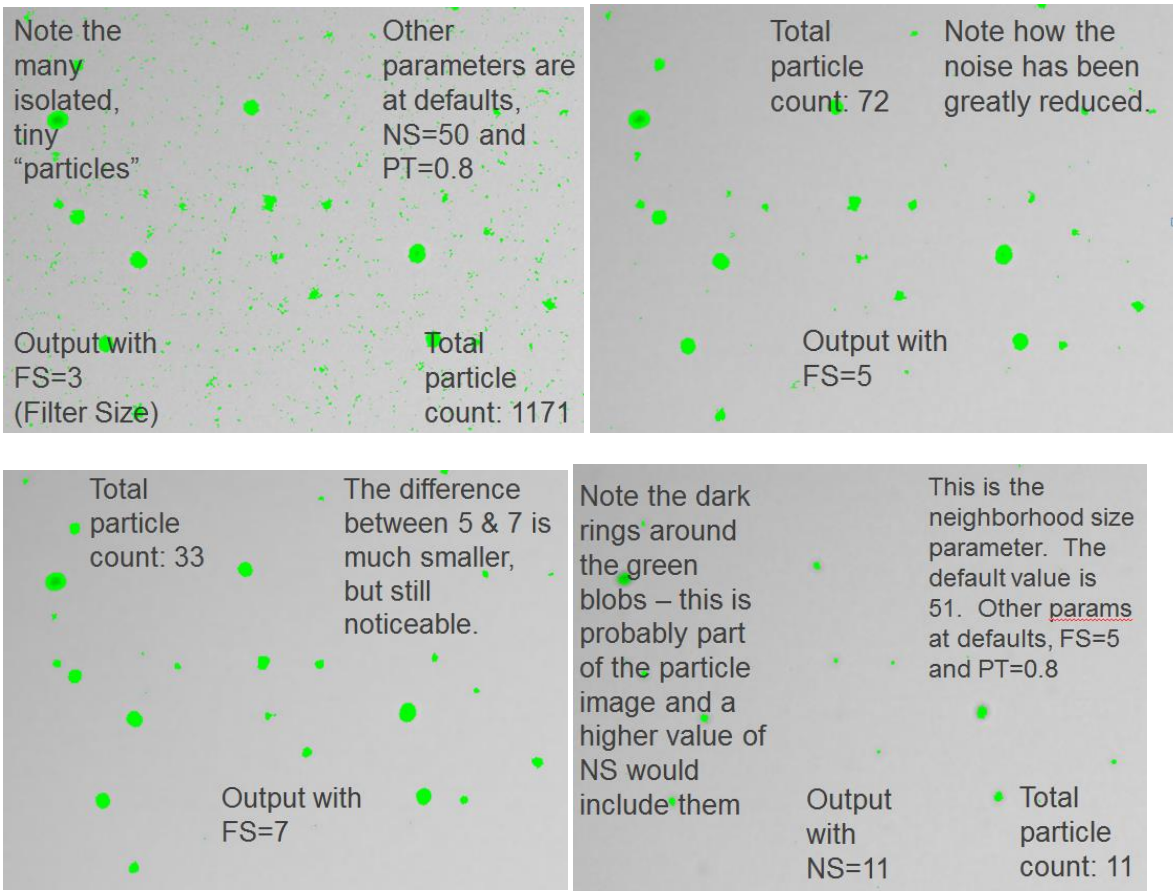
17. The parameter “PM” is technically part of the prior processing, in that it filters out particle candidates whose size is less or equal to the value of PM, which defaults to But its effects on the algorithm are more like these threshold values

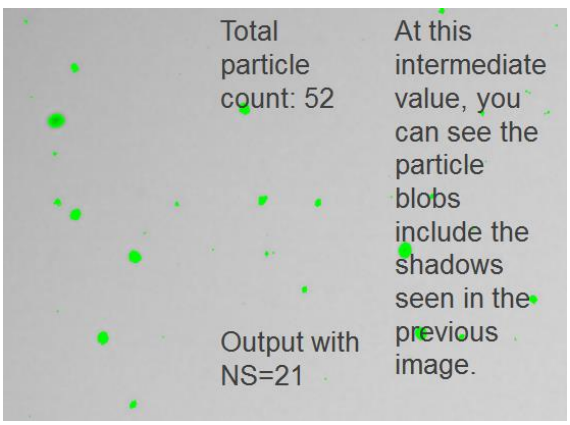
18. BD specifies a border size, units of pixels, default 0.

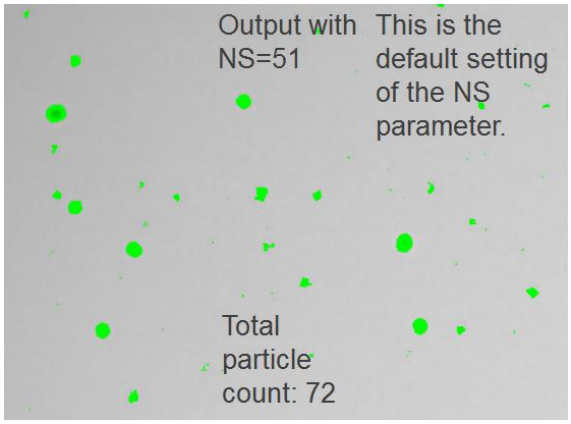
    a. This parameter is somewhat analogous to the inner/outer region settings, but it crops out specified number of pixels from the image borders before any further processing is done

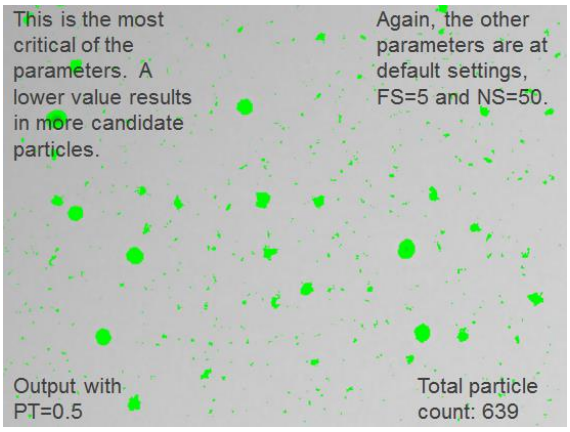
    b. No data on the cropped image region is available.

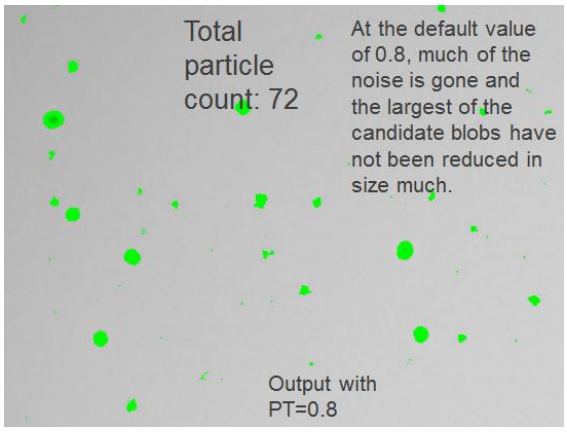
19. RE : [optional, default=1] report each particle - if this parameter is non-zero the size and location of each particle will be reported in the output result file

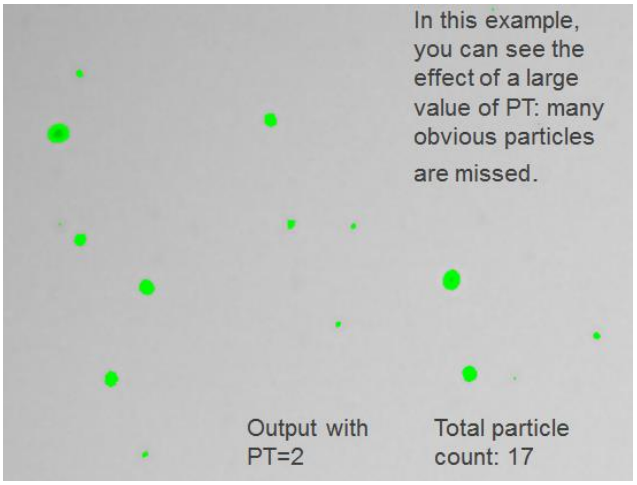












9.11 Dark signal test (optional)

Dark signal is caused by current leakage from the sensor pixels in the absence of light. The Pixid systems test for dark signal by returning pixel intensity while in darkness. Dark Signal Non-Uniformity (DSNU) is the dark signal non uniformity (DSNU) measures the variation of dark signal across the whole sensor. The Pixid systems test the DSNU by calculating the standard deviation of the pixels across the sensor.

9.11.1 Test setup

1) Setup camera registry settings to execute dark signal test.
2) Setup the test parameters as the following example:
    i) DSTNNumFrames : Number of image frames to acquire to create the temporally averaged image that is used by the tests to determine the average pixel intensity value, from 1 to 10.
    ii) DSTNBorderSize : The size of the border in pixels of the temporally averaged image to be excluded from the tests.
    iii) DSExecuteTest : 0 to not execute the dark signal test, 1 to execute it. The test will be performed at the dispense station.
    iv) DSThreshold : The threshold percentage from 0 to 100 % used to evaluate the calculated average pixel intensity value which is normalized as a value from 0 to 100. The camera fails this test if the normalized average pixel intensity value exceeds the threshold percentage.

9.12 Temporal noise tests

Temporal noise is the 'speckle' of the image that varies from frame to frame. Temporal noise is calculated by sampling pixels multiple instances over time. The RMS (root mean square) value of all measurements is calculated to give the temporal noise.

9.12.1 Test setup

1) Setup camera registry settings to execute dark signal test.
2) Setup the test parameters as the following example:
    a) TNExecuteTest : 0 to not execute the temporal noise test, 1 to execute it. The test will be performed at the dispense station.
    b) TNThreshold : The threshold percentage from 0 to 100 % used to evaluate the calculated temporal noise value which is normalized as a value from 0 to 100. The camera fails this test if the normalized temporal noise value exceeds the threshold percentage.

9.13 Hot/dead/defective pixel test

The Pixid tests for stuck or dead pixels by using a uniform target and looking for significant pixel deviations from the uniform target

9.13.1 Test setup

1) Setup camera registry settings to execute dark signal test.
2) Setup the test parameters as the following example:
    a) DPExecuteTest : 0 to not execute the hot/dead/defective pixel test, 1 to execute it. The test will be performed at the dispense station.
    b) DPBorderSize : The size of the border in pixels to be excluded from the test.
    c) DPLightPct : The percentage of power applied to the test light, from 0 to 100%. The dispense Station particle light will be used for illumination.
    d) DPHotDeadThreshold : Hot and Dead pixels are any pixels that deviate from the average pixel intensity value calculated for the entire image by this threshold expressed from 0 to 100%.
    e) DPDefectiveThreshold : Defective pixels are any pixels that deviate from their local region average by this threshold expressed from 0 to 100%

f) DPDefectiveRegionSize : Defective pixel region size from 3 to 201 where region is a square zone about each pixel. The region size must be an odd number to ensure that the region is symmetric about each pixel and will be forced up to the next nearest odd value if necessary. When evaluating defective pixels each pixel is compared to the average pixel calculated from the local region pixels.

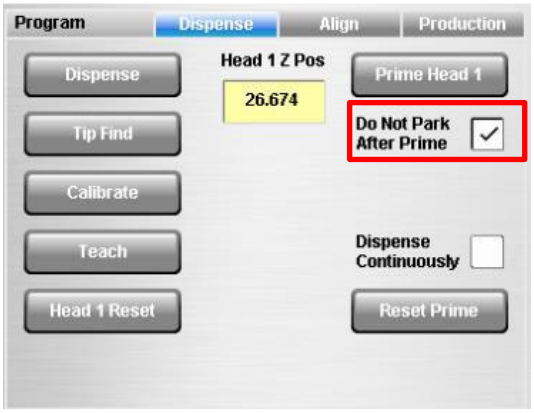
g) DPBadPixelsMaxCount : The maximum combined number of hot, dead, and defective pixels allowed.

9.14 Calibration

9.14.1 Tipfind calibration

Tip find calibration is used to define dispense tip position from with respect to home position. Follow the instruction below to execute tip find function.

1- Open main GUI screen; under Dispense tab- click tip find button.



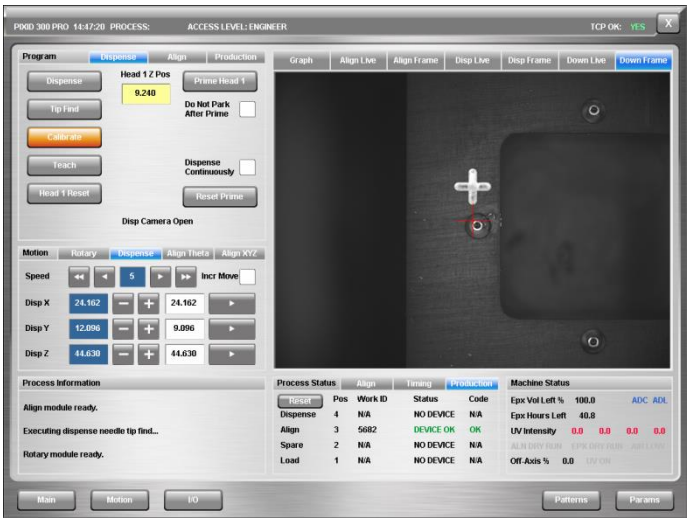
9.14.2 Dispense Calibration

1- Open main GUI screen; under Dispense tab- click Calibrate button.

2- Follow instructions on main GUI screen to calibrate dispense

3- It runs a tip find first.

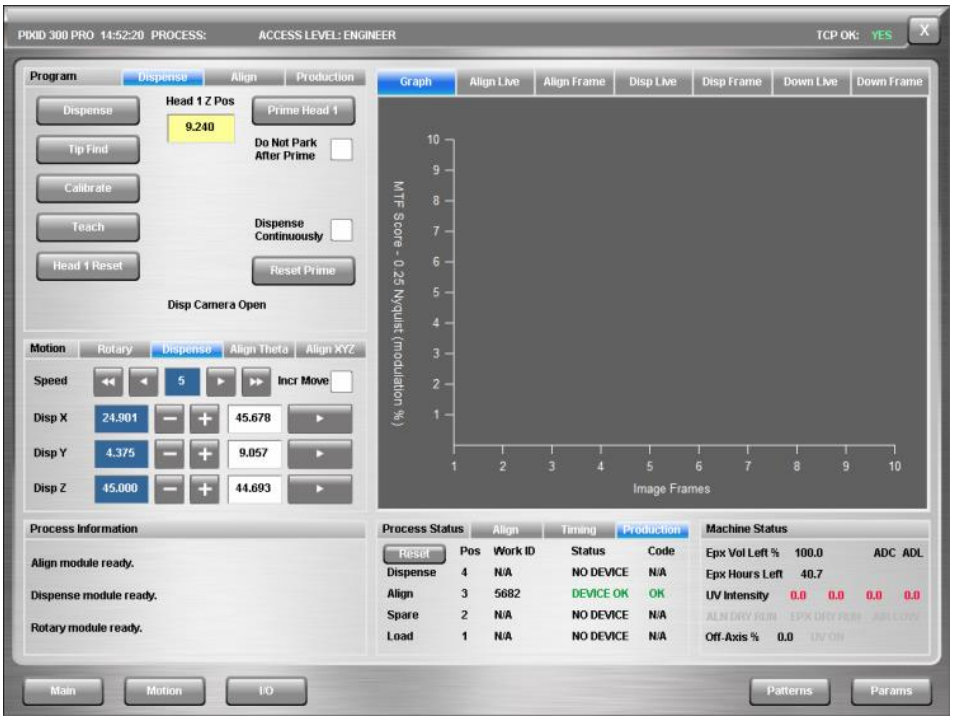
4- The down viewing camera locates the two fiducial on the tip find assembly, moving the dispense stack into position with red cross hairs over the white fiducial.



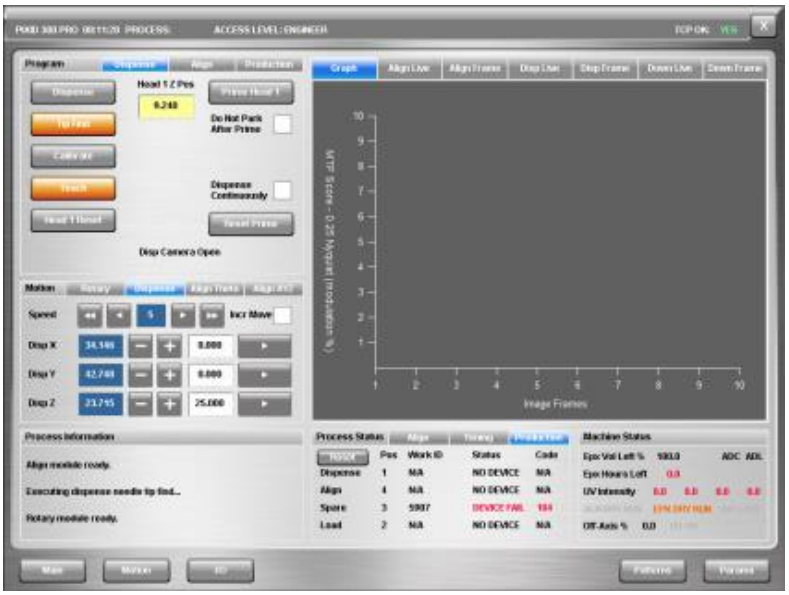
9.15 Dispense Teach

1- Open main GUI screen; under Dispense tab- click Teach button as shown below.

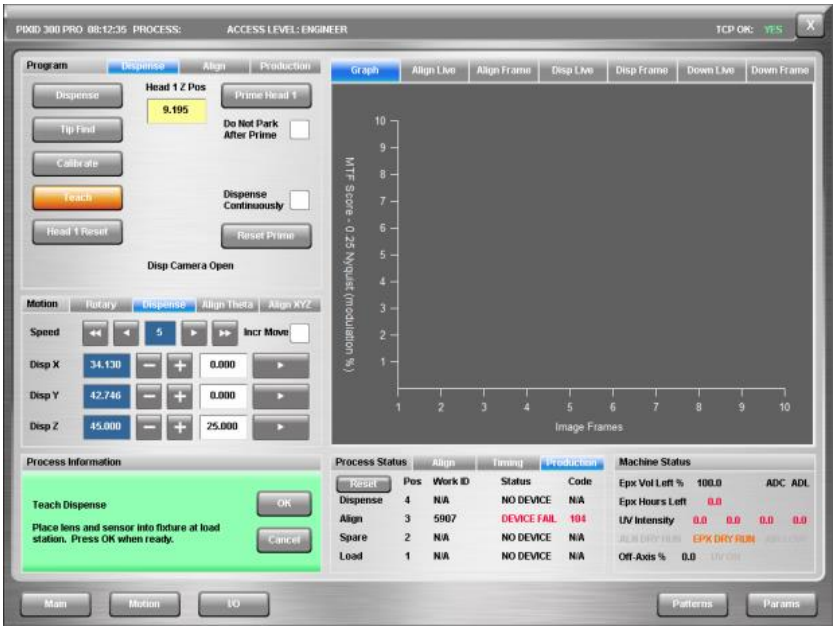
2- Follow the instructions on main GUI screen to set up dispense.



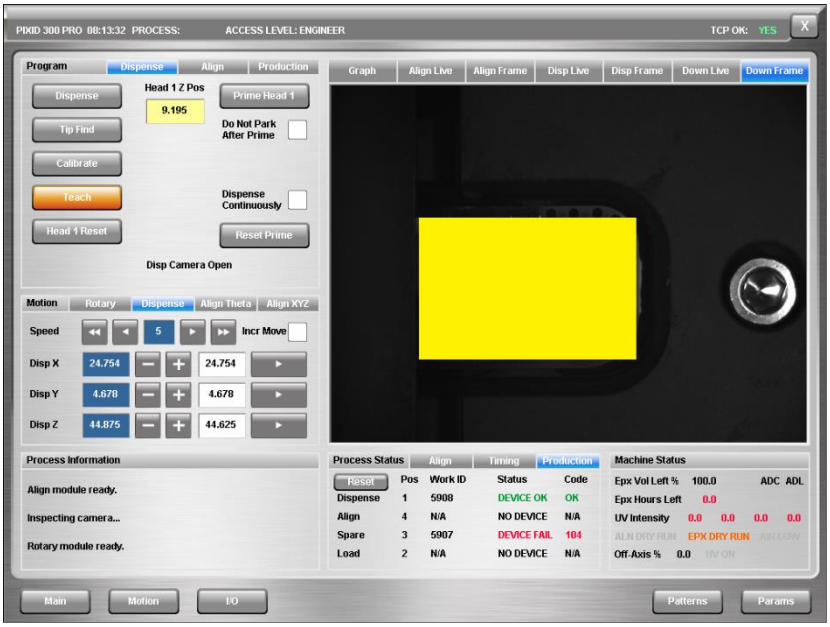
3- Tip Find is started.



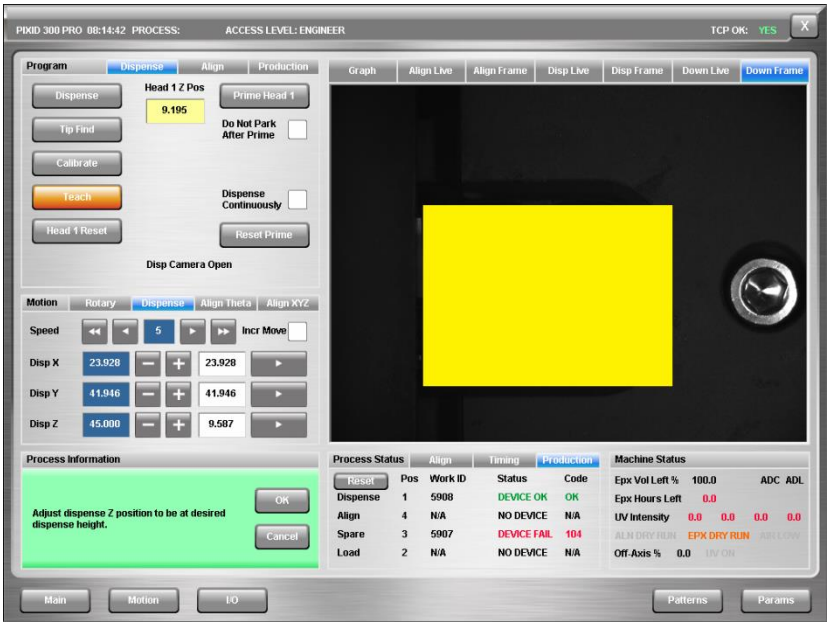
4- Load part at load station. Press OK button to continue, Cancel to quit.



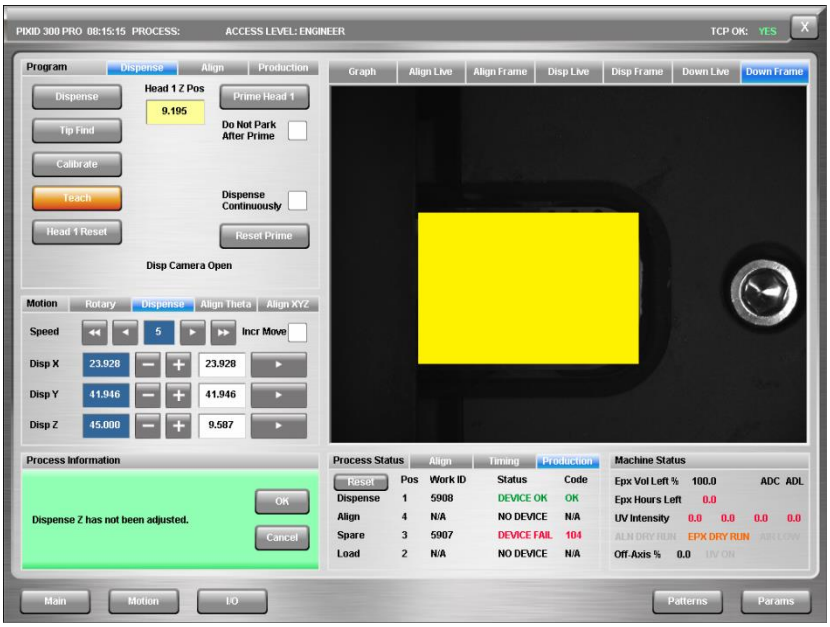
5- Part Per Inspection is executed.



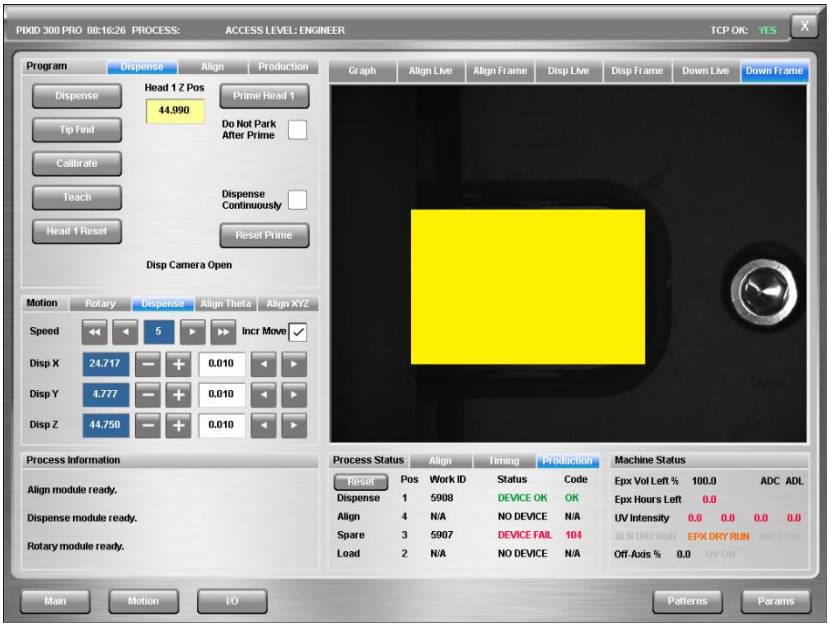
6- Adjust Z axis position. Use the Z axis Dispense arrow buttons to move the axis into initial position over the sensor. Press OK button to continue, Cancel to quit.



7- Adjust Z axis position not moved. If the Z axis has not been moved and the OK button was pressed the user will be prompted to move Z axis before continuing. Press OK button to continue, Cancel to quit.



8- Dispense Teach complete.



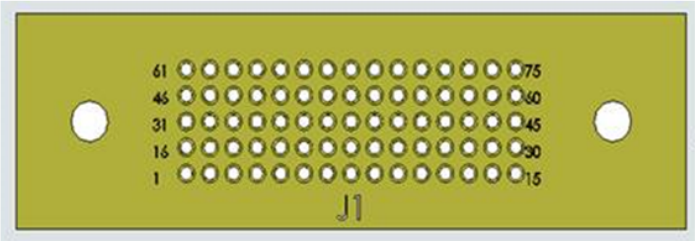
9- The new pre-inspection positions X, Y and Z will be updated in Ks.dispense file as XA,YA and ZA

9.16 Dispense station pogo Pins

Pogo pins are little spring-loaded contacts that establish a temporary connection with image signal processor (ISP) and fixture electronics. Dispense station pogo pins must be aligned with fixture electronics on order to grab and process an image from the camera.

Image signal processing Diagram

Parallel Signal to Laon People				*Common Signals*		*MIPI Signals to Laon People*			
J1 (Pogo)	*Signal*	*J1 (Pogo)*	*Signal*	*J1 (Pogo)*	*Signal*	*J1 (Pogo)*	*Signal*	*J1 (Pogo)*	*Signal*
1	S_D14	16	S_D15	31	PCH1_PIN	46	N/C	61	N/C
2	S_D3	17	GND	32	PCH2_PIN	47	GND	62	GND
3	S_PCLK	18	S_D2	33	PCH3_PIN	48	DATAP_2	63	DATAN_2
4	S_D6	19	GND	34	PCH4_PIN	49	GND	64	GND
5	S_D10	20	S_D5	35	S_SCL	50	DATAP_0	65	DATAN_0
6	S_D7	21	GND	36	S_SDA	51	GND	66	GND
7	S_GPIO0	22	S_D4	37	S_ENB	52	DATAP_3	67	DATAN_3
8	S_GPIO1	23	GND	38	S_RESET	53	GND	68	GND
9	S_D0	24	S_VSYNCH	39	PCHS_PIN	54	DATAP_1	69	DATAN_1
10	S_HSYNCH	25	GND	40	PS_12	55	GND	70	GND
11	S_D8	26	S_D1	41	MISO	56	CLKP	71	CLKN
12	S_D9	27	GND	42	MOSI	57	GND	72	GND
13	S_D13	28	S_D12	43	SCLK	58	S_CNT0	73	S_CNT1
14	S_GPIO2	29	GND	44	SS	59	S_CNT2	74	S_CNT3
15	S_D11	30	S_GPIO3	45	S_MCLK	60	NTSC+	75	NTSC-



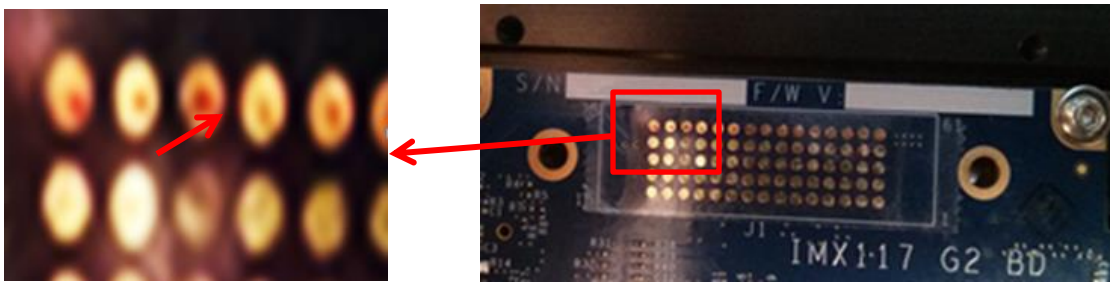
9.16.1 Pogo pins alignment

Pogo pins must be aligned to fixture electronics in order to grab an image frame. The following steps may be followed in order to align the pogo pins.

1) Power off MIU (Loan People) unit.
2) Load one fixture at table position#3
3) Index the table so that table position #3 to Dispense station
4) Bring the Pogo pins down at dispense station from I/O screen



5) Manually remove the fixture from table position#3 and stick a piece of double sided tape on the pogo pin pads in fixture electronics.



6) With the marker apply colorful ink on the tips of the needle pins
7) Re-load the fixture with the tape on the table position#3 at dispense station.
8) Bring the pogo pins up using I/O screen.

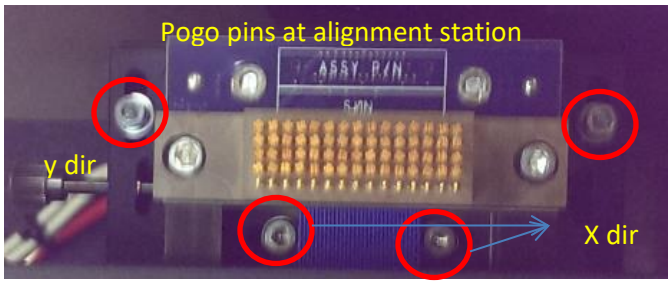


9) Unload the fixture from table position#3
10) Look at the impression left by the pogo pins on the tape under the microscope.



11) If the pogo pins are not align with pogo pins pads on the fixture electronics, adjust the pogo pins in X&Y, proceed with the following steps

i) Loose the two screws to move the POGO block in X direction
ii) Redo steps 4 through 11 until pogo pins are pointing at the center of pogo pins pads in fixture electronics
iii) Loose the two screws to move the POGO block in X direction
iv) Redo steps 4 through 11 until pogo pins are pointing at the center of pogo pins pads in fixture electronics



9.16.2 Pogo pins replacement

The pogo pins can be replaced with different type of pogo pins as needed. The procedure to replace the pogo pins the following steps must be followed

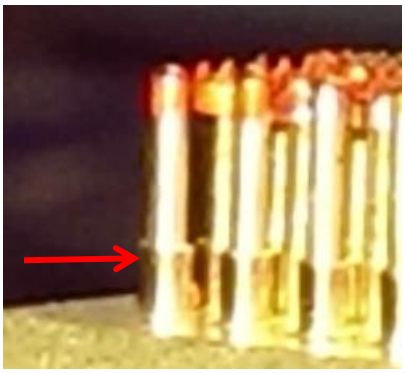
1. E-stop the system from E-stop button

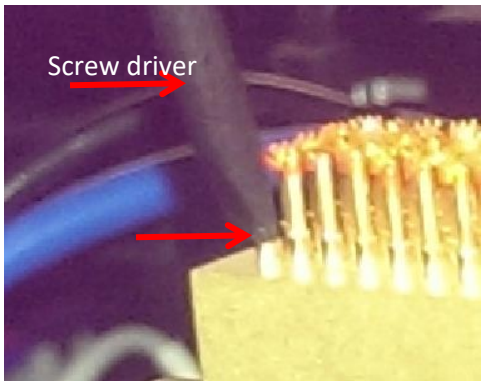


2. Use a plier to pull the pins out of the socket



3. Insert the new pins into the socket and push down the pins until is full inserted

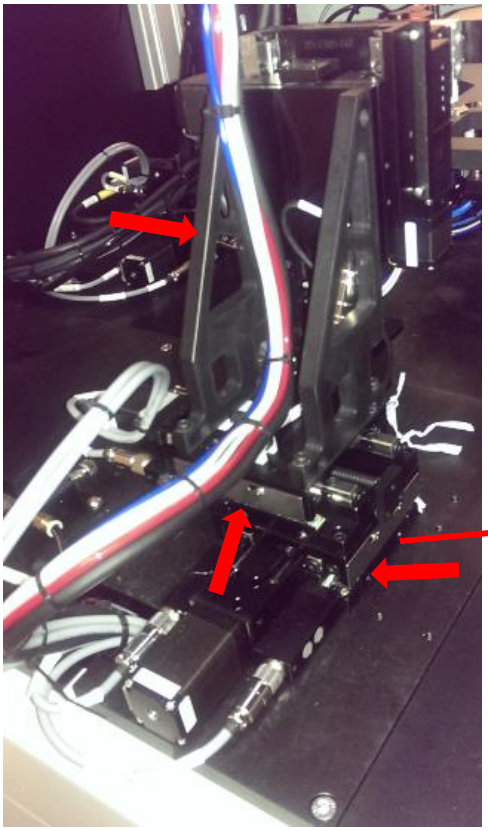


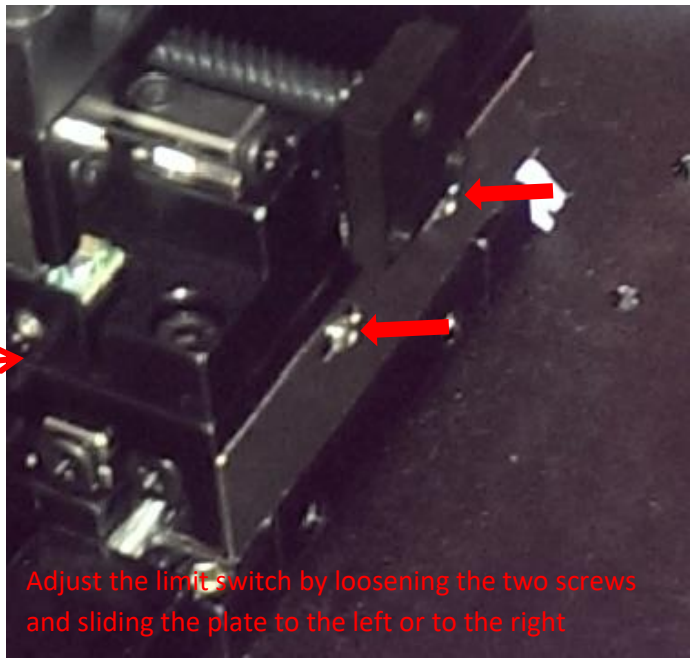


9.17 Dispense station limit switches

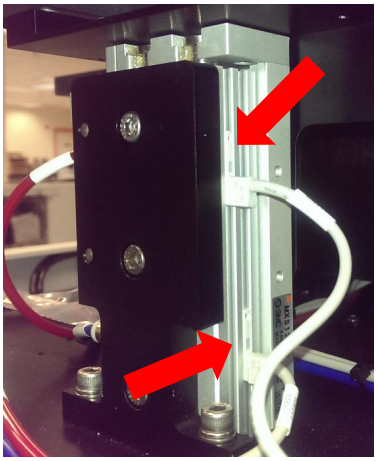
Dispense station has multiple limit switch to identify the position of stage axis and/or cylinder motion.

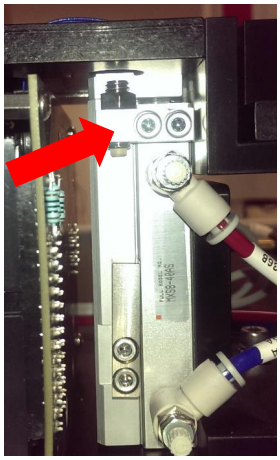
XYZ stage limit switches





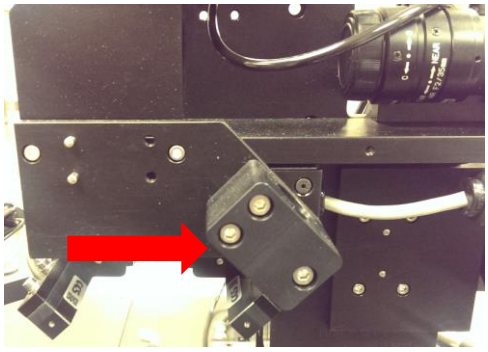
Air cylinder limit switches

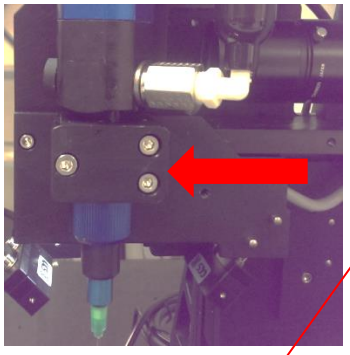




9.18 Mount dispense dosing unit

Dispense dosing unit can be mount it in two configuration 90° and 45°. The configuration depends on the field of view of the camera where narrow angle of view camera uses 90° configuration but wide angle of view uses and 45° configuration.

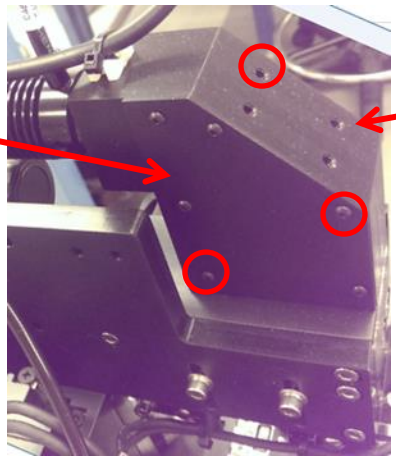






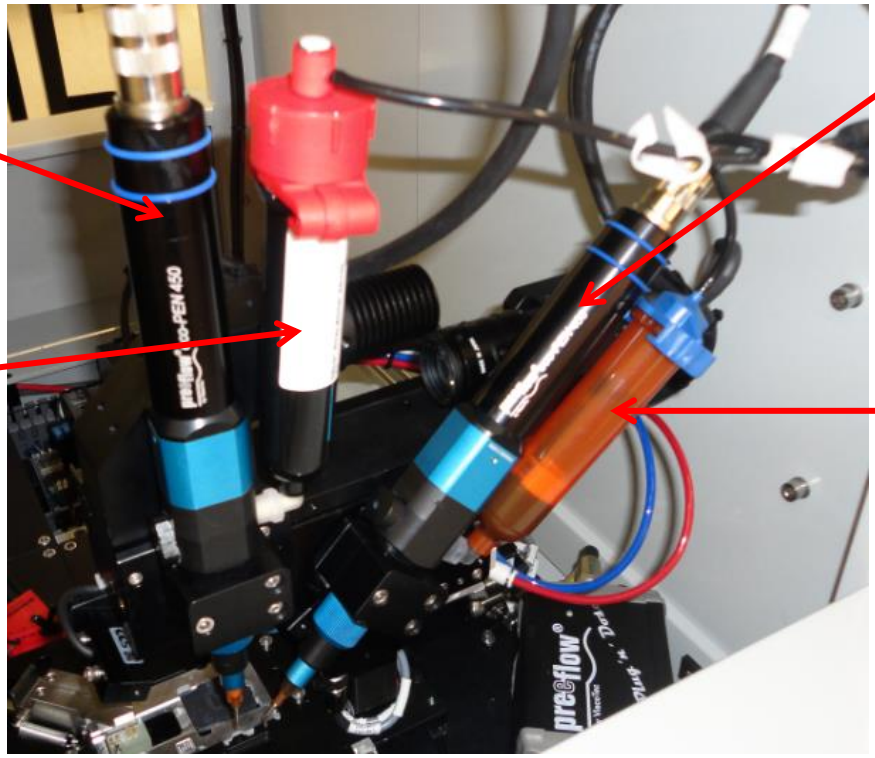
9.19 Dispense Prism

Dispense modules has a 60° prism to guide the light through during pre-inspection, post-dispense inspection…etc.



9.20 Dual head dispense module (Optional)

Pixid500 system can be configured for dual dispense by adding a second dispense head.



9.21 Dispense module check list

When the system is running in production mode there is a few tasks need to be checked by technician or engineer. The summary of the tasks is listed as shown below

1) If the machine is running 24/7, there is no need to change dispensing tip. If production stops overnight, the technician would need to replace dispense tip with a new tip
   a) Verify that the glue is flowing out of the tip by executing PrimeHead1 function
   b) Execute tip find after dispense tip is replaced.

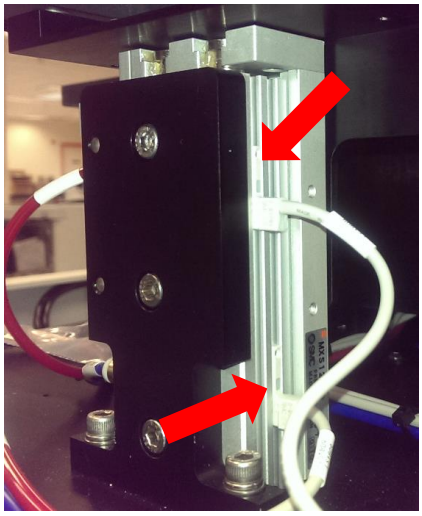
2) Do dispense calibration by clicking on calibrate button under dispense tab, when dispense unit (Eco-Pen) is removed from the system for cleaning purposes or MachineVisionCamera is moved
3) The dispense unit should not be cleaned often, it is very high precision unit and does not need cleaning unless
   a) There are particles showed up in Post Dispense Inspection image
   b) The production is down during the weekend, if the production is running 24/7 there is no point of cleaning the dispense unit because it is been running 24/7 (Check with adhesive vendor about adhesive Potlife)
4) Frequent cleaning process of dispense unit will wear up the stator inside dispense unit and eventually will need to be replaced.
5) E-stop the system before load a new glue cartridge.

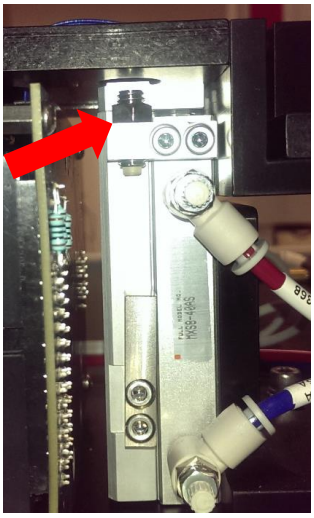
9.22 Dispense module troubleshooting

9.22.1 Tipfind troubleshooting

If tipfind fail check:

1. Error: waiting for tipfind to extend; check limit switch in the air cylinder





2. Error: fail Z position: check dispense Z position in .ks global file

```
TipFind1X=
TipFind1Y=
TipFind1Z=
```

9.22.2 Dispense calibration troubleshooting

If dispense calibration fail due two fiducial, check

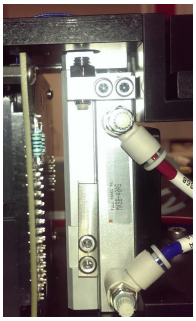
1. Check if the image is in good focus; Check `TipFindFiducialPos1Z`
2. Check DOAL light level
3. Check XYZ in .KSglobal file

```
TipFindFiducialPos1X=
TipFindFiducialPos1Y=
TipFindFiducialPos1Z=
```

9.23 Particle test troubleshooting/Camera open

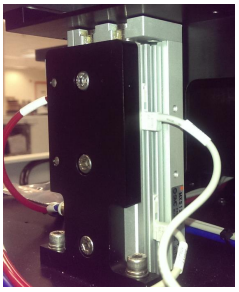
9.23.1 If particle test is fail because the device camera is not open,

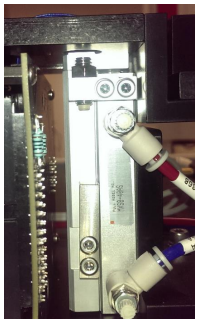
1. Check pogo pins alignment at dispense station
2. Check pogo pins height against the fixture electronics
3. Check particles detection light level
4. Check test thresholds in .KSdispense file
5. Check if the pogo pins are compressed against fixture electronics. Adjust air cylinder mechanical stop for higher stroke



9.23.2 If error code is “waiting for dispense station to extend”

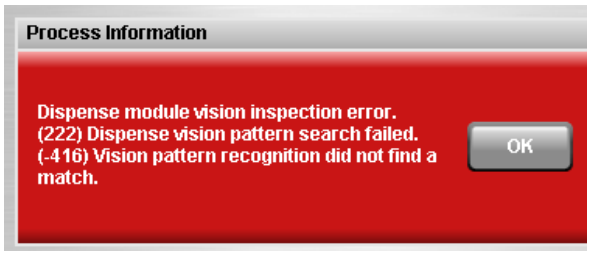
1. Check air cylinder limit switch





9.24 Vision Pattern troubleshooting

If the part failed dispense due pre inspection error.



Follow the instruction below for troubleshooting

1. Check if the right pattern is selected and try to dispense again
   1.1. Pattern file is located in .ksdispense file

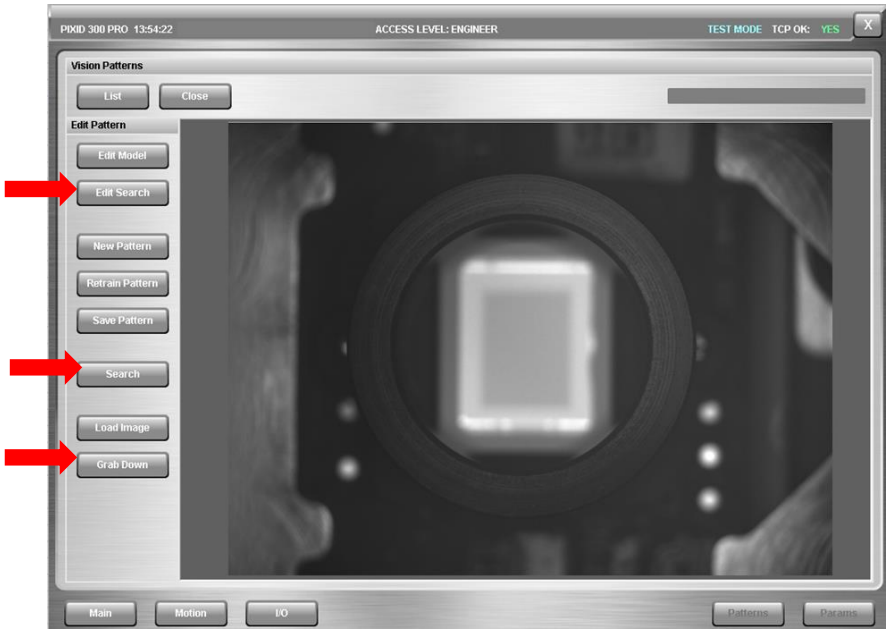
```
PAT PRF=1,FIL="SV210\SV210 Pre-Dispense.hdb"
PRE HD=1,ID=1,IN=1,PRF=1,LW=0,LD=50,LL=0,EXP=20.000,XA=7.960,YA=32.174,ZA=38.000
```

1.2. If the correct file is check and the dispense fail again

2. Check that the Doal light setting in the dispense file is the same setting that has been used when the pattern is designed (in this case it is 50). Make sure that the line light setting is set up to ZERO

```
PAT PRF=1,FIL="SV210\SV210 Pre-Dispense.hdb"
PRE HD=1,ID=1,IN=1,PRF=1,LW=0,LD=50,LL=0,EXP=20.000,XA=7.960,YA=32.174,ZA=38.000
```

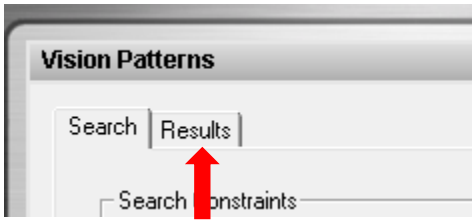
3. If the dispense fail again, use search function to check for image selected image features.
4. Click on the pattern button and double click on pattern file
5. Click on Grab Down button
6. Click on Search button then click on Edit Search



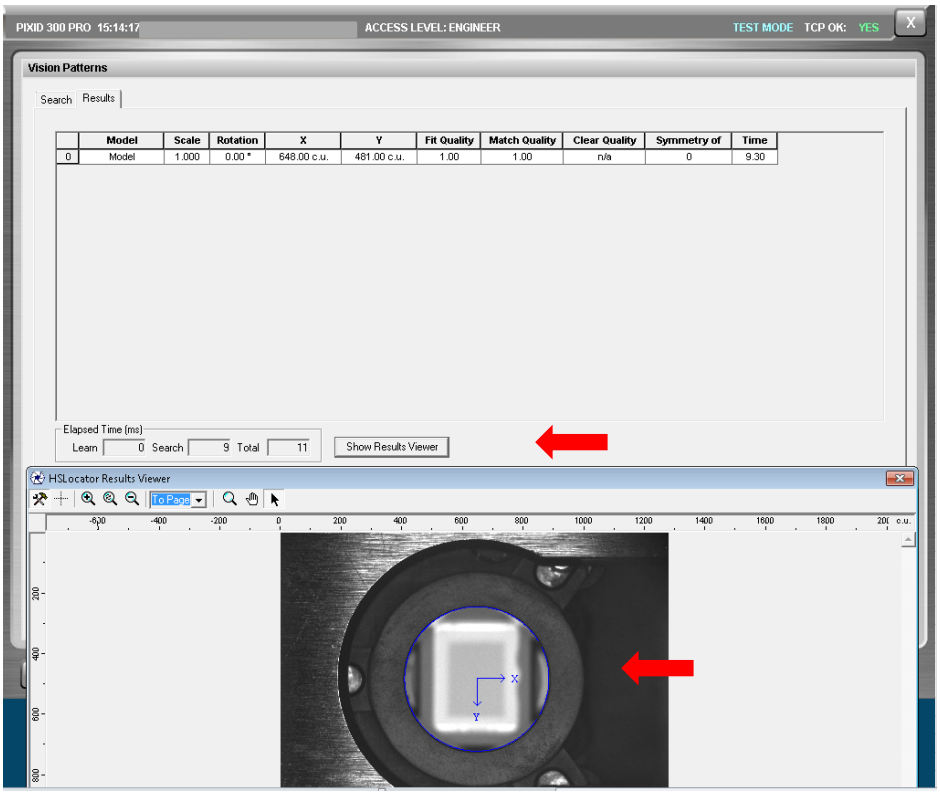
7. Click on Execute button at the bottom right of the screen



8. Click on the results button at the top left of the screen



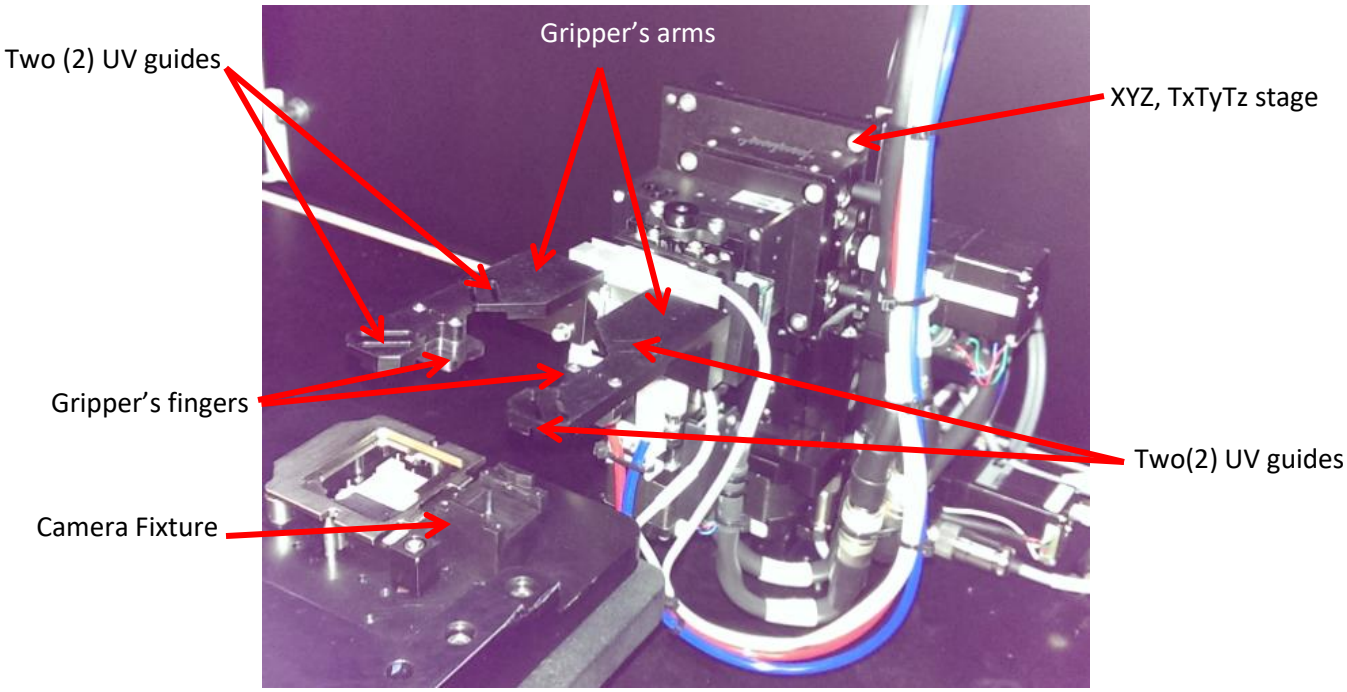
9. Click on Show Results Viewer to check for the features that the machine vision camera recognize it.



10. If there is no features are selected,
11. Repeat previous steps using different camera and check for pattern features
12. It the problem repeated with different cameras, re redesign dispense pattern using different camera.

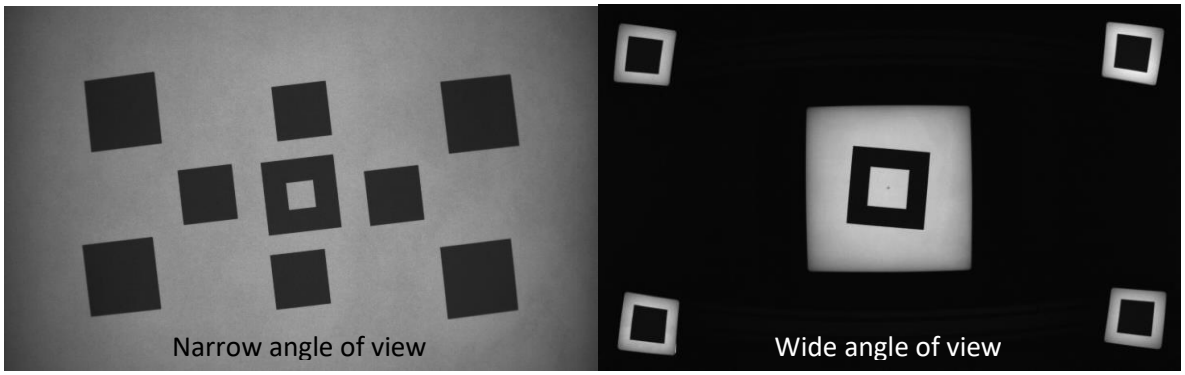
10 Alignment Module

Alignment module is assembled using XYZ stage, gripper's arms, gripper fingers, UV mounts and fiber optics UV guides.



10.1 Target features

The alignment/test target is placed at desired height from lens surface to the target. The target contains features for performing standard optical tests. The features are five slanted edge targets; 1 on-axis (center) and 4 off-axis (corner) target features. Slanted square target feature is used to calculate MTF based on the ISO 12233 standard. MTF feedback is used for optimizing focus and for verifying that the camera module's focus satisfies pass/fail criteria after assembly.



10.2 Optical tests

Test data acquired by the system is fed into the Adaptive Intelligence™ system allowing for automatic, real-time process improvements. Pass/fail criteria are configurable, allowing for the automatic rejection of failed devices to improve efficiency of upstream processes. Optical tests available for Pixid systems include:

10.2.1 Standard Camera Tests

10.2.1.1 MTF: Image Sharpness

The Pixid system reports image sharpness as an MTF (Modulation Transfer Function) score. That is, how quickly an edge transitions from light to dark. MTF is a measurement of sharpness of the image. Kasalis uses MTF algorithms which comply with ISO12233.

10.2.1.2 Focus Symmetry

When a perfect lens is properly aligned in rotational axes (theta-x, theta-y) to a sensor, the performance it produces is symmetric about the optical axis. However, even if the lens is aligned to yield the best possible performance out of each corner, due to optical imperfections in lens, it is possible that the MTF in the corners will not be uniform relative to one another. This test measures the non-uniformity of the off-axis lens performance.

10.2.1.3 Field Curvature

A lens exhibits field curvature when it is not possible to bring the center and the corners of the field of view into focus at the same time. When the center is in sharp focus, the corners are out of focus, and when the corners are sharp, the center is out of focus. The field curvature is captured and reported on the Pixid systems as the difference between the on-axis and off-axis through focus curves. It is an important metric in understanding the quality of the batch of lenses, and is one of the metrics measured by the Adaptive Intelligence system.

10.2.1.4 Exposure Verification

Exposure verification validates that the camera can detect differences in intensity between a series of light to dark patches in the image.

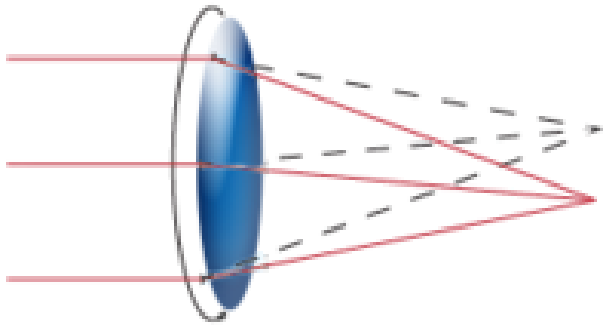
10.2.1.5 Image Contrast

Image contrast is the degree to which a sensor can differentiate between various tones in an image. It is defined by the system's tone curve using and ISO 14524 shades of grey test chart.



10.2.1.6 Centration

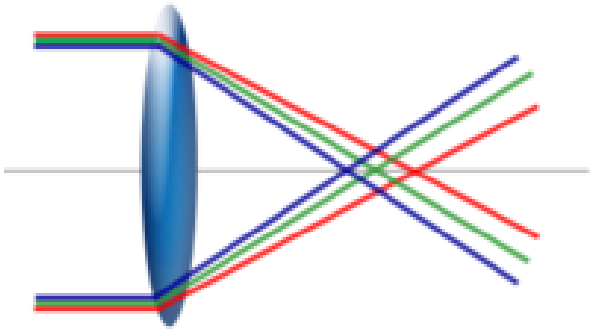
Centration (also known as centering) is defined as the beam deviation of a light beam through a lens. The Pixid assembly systems measure the pointing error of the lens relative to the center of the sensor. During alignment, this measurement is used to adjust the center of focal plane of the lens onto the center of the sensor. After alignment, this measurement is used to verify and report the centration error of the final assembly.



10.2.2 Advanced Lens Tests

10.2.2.1 Chromatic Aberration

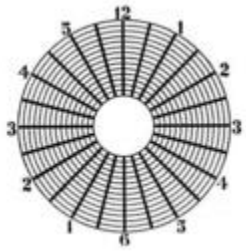
Chromatic aberration is a type of distortion in which different colors on the light spectrum focus on different points. It occurs because lenses have a different refractive index for different wavelengths of light (i.e. a lens bends different colors by differing amounts - the dispersion of the lens). Chromatic aberration results in "fringes" of color along boundaries that separate dark and bright parts of the image – a rainbow effect.



10.2.2.2 Astigmatism

MTF data for a lens can be separated into meridional and sagittal curves. Meridional MTF curves are the curve response for lines that are tangential to concentric circles around the center of the target. Sagittal

MTF curves are for lines that radiate out from the center of the target. Astigmatism occurs when the meridional and sagittal MTF curves differ. The degree by which they differ gives the astigmatism value.

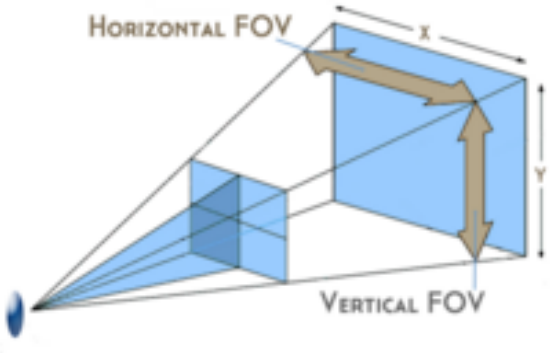


10.2.2.3 Field of View

There are 3 measurements for field of view:

1. Horizontal field of view
2. Vertical field of view
3. Diagonal field of view

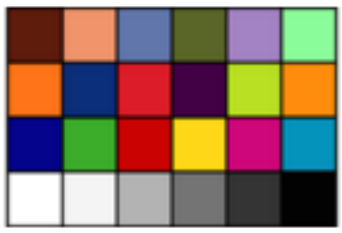
The field of view data are calculated from known reference points on the outer edges of the target.



10.2.3 Advanced Camera Tests

10.2.3.1 Color Reproduction

Color reproduction is checked using a color chart such as the X-Rite (Gretag MacBeth) chart. This test is used to verify that colors measured by the image sensor are correct.



10.2.3.2 Dynamic Range

The dynamic range is a measure of the highest and lowest amount of light that the sensor can capture. With the sensor in fixed exposure mode, a series of images are taken while the target illumination is steadily increased from off to highest light level. This establishes 2 thresholds – the lowest light level at which the camera can distinguish features, and the highest light level at which the camera can distinguish features before saturation. From this, the Pixid calculates the dynamic range.

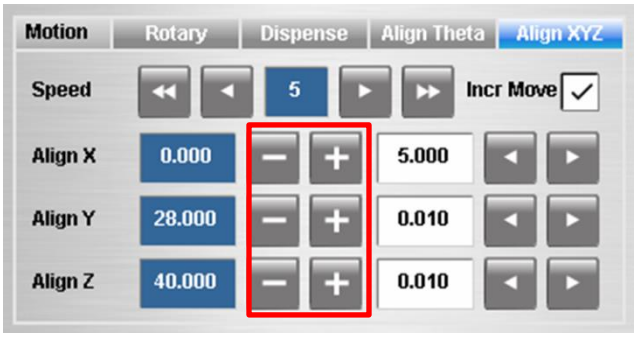
10.3 Lens pick and place positions

Griper arms at align station grip the lens and place it on the top of the image sensor to perform active alignment process. The process of teach pick and place positions is simplified by six (6) parameters in Ks.process file; Pickposition X. Pickposition Y, Pickposition Z, Place position X, place positon Y, Place position Z.

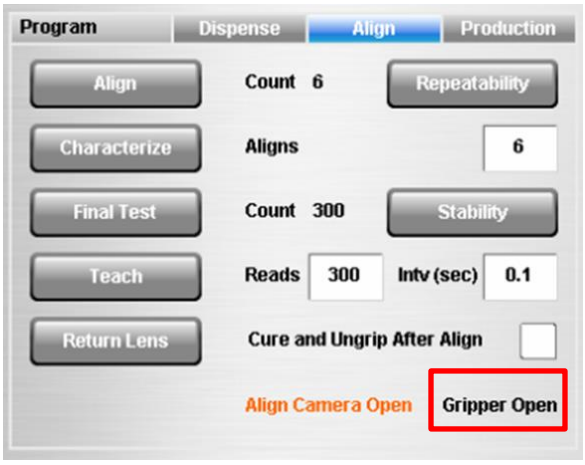
10.3.1 Teach pick and place positions

Pick and place positions can be setup using the following steps

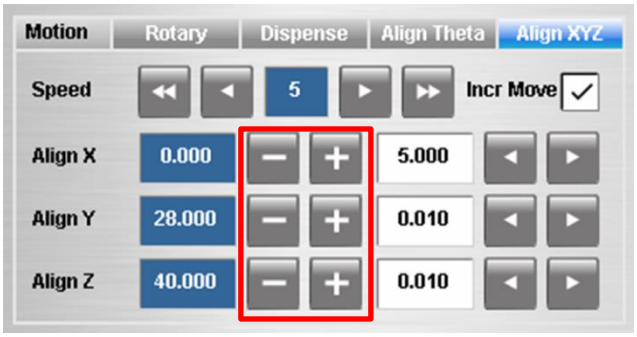
1. Load one camera and one lens in station number one (1)
2. Index station number one (1) to align station
3. Manual from GUI screen move X,Y and Z axis until the lens is griped properly



4. Close and open the grippers at pick positions to ensure that the lens is gripped properly



5. Once the lens is gripped properly, Record XYZ positions from motion tab in GUI screen

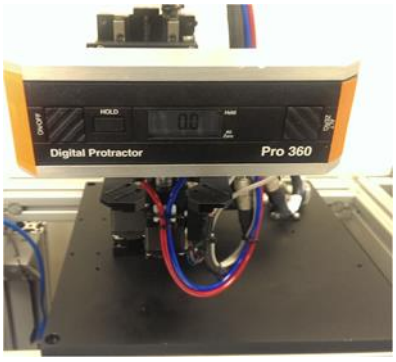


6. Open ksproces file and replace the current Pickposition X. Pickposition Y, Pickposition Z with the new positions from step
7. Move the grippers in Z after the gripper is close to ensure that the lens dose not interfering with any mechanical components.
8. Manually move the gripper (with the lens) in XYZ until the lens is place properly on the top of image sensor
9. In GUI screen under align tab, click on open camera to grab an image from image sensor
10. Manually move the gripper (with the lens) in XYZ until the center target is aligned with the center of the image
11. In GUI screen under align tab run alignment
12. After alignment process is completed, record XYZ positions from motion screen
13. Open ksproces file and replace the current Placeposition X. Placeposition Y, Placeposition Z with the new positions from
14. IN GUI screen under align tab click on return lens then run alignment again to ensure that pick and place positions as correct.

10.4 Home position for theta Y(TY)

The home position for Ty at alignment station must be setup properly to enhance tip/tilt routine during active alignment process. The following steps must be followed in order to setup Ty home positions:

1. Place digital level on the top of the grippers.



2. If the digital level is not ready Zero, record the reading from the digital level

3. From GUI screen, under params button, open .ks global file and find a parameter called “AlignHomeOffsetTY”



4. Adjust “AlignHomeOffsetTY” according the reading from the digital level

```
# AlignHomeOffsetTX, AlignHomeOffsetTY,
# switch that the align rotational axis
# its zero position.  Adjust these offs
# are at their zero postions.
AlignHomeOffsetTX=5.950
AlignHomeOffsetTY=10.280
AlignHomeOffsetTZ=5.050
```

5. Save and close .Ksglobal file
6. Remove the digital level from the top of the grippers
7. Initialize the system
8. Repeat steps 1 -7 until the reading from digital level is ZERO.

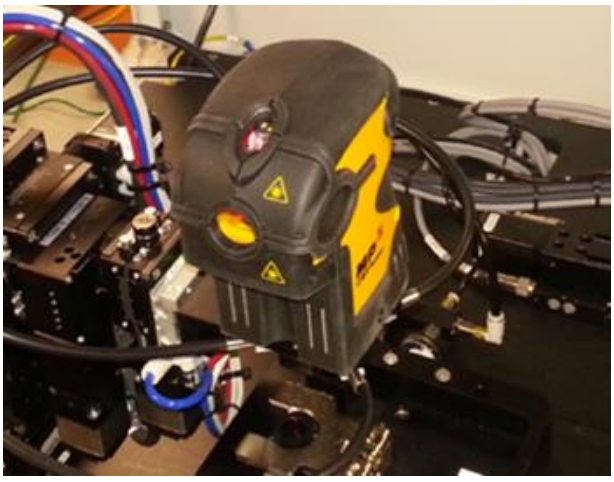
10.5 Center target setup

The optical axis of the image sensor must be aligned with “on axis target feature”, where image sensor optical axis must be pointing at the center of “on axis target feature”. The mechanical setup for system centration enhances centration process during lens active alignment process

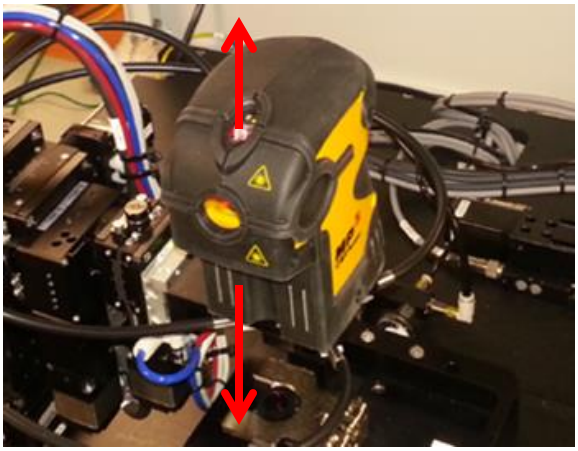
10.5.1 Wide angle of view

The centration procedure for wide angle of view camera is summarized in the following steps.

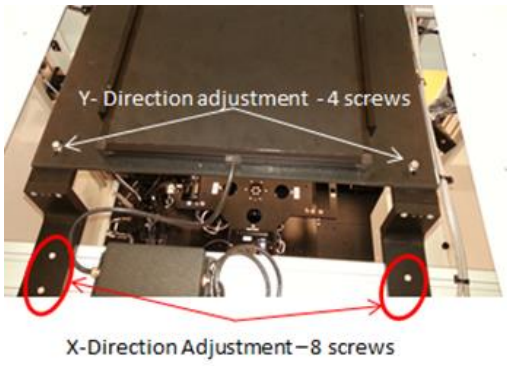
1. Place laser bulb on the top of the grippers

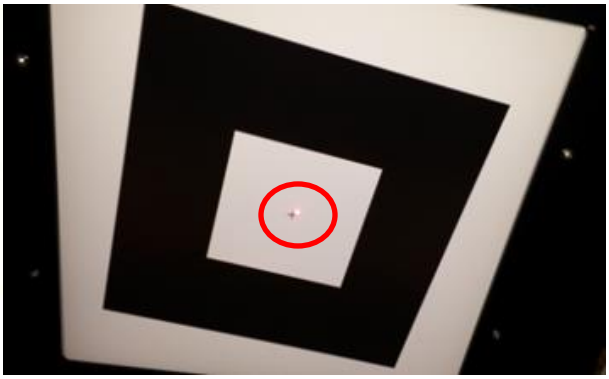


2. Set up the laser where the bulb is pointing at the center of the image sensor



3. Adjust the center target in X and Y until the laser is pointing at the fiducial feature in the “on axis target feature”

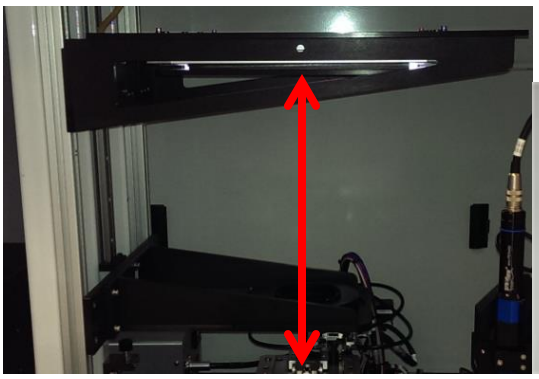




10.5.2 Narrow angle of view

The centration procedure for wide angle of view camera is summarized in the following steps.

1. Remove the diopter
2. Place a laser bulb on the top of diopter bracket





3. Ensure the laser is pointing at the center of image center
4. Adjust the target in X and Y until the laser is pointing at the fiducial feature in the “on axis target feature”

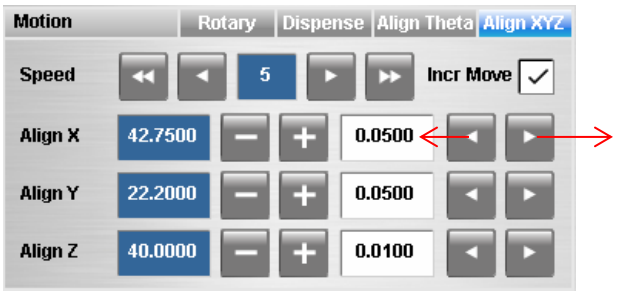
10.6 Alignment Stage Backlash

The backlash in leadscrews is lost motion when the leadscrew reverses direction due the gap between leadscrew threads and nut. The Pixid software compensates for this the lost motion with the ksglobal file BacklashOffsetX, BacklashOffsetY, and BacklashOffsetZ parameters. This section describes how to determine the proper values for these parameters.

10.6.1 Alignment Stage XY Backlash

Determine the proper values for the BacklashOffsetX and BacklashOffsetY parameters by performing the following procedure:

1. Zero out the BacklashOffsetX and BacklashOffsetY values in the ksglobal file.
2. Run a manual dry alignment of a camera.
3. Move the Align X axis 50 microns in the positive direction, immediately followed by moving back 50 microns in the negative direction by setting the incremental move distance to 0.0500 in the Motion jog panel and using the “Incr Move” mode.



4. Execute final test and record the X Adjust value in Notepad.



5. Repeat steps 3 and 4 above five times and calculate the average X Adjust value when moving positive then negative.
6. Move the Align X axis 50 microns in the negative direction, immediately followed by moving back 50 microns in the positive direction.
7. Execute final test and record the X adjust value in Notepad.

8. Repeat steps 6 and 7 above five times and calculate the average X Adjust value when moving negative then positive.
9. Calculate the difference between the average X Adjust value calculated in in step 5 and the average X Adjust value calculated in step 8.
10. Subtract 0.001 from the difference value calculated in step 9, and use this value as the BacklashOffsetX value in the ksglobal file.

    Example:

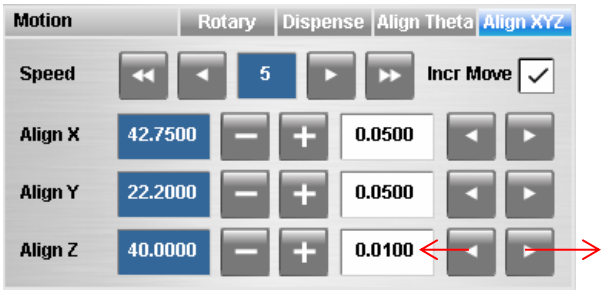
average X Adjust value calculated in step 5:	-0.0014
average X Adjust value calculated in step 8:	+0.0128
difference	+0.0142
subtract .001 from difference	+0.0132
BacklashOffsetX	+0.0132

11. Perform steps 1 through 10 above for the Y axis by moving the Y axis in the same manner and calculating the difference between the average Y Adjust values less 0.001 to determine the BacklashOffsetY value.

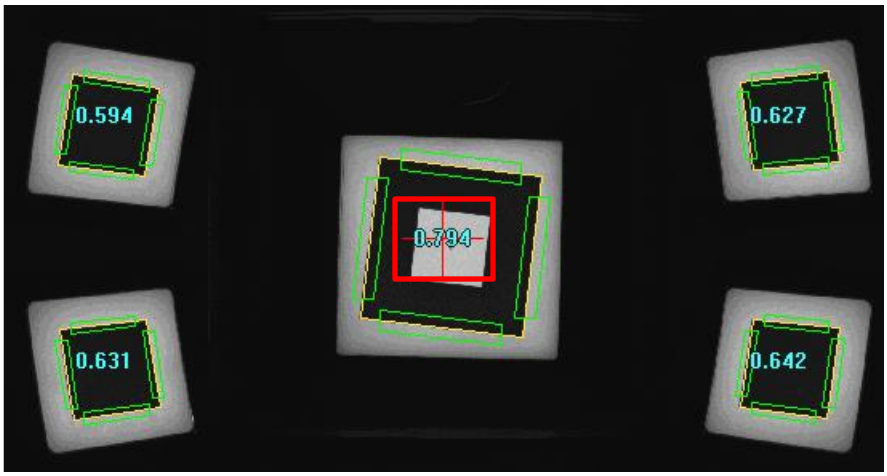
10.6.2 Alignment Stage Z Backlash

Determine the proper value for the BacklashOffsetZ parameter by performing the following procedure:

1. Zero out the BacklashOffsetZ value in the ksglobal file.
2. Run a manual dry alignment of a camera.
3. Move the Align Z axis 10 microns in the positive direction, immediately followed by moving back 10 microns in the negative direction by setting the incremental move distance to 0.0100 in the Motion jog panel and using the “Incr Move” mode.



4. Execute final test and record the center MTF value in Notepad.



5. Repeat steps 3 and 4 above five times and calculate the average center MTF value when moving positive then negative.
6. Move the Align Z axis 10 microns in the negative direction, immediately followed by moving back 10 microns in the positive direction.
7. Execute final test and record the average center MTF value in Notepad.
8. Repeat steps 6 and 7 above five times and calculate the average center MTF value when moving negative then positive.
9. Calculate the difference between the average center MTF value calculated in in step 5 and the average center MTF value calculated in step 8.
10. Add 0.0005 (1/2 micron) to BacklashOffsetZ value in the ksglobal file and repeat steps 1 through 9 until the difference calculated in step 9 is as close to zero as you can get it, then subtract 0.0005 from this value and enter it as the final BacklashOffsetZ value.

10.7 AlignVPPOffsetZ

Six DOF stage has a mechanical pivot point where Tx,Ty &Tz rotate. Once alignment process is started, Tx,Ty &Tz rotate around lens pivot point. Align virtual pivot pointZ (VPP) is the offset between virtual pivot point of the lens and mechanical pivot of alignment stack. During tip/tilt adjustment process, XY axis moves slightly to compensate the adjustment in Tx and Ty without losing lens centration point.

VPP setup process depends on the lens optical feature. The steps below are recommended to setup VPP for new lens

1. Align one camera using the current process recipes
2. In GUI screen under motion tab click on Align Theta



3. Check incr-move box 

4. Enter 0.50 in align Ty then click on 
5. Open AlignLive screen and notice if the center of the image is shifted when  is clicked.



6. If the center of the image is shifted, Open ksprocess file and change VPP until there is no shift
7. Repeat previous steps for Tx

10.8 Alignment station limit switches

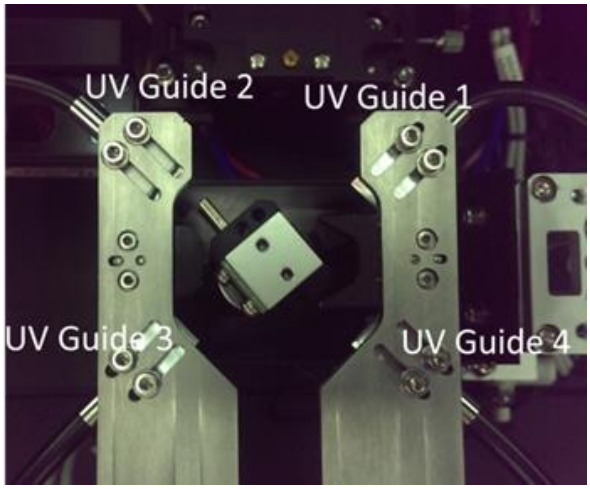
See section 5.17 for details.

10.9 Alignment station pogo pins

See section 5.16 for details

10.10 UV Cure

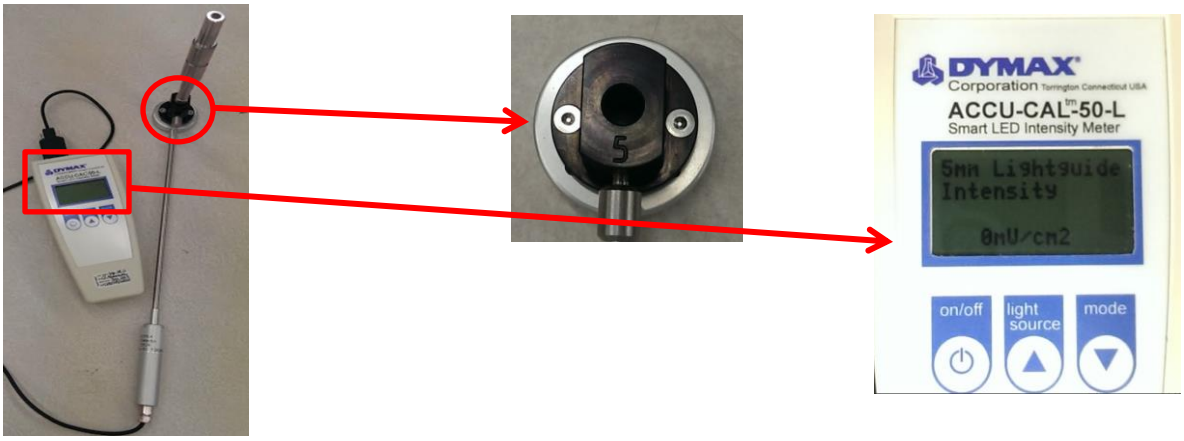
UV light source is used to cure the adhesive after alignment process is completed. UV wands fiber optics is attached to the grippers arms at all the time during alignment process.



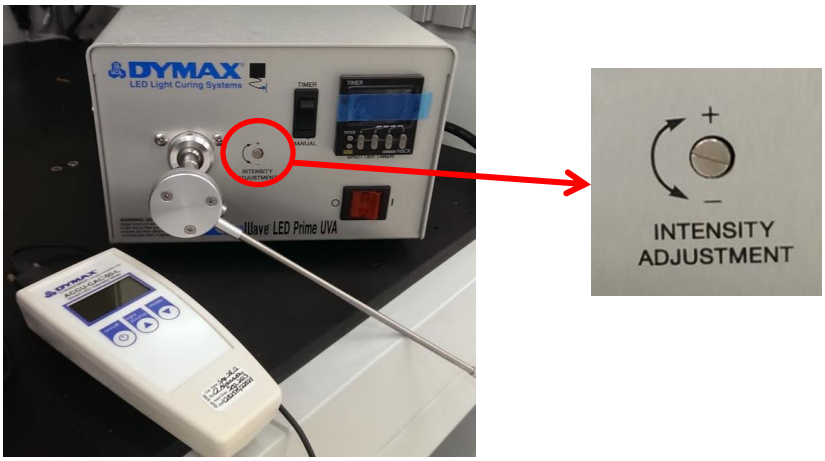
10.10.1 UV Cure calibration

The UV source is calibrated by measure the intensity of the UV light at the source using radiometer. Calibration steps are:

1. Use the radiometer to calibrate UV light
2. Make sure the diameter of the adapter is (5 mm)
3. Make sure radiometer output to be equal to (5 mm)
    a. Note: If 8 mm adapter is used, radiometer output must be equal to 8 mm.



4. Attached the radiometer to UV light source
5. Increase light Intensity to the max by turning the screw in clockwise

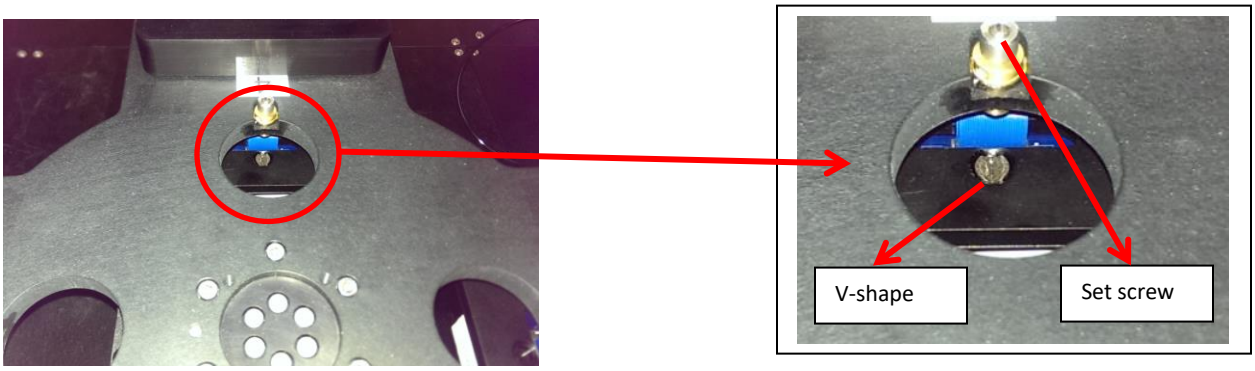


6. Set up UV light intensity to desired intensity by turning the screw counter clockwise

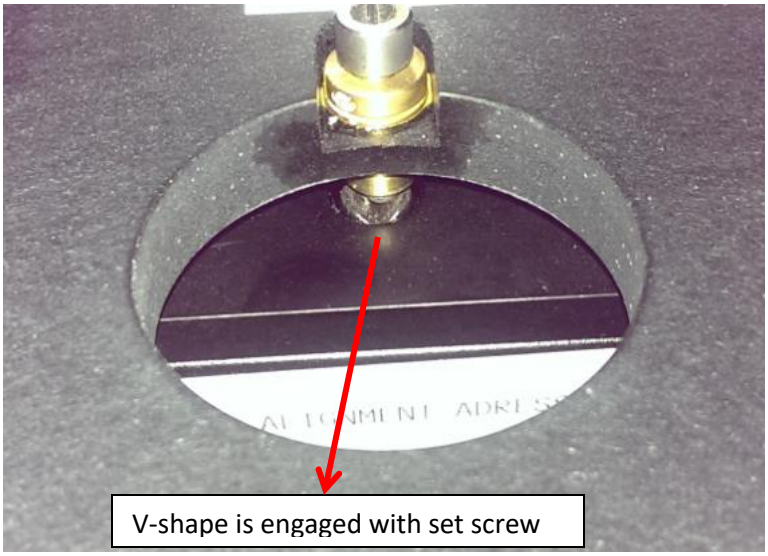
10.11 Rotary table home offset

The purpose of rotary table home offset is to lock index table in place during alignment and dispense process. The repeatability of index table position prevents failure due streaming images. The process of setting up the rotary table home offset is as follow:

1) Initialize the system and check if the set screw is aligned with V-shape fittings at station number three



2) Setup “rotarytablehomeoffset” in ksGlobal file counterclockwise (-) or clockwise (+)
3) Re-initialize the system and check if
   a) The set screw is aligned with V-shape fitting at station number three
   b) The rotary table dose not moves when V-shape is engaged with set screw



4) Index rotary table and check station number four, station number two and station number one
5) Repeat steps 1 -4 (if necessary)

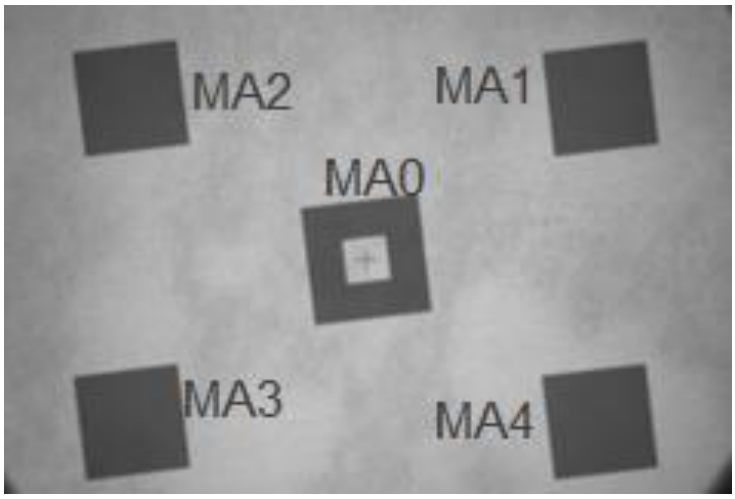
11 Target module

The target module includes a backlit targets for feedback during the alignment process as well as for functional testing of the camera module. The backlit targets provide uniform illumination across the entire target area for optimal alignment and testing measurements. The alignment/test target is placed at desired height from lens surface to the target.

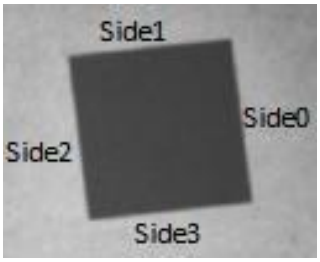
11.1 Target features

Target features are five slanted edge targets; 1 on-axis (center) and 4 off-axis (corner) target features. Pixid software uses target map to report the data for each individual target. The targets are mapped as

1) MA0: is the center target
2) MA1: is the back right target
3) MA2: is the back left target
4) MA3: is the front left target
5) MA4: is the front right target



The target contains features for performing standard optical tests. Slanted square target feature is used to calculate MTF based on the ISO 12233 standard. MTF feedback is used for optimizing focus and for verifying that the camera module's focus satisfies pass/fail criteria after assembly. The angle of each side of the slanted target is reported in "FinalTestPeak_MTFData" file. The sides of slanted edge target are mapped as side0, side1, side2 and side 3. Each side has an angle for 5°-11°.



12 Rotary module

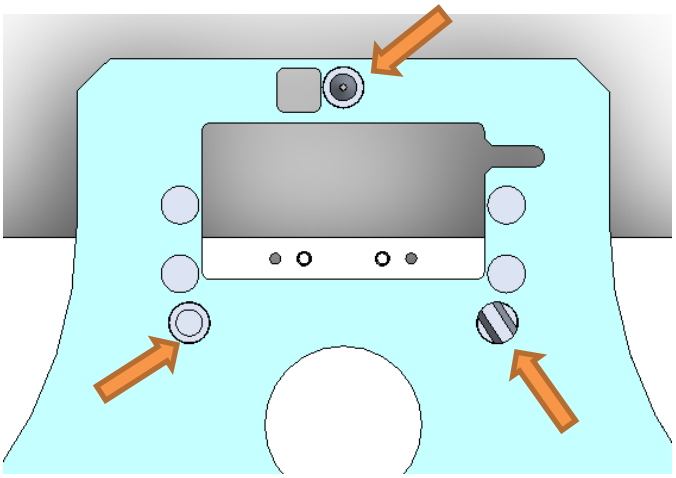
The rotary module is a motorized rotation table that moves the camera module from one process module to another. Attached to the indexing module are custom designed camera fixtures that are used to mechanically and electrically engage camera modules loaded into the machine.

12.1 Rotary table home offset

See section 6.11

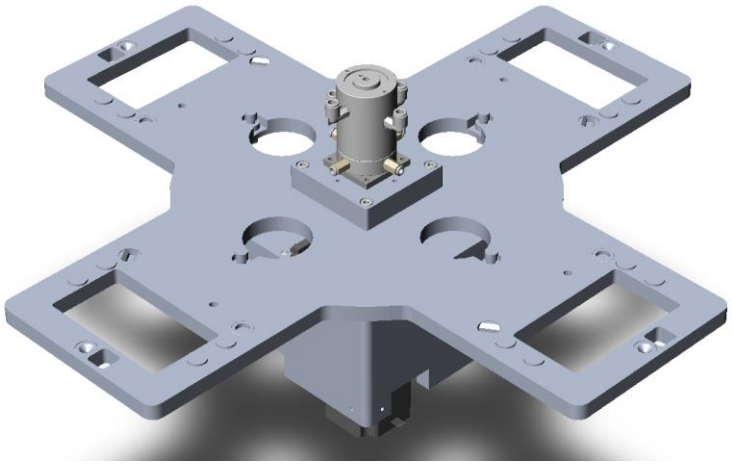
12.2 Kinematic coupling

The index table has a kinematic coupling that guides the fixture in place using loading and unloading process. The kinematic couplings guide the fixture in X, Y positions.



12.3 Vacuum (Optional)

The image sensor or the lens can be held down my vacuum. The vacuum is enabled during alignment and dispenses process but it disabled at load station to make loading and unloading process more robust.



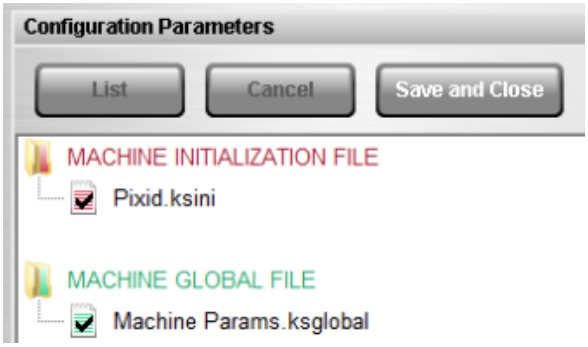
12.4 Rotary table configuration

The rotary table has four fixtures. The system can be run using only three, two or one fixture if the other fixtures need maintenance. To define the active stations do the following:

1) Press the Params button at the bottom right corner of the main user interface. Note that this button is not available while in Operator mode.



2) Press the List button at the upper left corner of the Configuration Parameters view, and then double-click the machine global file (ksglobal file).



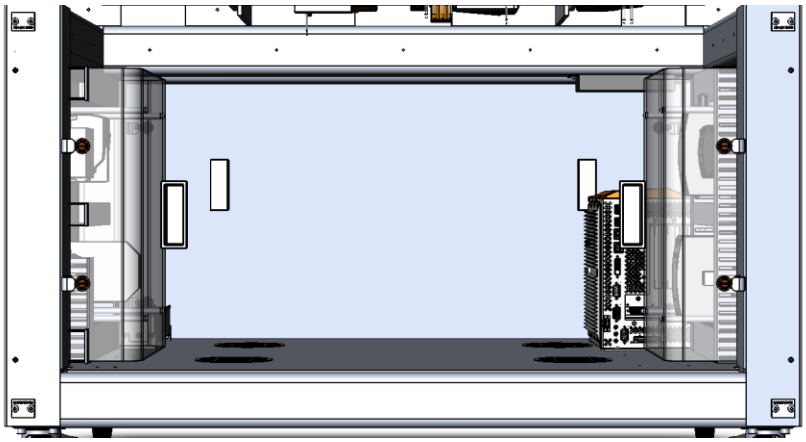
3) Set the TablePositionEnabledMask parameter near the top of the file in the GENERAL section as needed according to the instructions preceding this parameter in the file.
   a) TablePositionEnabledMask=15; the parameter is bit1+bit2+bit4+bit8=15 (table position 1 + table position2 + table position 3+ table position4) are active
   b) TablePositionEnabledMask=12 ; the parameter is bit 8 + bit 4 (table position 3+ table position4) are active

```
[GENERAL]
# TablePositionEnabledMask : A bit mask that allows enabling or
#disabling individual table positions on
# the rotary index table.  Bit values are as follows:
# Bit 0 [1] If bit is set table position 1 is enabled, otherwise it is disabled.
# Bit 1 [2] If bit is set table position 2 is enabled, otherwise it is disabled.
# Bit 2 [4] If bit is set table position 3 is enabled, otherwise it is disabled.
# Bit 3 [8] If bit is set table position 4 is enabled, otherwise it is disabled.
TablePositionEnabledMask=15
```

4) Press the Save and Close button near the upper left corner of the Configuration Parameters view to commit the parameter change.

13 Base and Enclosure module

The base station incorporates the AC Distribution and DC distribution panels and the industrial control PC and the touchscreen monitor.



The safety enclosure protects the operator from moving parts and UV light exposure and also prevents external light from affecting the alignment and testing.

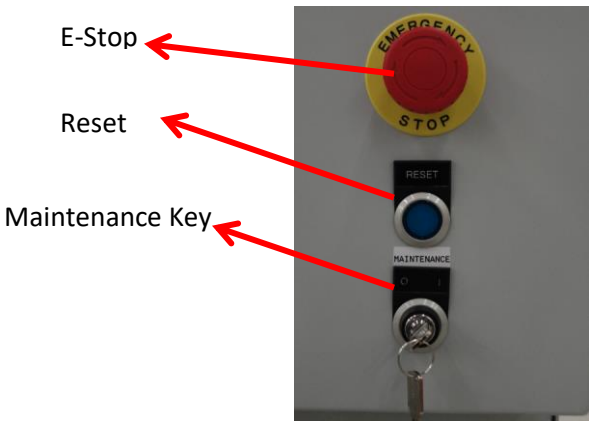
13.1 Light Tower

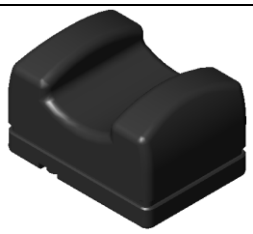
The light tower indicate system status during production

Machine State	Red	Yellow	Blue	Green	Buzzer
E-Stop	ON	OFF	OFF	OFF	OFF
Machine running normal cycle in run Mode	OFF	OFF	OFF	ON	OFF
Machine Running in Maintenance Mode	OFF	FLASHING	OFF	ON	OFF
Access Door Open	OFF	OFF	ON	OFF	OFF
Adhesive Low – Warning	OFF	ON	OFF	OFF	OFF
Adhesive Low – Critical	OFF	ON	FLASHING	OFF	ON
Adhesive Pot Life – Warning	OFF	ON	OFF	OFF	
Adhesive Pot Life – Critical	OFF	ON	FLASHING	OFF	ON
Machine Idle, waiting for parts	OFF	OFF	FLASHING	OFF	OFF
Robot Communications error	OFF	OFF	OFF	OFF	OFF
Critical error (non-recoverable, requires init.)	FLASHING	OFF	OFF	OFF	OFF
Non-critical error (waiting for operator response)	OFF	ON	FLASHING	OFF	OFF

13.2 Controls

The safety enclosure protects the operator from moving parts



Control	Description
	***Emergency Stop:*** This button disconnects all power to the machine and stops all operations. It must be released before turning the machine on again
	***Power Button:*** This button turns on the machine or resets the machine after an emergency stop.
	***Key Switch:*** Turn the key to the O position sets the machine into operation mode, while turning the key to I will set the machine into maintenance mode.
	***Optical Touch Button:*** This button is used to indicate that the part has been loaded and is ready for the index table to rotate and perform the next operation.

14 Operations

14.1 Production mode

When a part is ready to be loaded for the first time or unloaded after a complete cycle, a green light will appear in the optical touch button after the operator click on build button in GUI screen.

The part should be removed properly and then next part should be loaded in properly. Once the part is loaded, the operator touches the optical touch button to trigger the next cycle (this will not happen immediately upon touching the button, but rather when the machine has prepared for the new part).

The rotary table moves the fixture from the load station to the dispense station to execute dispense sequence and dispense inspection. The rotary table moves the fixture from the dispense station to the align station to perform alignment process. Once the best focus is achieved, the UV system is activated to tack cure the lens to the camera. A final test image is taken and the data stored.

The rotary table moved the fixture from the align station through the spare station to the load/unload station. The completed camera is unloaded from the fixture. All operations at each station are carried out in parallel.

14.2 Manual alignment operation

Alignment tests can be run independently from the production mode on the test button under align tab. For instance;

To run Alignment, index the rotary to alignment station. Then select the Align tab at the upper left area of the GUI press the Align button.

To run characterization, index the rotary to alignment station. Then select the Align tab at the upper left area of the GUI press the Characterize button

Same methodology can be applied for dispense for instance;

To active dispense sequence; index the rotary to dispense station. Then select the dispense tab at the upper left area of the GUI press the dispense button

14.3 Test mode and Dry run operation

Dispensing adhesive on a module can be disabled by setting a value of DryRunMask to 1 instated of 0. DryRunMask is exist in the first parameter in the DISPENSE section of the ksglobal file. It is typically set to a value of either 0 to enable activation of the dispenser, or 1 to disable activation of the dispenser The second bit in this parameter can also be set to disable vision inspection, but this is not typically used and is not required to successfully operate the machine or adjust the dispense parameters.

15 Data logging

15.1 Data folder

Pixid software save the data in DATA folder in D:drive. Parameters to save CSV files and/or images are existing in .Ksini files. The files reported may be different depends on customer specifications. The summary of data files are shown in table below as:

File	Data
Pre-dispense insepection.bmp	An image of the module before dispense
Post dispense inspection_01.csv	A csv file with post dispense inspection data
PostDispenseInspectSource_01.bmp	An image of post dispense glue before analysis
PostDispenseInspectResult_01.bmp	An image of post dispense glue after analysis
AlignmentScan01.csv	A csv file for 1st alignment scan
AlignmentScan02.csv	A csv file for 2nd alignment scan
FinalTestPeak_MTFData.csv	A csv file for SFR curves and slanted edge angle
FinalTestPeak.csv	A csv file for final scan data
FinalTestPeak.bmp	An image of final test peak
FinalTestExtraOffset_MTFData.csv	A csv file for SFR curves and slanted edge angle after applying extra offset
FinalTestExtraOffset.csv	A csv file for extra offset scan data
FinalTestCureOffset_MTFData.csv	A csv file for SFR curves and slanted edge angle after apply shrinkage offset but before applying UV cure
FinalTestCureOffset.csv	A csv file for scan data after apply shrinkage offset but before applying UV cure
FinalTestUngrip_MTFData.csv	A csv file for SFR curves and slanted edge angle after open the grippers
FinalTestUngrip.csv	A csv file for scan data after open the grippers
FinalTestPostCure_MTFData.csv	A csv file for scan data after apply shrinkage offset and after applying UV cure
FinalTestPostCure.csv	A csv file for scan data after apply shrinkage offset and after applying UV cure
FinalTestRetest.csv	A csv file for final scan data when modules are retested after thermal cure
FinalTestRetest_MTFData.csv	A csv file for SFR curves and slanted edge angle when modules are retested after thermal cure
FinalTestRetest.bmp	An image of retest when modules are retested after thermal cure
MasterProcessLogData.csv	A csv file records data for modules during production

15.2 MasterProcessLogData

Pixid software export one file during production that contains all data parts serial number and production data. The file is csv which can be opened in excel. The heading definition for each column is summarize in table below

15.2.1 General information section

Process name	The name of the currently loaded process file, without the file extension.
Operator ID	[reserved for future use]
Work ID	A monotic unique ID number assigned to each camera that is processed by the machine during either production or manual align or dispense operations.
Camera ID	The identification string assigned to the camera by the operator at the load parts prompt (via text entry or bar-code scan). Can also be assigned automatically by reading from the device itself for devices which support this feature.
Lens ID	The identification string assigned to the camera lens by the operator at the load parts prompt (via text entry or bar-code scan).
Lot ID	The identification string assigned to the camera lens lot by the operator at the load parts prompt (via text entry or bar-code scan).
Fixture ID	The ID assigned to the fixture in the ksprocess file.
Fixture position	The position on the rotary table where the fixture is mounted.
Focal Plane	[reserved for future use]

15.2.2 Build Status Information

Build mode	The operation performed on the device, either BUILD (glue and AA process) or RETEST (retesting of a previously built device).
MNTC Key	State of the maintenance key (ON/OFF).
Results	Pass or fail results
Error station	If the device failed during build or retest, specifies the machine station (LOAD, DISPENSE, ALIGN, SPARE) where the error occurred.
Error Category	[reserved for future use]
Error code	If the device failed during build or retest, specifies the error code for the failure.
Error Extra	If the device failed during build or retest, specifies the extended error code for the failure (not always populated depending on the error).
Align extra offset Align extra offset2	Additional distance in microns to move z after alignment has been completed but before executing the “Extra Offset” final test. Negative value moves lens toward sensor, positive moves lens away from sensor.
Bond Gap	
UV cure Offset	Distance in microns to move z after alignment has been completed but before applying UV light to cure the adhesive. Negative value moves lens toward sensor, positive moves lens away from sensor.
UV Measure Head 1,2,3 & 4 Value	UV light intensity at each UV wand (optional: automated UV intensity measure)
Cam Open Retries Align/Dispense	Indicates the number of tries (pogo pins down then up=1 try) before stream image from image sensor at align station/Dispense Station
Target Lux Center, BackLeft, BackRight, FrotLeft, FrontRight	Target light intensity

15.2.3 Cycle Time Information

UV cure time	Time in milliseconds to apply UV light when curing the adhesive at the align station.
UV Cure Time Secondary	Time in milliseconds to apply UV light when curing the adhesive at the spare station. Only applies to systems which have the secondary cure system installed.
Index time	Time in seconds that it took for the rotary table to index between the dispense and align stations.
Dispense time	Time in seconds that the dispense station process took to complete.
Align time	Time in seconds that the align station process took to complete.
Process time	Total time in seconds that the process took (Index Time + longer of Dispense Time or Align Time)
Cycle time	Total time in seconds between successive indexes of the rotary table, including operator load time.

15.2.4 Dispense Information

Dry run	Device was processed with the dispense Dry Run parameter enabled, disabling dispensing of adhesive.
Lens wet	Lens came into contact with adhesive.
Sensor wet	Device came into contact with adhesive.
Adhesive type	Type of adhesive entered into the Head1 Reset screen.
Adhesive batch ID	Batch ID of adhesive entered into the Head1 Reset screen.
Adhesive Expiration Date	Expiration date of adhesive entered into the Head1 Reset screen.
Adhesive volume left%	The percentage of adhesive volume left in the adhesive reservoir. Total reservoir size is configured in the ksdispense file.
Adhesive hours left	The number of hours left before the adhesive expires. Total expiration hours is configured in the ksdispense file
Adhesive Weight	Measure adhesive weight at dispense station (Optional Test)
Sealant Type	[reserved for future use]
Sealant Batch ID	[reserved for future use]
Sealant Expiration Date	[reserved for future use]
Sealant Volume Left %	[reserved for future use]
Sealant Hours Left	[reserved for future use]

15.2.5 Sensor Test Results

Sensor tests are performed at the dispense station prior to dispensing adhesive on the device.

Sensor user test data	[reserved for future use]
Sensor User Test Result	[reserved for future use]
Dark Signal Value	Dark signal value measured when performing the sensor tests at the dispense

	station.
Dark Signal Test Result	Dark signal test result (Pass / Fail – Test threshold configured in the ksprocess file).
Temporal Noise Value	Temporal noise value measured when performing the sensor tests at the dispense station.
Temporal Noise Test Result	Temporal noise test result (Pass / Fail – Test threshold configured in the ksprocess file).
DP Hot Count	Defective pixel test count of hot pixels. To specify specific camera registers to set prior to performing the defective pixels test include a section in the CameraFile called [Pixel Test Pre], and to specify registers to set after performing this test include a section in the CameraFile called [Pixel Test Post]
DP Dead Count	Defective pixel test count of dead pixels.
DP Defective Count	Defective pixel test count of defective pixels.
DP Total Count	Defective pixel test total count of hot+dead+defective pixels.
DP Test Result	Defective pixel test result (Pass / Fail – Test threshold configured in the ksprocess file).

15.2.6 Particle Test Results

Particle tests are executed during the dispense sequence. Up to three tests can be performed at any point before or after dispense, based on where the test operation is defined in the dispense sequence

Particle-[1..3] Inner Count	Number of particles in the configured inner region of the image.
Particle-[1..3] Inner Largest Size	Size (in pixels) of the largest particle located in the configured inner region of the image.
Particle-[1..3] Inner Total Size	Total combined size of all particles (in pixels) in the configured inner region of the image.
Particle-[1..3] Outer Count	Number of particles in the configured outer region of the image.
Particle-[1..3] Outer Largest Size	Size (in pixels) of the largest particle located in the configured outer region of the image.
Particle-[1..3] Outer Total Size	Total combined size of all particles (in pixels) in the configured outer region of the image.
Particle-[1..3] Test Result	Particle test result (Pass / Fail – Test thresholds configured per test operation in the ksdispense file).

15.2.7 Lens Pick & Place Information

Pick X	X axis position when picking the lens from the fixture lens holder.
Pick Y	Y axis position when picking the lens from the fixture lens holder.
Pick Z	Z axis position when picking the lens from the fixture lens holder.
Pick TX	TX axis position when picking the lens from the fixture lens holder.
Pick TY	TY axis position when picking the lens from the fixture lens holder.
Pick TZ	TZ axis position when picking the lens from the fixture lens holder.

Place X Nominal	X axis position when placing the lens above the device.
Place X Applied	Applied X axis position when placing the lens above the device when AI logic is turned on
Place Y Nominal	Y axis position when placing the lens above the device.
Place Y Applied	Applied Y axis position when placing the lens above the device when AI logic is turned on
Place Z Nominal	Z axis position when placing the lens above the device.
Place Z Applied	Applied Z axis position when placing the lens above the device when AI logic is turned on
Place TX Nominal	TX axis position when placing the lens above the device.
Place TX Applied	Applied TX axis position when placing the lens above the device when AI logic is turned on
Place TY Nominal	TY axis position when placing the lens above the device.
Place TY Applied	Applied TY axis position when placing the lens above the device when AI logic is turned on
Place TZ	TZ axis position when placing the lens above the device.

15.2.8 Active Alignment Information during Alignment

For each alignment scan (up to the first 5 scans) the calculated tip/tilt errors as well as the polynomial equation coefficients used to interpolate the measured data are reported.

Align Scans	Total number of alignment scans required to align the device.
Align Step Size	The configured size of each step of an alignment scan, in microns.
Align Scan-[1..5] Steps	Total number of steps for the scan.
Align Scan-[1..5] VTX	Calculated remaining tip/tilt error in the Theta-X dimension after the scan was completed. This value must be lower than configured angle threshold in the ksprocess file. If it is not the software will execute another scan.
Align Scan-[1..5] VTY	Calculated remaining tip/tilt error in the Theta-Y dimension after the scan was completed. This value must be lower than configured angle threshold in the ksprocess file. If it is not the software will execute another scan.
Align Scan-[1..5] RSQ0	The r-squared value of the polynomial equation used to curve fit the data points collected for the center region (region 0).
Align Scan-[1..5] RSQ1	The r-squared value of the polynomial equation used to curve fit the data points collected for the upper-right corner region (region 1).
Align Scan-[1..5] RSQ2	The r-squared value of the polynomial equation used to curve fit the data points collected for the upper-left corner region (region 2).
Align Scan-[1..5] RSQ3	The r-squared value of the polynomial equation used to curve fit the data points collected for the bottom-left corner region (region 3).
Align Scan-[1..5] RSQ4	The r-squared value of the polynomial equation used to curve fit the data points collected for the bottom-right corner region (region 4).
BoresightOffsetX BoresightOffsetY BoresightOffsetTZ	Identify BoresightOffsetX as specified in ksprocess file

15.2.9 Active Alignment Information during Prefocus routine

Prefocus Noise Count	Count of noise pixels as those pixels that deviate from thei r neighbors
Prefocus Noise Variance	The amount of variance allowed in the image histogram.
Prefocus Noise Test Result	Pass / fail Prefocus Noise Test
Prefocus Num Scan	Total number of scan before determine the prefocus position
Prefocus Start Z	Position of Z axis before prefocus is started
Prefocus End Z	Position of Z axis after prefocus is ended
Prefocus Expected Peak	The expected peak MTF score as specified in ksprocess file If adaptive intelligence is turned on
Prefocus Applied Peak	The applied peak MTF score after prefocus routine is completed
Prefocus Expected FWHM	The expected full width at half maximum, in mm, of the MTF curve as specified in ks process file
Prefocus Applied FWHM	The applied full width at half maximum, in mm, of the MTF curve after prefocus routine is completed

15.2.10 Active Alignment Information after Alignment

Peak Z On Axis	The Z position recorded where the interpolated peak MTF value was calculated for the center region of the image.
Peak Z Off Axis	The Z position recorded where the interpolated peak MTF value was calculated for the average of the corner regions of the image.
Peak DZ	Delta between (Peak Z On Axis- Peak Z Off Axis).
Peak DZ Test Result	Peak DZ test result (Pass / Fail – Test thresholds configured in the final test section of the ksprocess file.
Peak Noise Count	Count of detected noise pixels when performing the peak noise test at the alignment station.
Peak Noise Test Result	Peak noise test result (Pass / Fail – Test threshold configured in the final test section of the ksprocess file).
Peak Saturation Actual Min	The min intensity of image pixels during final test peak
Peak Saturation Actual Max	The max intensity of image pixels during final test peak
Peak Saturation Test Result	Pass/ fail results
Peak Z Weight Mode	The manner in which the final align position is determine based on the on-axis versus off-axis performance of the camera as follows: Mode 0 - Use specified AlignOffAxisWeightPercent value to determine focus position as a percentage of the distance from the on-position peak position to the off-axis peak position where 0 % = on-axis peak, 100% = off-axis peak. Mode 1 - Calculate values at each focus step using the following formula: **value = (center_focus + ((corner_1_focus + corner_2_focus + corner_3_focus + corner_4_focus) / 4)) / 2** These calculated values are

	then interpolated using the specified AlignPolyOrder and the final focus position is the peak of the interpolated result.
Peak Z Weight Pct	The amount of weight given to the off-axis ROIs when determining the final peak z position of the alignment. Valid value is from 0 to 100 where 0 uses the peak on-axis z value exclusively, 100 uses the peak off-axis z value exclusively, and values between 0 and 100 use a compromise between the on and off axis peak z positions based on the weight percentage value
Peak Z Scan Enabled	Whether or not the optional post-alignment Peak Z scan was enabled (0=disabled, 1=enabled). Scan parameters and description exists in the alignment section of the ksprocess file.
Final Scan	The start position and polynomial equation coefficients of the last scan performed prior to completion of the lens alignment are recorded as follows. The coefficients are described by the standard polynomial equation: cN*x^N + cN-1*x^[N-1] + ... + c2*x^2 + c1*x + c0 where N is the configured poly order of the equation.
Final Scan Start Pos Z	The start position of the last scan performed prior to completion of the lens alignment
Final Scan Poly CXQY	The polynomial equation coefficient number 'X' for image region 'Y' where region 0 is center, 1 is upper right, 2 is upper left, 3 is bottom left, and 4 is bottom right.

15.2.11 Post-Alignment Final Test Results

Test results are recorded for each final test event that is enabled. Test events occur at Peak (immediately after alignment completes), Extra Offset, Extra Offset 2, Pre-Cure, Post-Cure, Ungrip, and Retest. Note that when performing Retest the results are recorded in the same fields used to record the "Peak" results – these are the fields prefaced by "Align" in the file. The Peak event is shown in the following table; these fields are repeated for the other final test events.

Align X	Final Test – Peak X axis position.
Align Y	Final Test – Peak Y axis position.
Align Z	Final Test – Peak Z axis position.
Align TX	Final Test – Peak TX axis position.
Align TY	Final Test – Peak TY axis position.
Align TZ	Final Test – Peak TZ axis position.
Align RefPos	[reserved for future use]
Align MA[0..8]	Final Test – Peak MTF score as average of horizontal and vertical sides of slanted edge target feature for image regions 0 through 9.
Align MH[0..8]	Final Test – Peak MTF score as average of horizontal sides of slanted edge target feature for image regions 0 through 9.
Align MV[0..8]	Final Test – Peak MTF score as average of vertical sides of slanted edge target feature for image regions 0 through 9.
Align MAC	Final Test – Peak MTF average of all corner region average scores.
Align MRC	Final Test – Peak MTF range max-min of all corner region average scores.

Align MTF Test Result	Final Test – Peak MTF test result (Pass / Fail).
Align VX	Final Test – Peak centration X error.
Align VY	Final Test – Peak centration Y error.
Align VTX	Final Test – Peak tip/tilt Theta-X error [informational only].
Align VTY	Final Test – Peak tip/tilt Theta-Y error [informational only].
Align VTZ	Final Test – Peak tip/tilt Theta-Z error [informational only].
Align Centration Result	Final Test – Peak centration test result (Pass / Fail).
Align Focus Symmetry Min	Final Test – Peak focus symmetry test min off-axis MTF value observed during test.
Align Focus Symmetry Max	Final Test – Peak focus symmetry test max off-axis MTF value observed during test.
Align Focus Symmetry Score	Final Test – Peak focus symmetry test score.
Align Focus Symmetry Thresh	Final Test – Peak focus symmetry test threshold.
Align Focus Symmetry Test Result	Final Test – Peak focus symmetry test result (Pass / Fail).
Align Field Curvature AvgOnAxis	Final Test – Peak field curvature test on-axis MTF score observed during test.
Align Field Curvature AvgOffAxis	Final Test – Peak field curvature test average of off-axis MTF scores observed during test
Align Field Curvature Score	Final Test – Peak field curvature test score.
Align Field Curvature Thresh	Final Test – Peak field curvature test threshold.
Align Field Curvature Test Result	Final Test – Peak field curvature test result (Pass / Fail).
Test results are recorded for each final test event that is enabled. Test events occur at Peak (immediately after alignment completes), Extra Offset, Extra Offset 2, Pre-Cure, Post-Cure, Ungrip, and Retest	

15.2.12 Post-Dispense Inspection Results

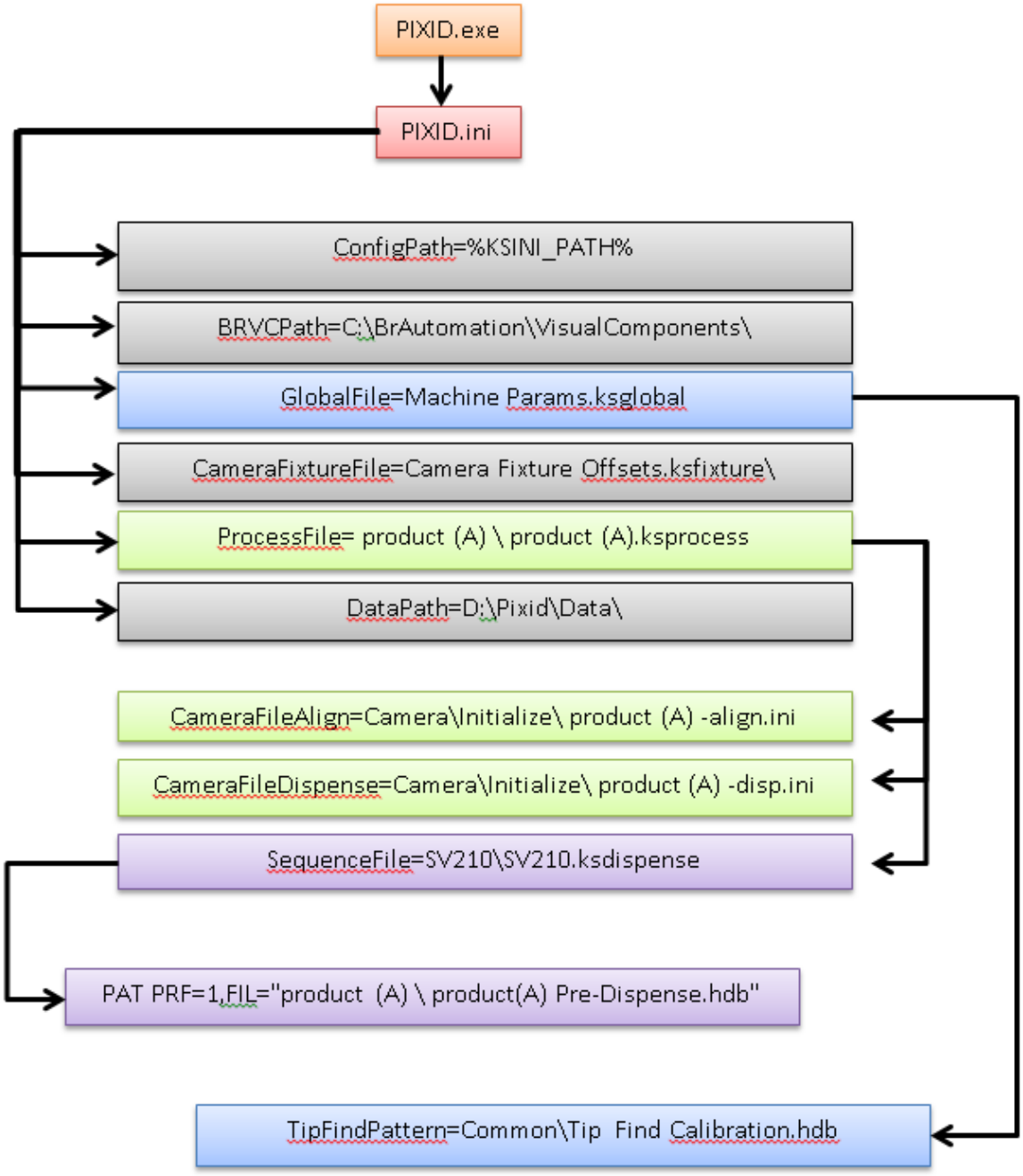
Results are recorded for up to the first five post dispense inspections.

Post Disp Test Result	The overall result of all post-dispense inspections (Pass / Fail).
Disp-[1..5] X	The center of mass position in X where the dispensed material was detected.
Disp-[1..5] Y	The center of mass position in Y where the dispensed material was detected.
Disp-[1..5] Area	The area of detected dispensed material (0 to 100% as a percentage of image area).
Disp-[1..5] BoxArea	The area of the box surrounding the entire detected dispensed material (0 to 100% as a percentage of image area).
Disp-[1..5] Perimeter	The perimeter of the detected dispensed material (pixels)
Disp-[1..5] MinWidth	The min measured cross-section of the dispensed material (pixels, applicable for dispense beads and only applies of cross-section analysis

	feature is enabled).
Disp-[1..5] MaxWidth	The max measured cross-section of the dispensed material (pixels, applicable for dispense beads and only applies of cross-section analysis feature is enabled).
Disp-[1..5] Test Result	The result of the post-dispense inspection (Pass / Fail).

16 Software Reference

The Pixid 500 system software is controlled by parameter settings in a number configuration files stored in the C:\Pixid\Config directory. It recommended that you only use the Main screen Params button and the configuration parameters screen to access these files. An overview of each of these files is given in section titled “Parameter Settings”.



16.1 Pixid.ksini configuration file

The Pixid 500 system expects to find the Pixid.ksini file in the C:\Pixid\Config directory. This file maintains the setting used at system startup and initialization. Managing the location of output data and image files, the camera fixture ksfixture file, process and dispense configuration file directory paths, global parameter settings, and adhesive barcode data and last work ID value.

The Pixid.ksini file is broken up into multiple sections.

Section Name	Short Description
CONFIG	Manage the location of camera fixture file, displayed text language, and main machine global file
DATA	Manage the data path location and what data output files are generated during the machine operation.
PROCESS	Manage the WorkID during the machines operation.

16.1.1 KSprocess Recipes

The Pixid process recipe parameter data is maintained in files with the extension ksprocess. The Configuration Parameters screen list view will display all available ksprocess files. The activated process has a check next to it. The directory location for these process files is managed in the main Pixid.ksini configuration file under the ProcessFile label. The process file has the following sections.

Section Name	Short Description
CAMERA	Manage camera and image types, and initialization settings.
ALIGNMENT	Updated when running Teach Alignment from user interface. Manage lens pick and place position, and alignment threshold settings.
DISPENSE	Manages the path to the dispense sequence file used during the process.
MTF	Settings used during the MTF image processing.
PREWET	Settings used during lens motion contact in adhesive.
PREFOCUS	Settings used during alignment focus.
CENTRATION	Settings to use during centration test.
FINALTEST	Settings to use during final test.
SENSORTEST	Specify specific camera registers to set prior to performing the hot/dead/defective test.
PARTICLETEST	Specify specific camera registers to set prior to performing the particle test.

16.1.2 KSdispense Dispense Sequence

The Pixid dispense parameter data is maintained in files with the extension Ksdispense. The directory location for these process files is managed in the main Pixid.ksini configuration file under the ProcessFile label. There can be multiple ksdispense files. The Configuration Parameters screen list view will display all available ksdispense files. The activated dispense has a check next to it. The dispense files has two sections, CONFIG and SEQUENCE. The config section maintains all the settings related to control dispense material, volume, and pot life. The sequence section maintains the different dispense sequence used in the build.

Section Name	Short Description
CONFIG	Settings used during the dispense sequence. Manage adhesive warning threshold level settings.
SEQUENCE	A pattern of commands to distribute the adhesive around the sensor surface to adhere lens to sensor.

16.1.3 Camera Fixture Offsets

The camera XY alignment offsets, the lens place offsets, and the dispense Z offset that are unique for each machine camera fixture are managed in this file.

16.1.4 Machine Global Parameters

The Pixid 500 machine maintains machine specific global variable settings in the ksglobal file found in the C:\Pixid\Config directory.

Section Name	Short Description
GENERAL	Manage enabled table and rotary park position, and rotary table home offset settings.
ALIGNMENT	Manage alignment station axis motion UV measurement settings.
DISPENSE	Manage dispense station axis motion, prime, and tip settings.

17 New device setup

Setup a new device on Pixid systems requires mechanical changes and software parameters setup. PIXID systems are configured according the field of view (FOV) of a device. Slanted edge target modules can be adjusted or replaced in order for PIXD system to work with a new device. For production purposes each device has custom gripper fingers, interface electronics and customer fixture.

Pixid software is developed to isolated system parameters from process parameters where system parameters are defined as .ksini file and .Ksglobal file but process parameters are defined as .ksprocess file and .ksdispense file.

For production purposes, each device has unique ksprocess file and ksdispense file that contain device process recipe. KSprocess file has the device process parameters that would be used for production. Once Ksprocess file is selected, ksdispense file is automatically selected to execute dispense sequence.

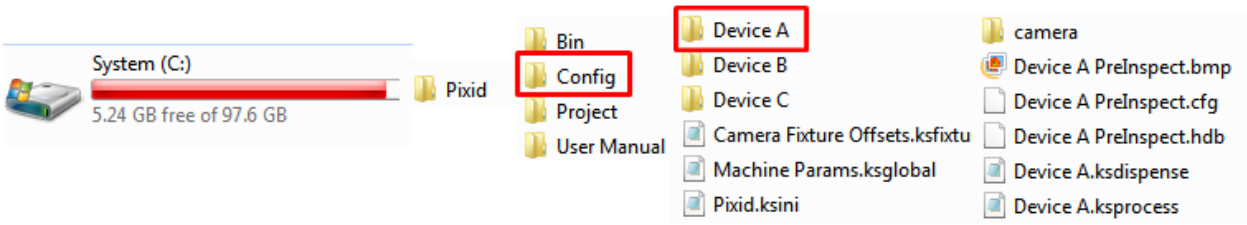
17.1 Mechanical setup

Prior setup a new camera module; a mechanical changes need to be checked and verified through 3D models and KASALIS engineer. Mechanical changes are itemized as shown below:

1. Custom fixture must be designed for the new camera
2. Custom set of gripper and gripper fingers must be designed for the new lens.
3. Custom set of UV mount must be designed
    a. Verify that there no mechanical interference between the gripper fingers, UV mount, the fixture and the camera modules.
4. Adjust/replace target module (center target and corner targets)
    a. Check the focus distance (from the lens to the center target)
    b. Confirm that the center of the center target is aligned with the center of the image sensor and diopter (diopter, if narrow angle of view camera is setup)
    c. Verify that the corner targets within the FOV of the new camera.
    d. Verify that slanted edge angle is between 6°-11°.
5. Alignment and dispense station pogo pins are aligned with fixture electronics to capture an image from image sensor.
6. Setup fixture offsets in ksfixtureoffset file for the new fixtures.

17.2 PIXID software setup

For production purposes, each device has a unique ksprocess file and ksdispense file that contain camera process recipe. In order to setup a new device for production, new ksprocess file must be created. Under C:\Pixid\configs\ a new folder need to be created with the new product name. The new folder holds ksprocess file, ksdispense file, dispense inspection files and image sensor ini files.



17.3 Critical parameters for new process development

When new ksprocess file is created for the new camera, some process parameters must be changed to develop a new process file for production. The parameters are summarize as shown below

17.4 Ksprocess file

Process file has process recipe that control camera active alignment process in terms of thresholds and executed tests.

17.4.1 Camera section

1. FixtureAtTablePos1, FixtureAtTablePos2, FixtureAtTablePos3, FixtureAtTablePos4,: This parameter must be changed according to the number of fixtures. For instance if there is 10 fixtures will be used and each fixture has a unique number, the parameter (FixtureAtTablePos) must change to match fixture ID on each station.
   i. ***Note:*** This parameter must be defined in the .Ksfixtureoffsets file
2. CameraFileALign: verify the path of camera initialize file.
3. CameraFileDispense: verify the path of camera initialize file.
4. MicronsPerPixelX/ MicronsPerPixelY : image sensor pixel size in microns
5. CameraTransform: A bit mask that determines how the image from the camera is transformed (rotation and mirror the image around specific axis). This parameter represents how the targets are physically mapped on streamed image. For instance if X axis is moved in positive X, the streamed image must shift in positive X also. If it is not, use this parameter to adjust image rotation/mirror
6. AlignVPPOffsetZ: Defines the offset to virtual pivot point of the lens from mechanical pivot of alignment stack (prevent the center of image from shifting X&Y due tip/tilt process
7. TargetLightPct: Center target backlight intensity, from 0 to 100%. Prevent the light from saturation (RGP255). The target light must be setup as RGB values are reported between 180-190
8. TargetLightBackLeftPct: same as 6. Corner targets are mapped according to live streamed image on screen
9. TargetLightBackRightPct: same as 6. Corner targets are mapped according to live streamed image on screen
10. TargetLightFrontLeftPct: same as 6. Corner targets are mapped according to live streamed image on screen
11. TargetLightFrontRightPct: same as 6. Corner targets are mapped according to live streamed image on screen

17.4.2 Alignment section

1. LensPickX, LensPickY, LensPickY, LensPlaceX, LensPlaceY, and LensPlaceY: Absolute lens pick and place positions in millimeters.
2. PickRetractDistance: The distance, in millimeters, that the alignment Z stage is retracted relative to the lens pick z position when picking and placing the lens. This value should be large enough to allow the lens to clear any obstructions when being moved from pick XY position to the place XY position.
   a. ***Note:*** confirm that the lens dose not interfere with the fixture when it moves from pick XY position to place XY position
3. AlignMinSteps, AlignMaxSteps : Minimum and maximum number of steps to perform during each scan to graph thru focus curve for each sweep
4. CentrationThreshXYAlign: Threshold in microns that iterative centration adjustment must achieve during alignment process (between each sweep) but prior to final alignment iteration.
5. ThresholdAngle: thresholdAngle is used to define the angular threshold in degrees at which point the alignment is completed and considered to be within the threshold tolerance. This threshold is compared with the determined required adjustment in theta-x and theta-y at the completion of each alignment scan. If the absolute value of the required adjustment in both theta-x and theta-y is less than the threshold angle the alignment will be completed.
6. CentrationThreshXYFinal: Threshold in microns that final iterative centration adjustment must achieve to satisfy alignment complete.
   a. ***Note:*** CentrationThreshXYFinal will be done after alignment process is completed but before execute final test
   b. ***Note:*** number 3,4 and 5 related to lens thru focus curves, lens distortion and field curvature.
7. AlignStepSize: Distance in microns to move for each step of an alignment sweep. Negative value moves lens toward sensor, positive moves lens away from sensor.
8. AlignStepDwell : Delay in milliseconds to wait after each alignment move step before acquiring and analyzing an image from the camera
9. AlignOffAxisWeightMode
   a. Mode 0 - Use specified AlignOffAxisWeightPercent value to determine focus position as a percentage of the distance from the on-position peak position to the off-axis peak position where 0 % = on-axis peak, 100% = off-axis peak.
   b. Mode 1 - Calculate values at each focus step using the following formula: value = (center focus + ((corner_1_focus + corner_2_focus + corner_3_focus + corner_4_focus) / 4)) / 2 These calculated values are then interpolated using the specified AlignPolyOrder and the final focus position is the peak of the interpolated result.
10. AlignOffAxisWeightPercent: The amount of weight given to the off-axis ROIs when determining the final peak z position of the alignment. Valid value is from 0 to 100 where 0 uses the peak on-axis z value exclusively, 100 uses the peak off-axis z value exclusively, and values between 0 and 100 use a compromise between the on and off axis peak z positions based on the weight percentage value.

11. UVCureOffset: Distance in microns to move z after alignment has been completed but before applying UV light to cure the adhesive. Negative value moves lens toward sensor, positive moves lens away from sensor.
12. UVCureTime: Time in milliseconds to apply UV light when curing the adhesive.
    a. UVCureOffset and UVCureTime are determined through shrinkage study

17.4.3 Dispense Section

1. SequenceFile: Full or relative path to dispense sequence file. This file must be existed in the root of the configuration path or somewhere in a sub-folder or nested sub-folder of the configuration path. See *.ksini file ConfigPath parameter

17.4.4 MTF section

1. SetPointMode: 0 = modulation percentage, 1 = frequency. This parameter specifies whether the MTF algorithm takes modulation percentage as its SetPoint and returns frequency, or takes frequency as its SetPoint and returns modulation percentage. For example,
    a. If SetPointMode = 0 and SetPoint = 0.5 the MTF algorithm will return the frequency response at MTF50.
    b. If SetPointMode = 1 and SetPoint = 0.125 the MTF algorithm will return the modulation percentage at 1/4 Nyquist.
2. SetPoint: the range is between 0.05 to 0.9
    a. if SetPointMode = 0 (SetPoint is modulation percentage such as 0.5 to calculate MTF at 50% modulation percentage commonly known as MTF50), or 0.01 to 0.5
    b. if SetPointMode = 1 (SetPoint is frequency such as 0.125 to calculate MTF at 1/4 Nyquist
3. EvaluationMode : A mode to report MTF score from Slanted edge target
    a. 0 = Average all sides: report the average of MTF score for all sides (4 sides of one slanted edge square).
    b. 1 = horizontal sides only: report MTF for horizontal sides only (2 horizontal sides of one slanted edge)
    c. 2 = vertical sides only: report MTF for vertical sides only (2 vertical sides of one slanted edge)
    d. 3 =combination of horizontal and vertical side scores where reported score is a weighted percentage based on EvaluationHorzPct (see parameter definition below)
4. EvaluationHorzPct: The amount of weight given to average of 2 horizontal sides of slanted edge when reporting MTF scores for slanted edge targets. Valid value is from 0 to 100 where
    a. 0 is vertical sides only, 100 is horizontal sides only, and values between 0-100 use a compromise between the horizontal sides and vertical sides of slanted edge target when reporting MTF score based on the weight percentage value.
    b. For instance; EvaluationHorzPct=30
5. CenterROISizeHorizontal, CenterROISizeVertical: Center ROI size as percentage of image width / height
6. CornerROISizeHorizontal, CornerROISizeVertical, CornerROIPosHorizontal, CornerROIPosVertical: Corner ROI size and position as percentage of image width / height

7. FilterAreaMin and FilterAreaMax: Area filter as a percentage of image size (width x height
   a. Blob analysis for MTF section to detect slanted edge features
   b. Use Ksvision utility to setup the threshold and filters

17.4.5 Prefocus section

1. MTFStopThreshold : At each step of the pre-focus scan the MTF score at the center ROI is measured, if possible, and if the measured value is greater than or equal to the configured MTFStopThreshold the scan will terminate
   a. This parameter must be setup according to the value of SetPoint&SetPointMode in MTF section. For instance, if MTFStopThreshold is setup to 0.20 and alignment process is using Nyquist frequency, the prefocus scan will be terminated. MTFStopThreshold value must be updated when SetPoint&SetPointMode is updated.
2. GradientStep: Distance in pixels at which gradient is measured. Valid from 1 to 20 pixels, or 0 to disable prefocus
   a. Verify that MTFStopThreshold is setup as best focus value.
   b. Only change this parameter if the alignment fail prefous scan due MTFStopThreshold.
3. GradientThreshold: minimum required change in pixel intensity to be included in focus measurement expressed as a percentage of image luminance resolution.
   a. For example, if the image is an 8 bit image the pixel intensity will vary between 0 (for black) and 255 (for white) and a GradientThreshold value of 10.0 % would equate to a minimum required change in pixel intensity of 255 * 0.1 or 25.5. If set to 0 the minimum required change in pixel intensity will be set to 1.
   b. Verify that MTFStopThreshold is setup as best focus value.
   c. Only change this parameter if the alignment fail prefous scan due MTFStopThreshold.
4. ROISizeHorizontal, ROISizeVertical: OI size as percentage of image width / height
   a. KSvision utility should be used to setup this parameter

17.4.6 Centration section

1. ROISizeHorizontal, ROISizeVertical: ROI size as percentage of image width / height
2. FilterAreaMin and FilterAreaMax: Area filter as a percentage of image size (width x height)
   a. KSvision utility should be used to setup this parameter

17.4.7 Final test section

1. MTFMinPeakCenter, MTFMinPeakCorners, MTFMinPeakNSEW: The min valid center, corners, and sides MTF scores expressed in cycles per pixel at the point in production when the lens has been aligned but prior to curing the adhesive. MTF scores range from 0 to 1. If this test is enabled all scores for active MTF regions must meet these minimum valid thresholds. 'A' is the average of all four sides of a given MTF feature, 'H' is the average of the two horizontal sides, and 'V' is the average of the two vertical sides. Same Applied for extra offsets, post cure offset, ungripoffset and retest final test.
   a. Those parameters are the threshold that defines pass /fail criteria
2. FSExecuteTest: the focus symmetry test

3. FSThresholdPeak, FSThresholdUngrip, FSThresholdUngrip, FSThresholdUngrip, FSThresholdRetest: The maximum variance in off-axis performance of the camera. The variance and thresholds are expressed as a percentage from 0 to 100% using the following formula: v = ((best_off_axis_mtf – worst_off_axis_mtf) / best_off_axis_mtf) * 100.0
    a. It is recommended to keep focus symmetry test off until and 1000 parts are assembled using PIXID system to determine the threshold values.
4. FCExecuteTest: the field curvature test
5. FCThresholdPeak, FCThresholdUngrip, FCThresholdUngrip, FCThresholdUngrip, FCThresholdRetest: The maximum variance in on-axis versus off-axis performance of the camera. The variance and thresholds are expressed as a percentage from 0 to 100% using the following formula: v = (abs (on_axis_mtf – off_axis_mtf) / best_mtf) * 100.0 where best_mtf is the greater of on_axis_mtf or off_axis_mtf.
    a. It is recommended to keep field curvature test off until and 1000 parts are assembled using PIXID system to determine the threshold values.

17.5 Ksdispense File

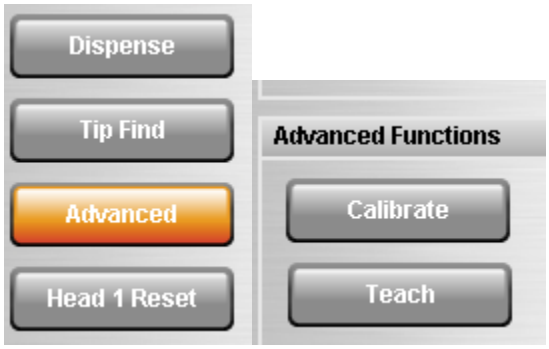
A new camera requires a new vision pattern recognition file for pre dispense inspection. The parameters are well defined in ksdispense file to help the user setup vision pattern recognition file. I addition, particle test, dispense sequence and post dispense inspection parameters must be redefined to match the new camera production settings.

1. VolumeHead1: The total volume in milliliters of the adhesive reservoir
2. VISON PATTERN RECOGNITION FILE: Inspect for adhesive dispense surface before executing dispense sequence.
    a. PAT PRF=1, FIL="camera A\camera Pre-Dispense.hdb"
    b. PRE HD=1,ID=1,IN=1,PRF=1,LW=0,LD=50,LL=0,EXP=0,XA=0,YA=0,ZA=0
3. PARTICLES TEST: Inspect for particles on image sensors before starting dispense sequence
    a. PRT LW=0,LP=0,BD=0,PT=0,IRH=0,IRW=0,ITC=0,ILS=0,ITS=0,OTC=5,OLS=3,OTS=10,RE=1
4. DISPENSE SEQUENCE: Dispense s single dot, line or arc of materials. For instance;
    a. ARC X=0,Y=0,XR=0,YR=0,SA=0,IA=0,VS=0,VA=0,LI=0,LO=0,DR=0,SW=0,SR=0,ST=00,ZO=00, ZR=0.0
5. POST DISPENSE INSPECTION: Inspect glue volume and shape using blob analysis
    a. VIS HD=1,ID=1,LW=0,LD=0,LL=0,EXP=00.,XO=0.000,YO=0.0,ZO=0,SM=1,ST=0,RL=0,RT=0, RR=0,RB=0,AMN=0,AMX=0,BMN=0,BMX=0,PMN=0,PMX=0,XMN=0,XMX=0,YMN=0,Y MX=0,CSA=1,CSR=3,CAW=1,CMN=50,CMX=100,CSO=50,CTW=200,CUT=3,COT=3

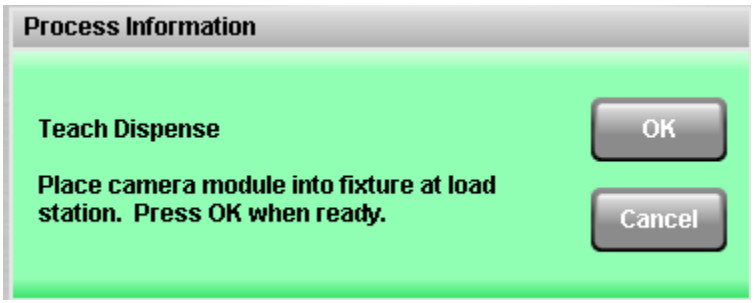
17.6 Dispense Teach

Dispense teach function can be used to teach the height of dispense tip with respect to dispense surface area. The procedure to setup dispense tip height is described below

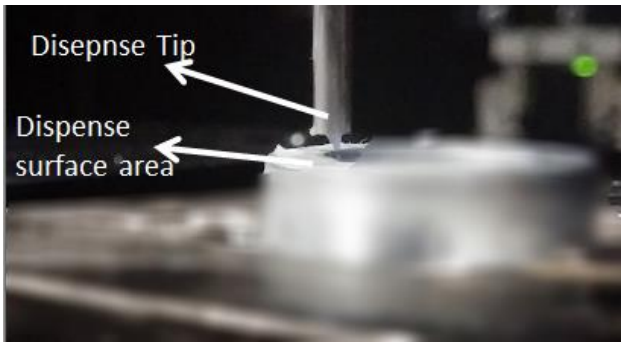
1. On the GUI screen, under dispense tab, click on advanced then teach



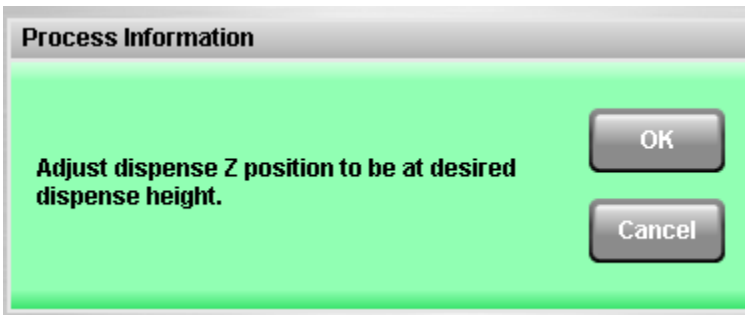
2. Follow the instruction on the screen by loading a camera modules into a fixture



3. Move dispense XYZ axis until dispense tip touches dispense surface area then move Z axis up about 500micron as starting position



4. Click ok after teach process is completed



5. Execute dispense sequence in dry run mode to check if the dispense teach working



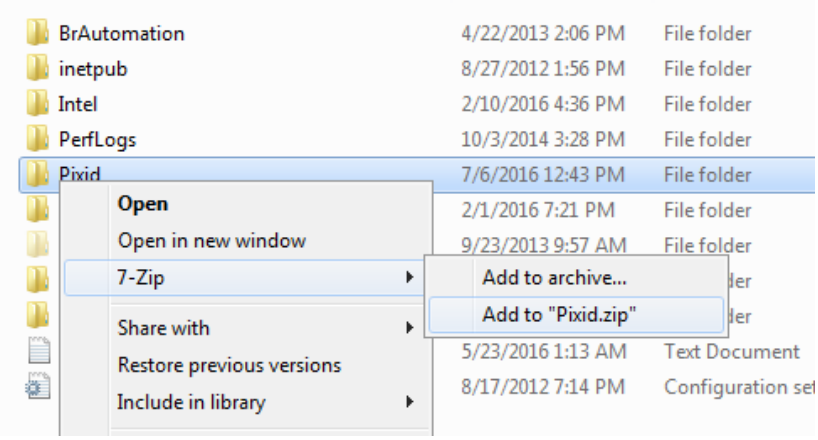
6. Adjust dispense parameters in Ksdispense file to dispense adhesive on dispense surface area
7. Adjust dispense parameters as needed according to required adhesive weight

18 Replace CFAST Boot Media Card

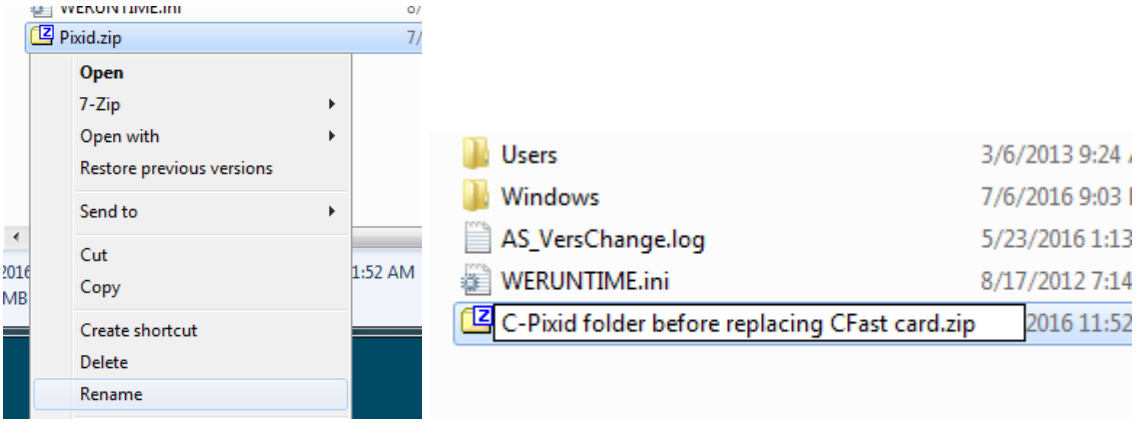
18.1 Backup Software and Configuration Files

Procedure defined steps that need to be taken in order to backup the entire C:\Pixid folder as a zip archive.

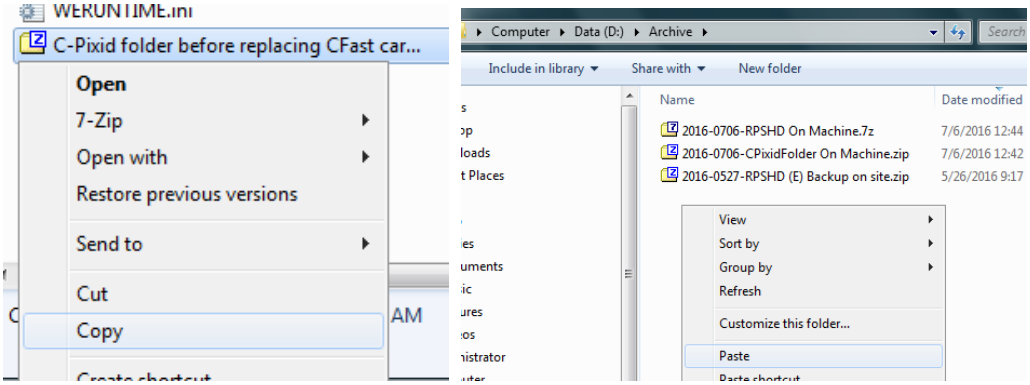
1. Zip C:\Pixid folder



2. Rename the Zip file “C-Pixid folder before replacing CFAST card”



3. Copy zip file “C:\Pixid folder before replacing CFast card” to D:\Archive



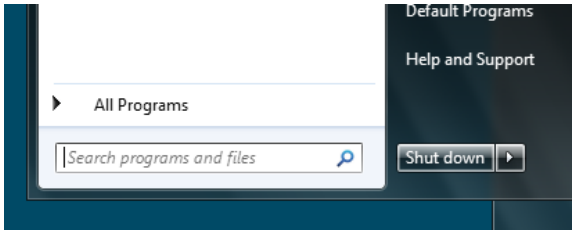
4. Repeat steps 1-3 for RPSHD (E:) and RPSHD (F:)



18.2 Replace CFAST Boot Media Card

See procedure below on how to replace CFAST card

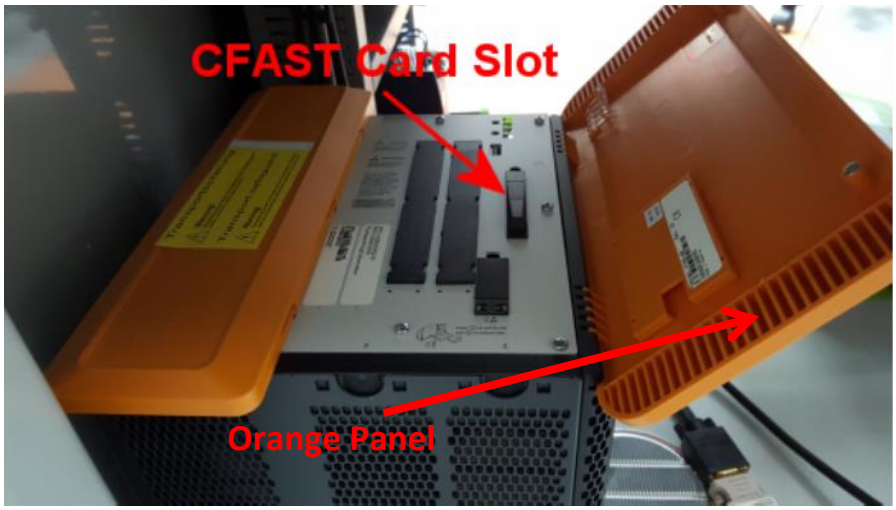
1. Shutdown the PC from start menu

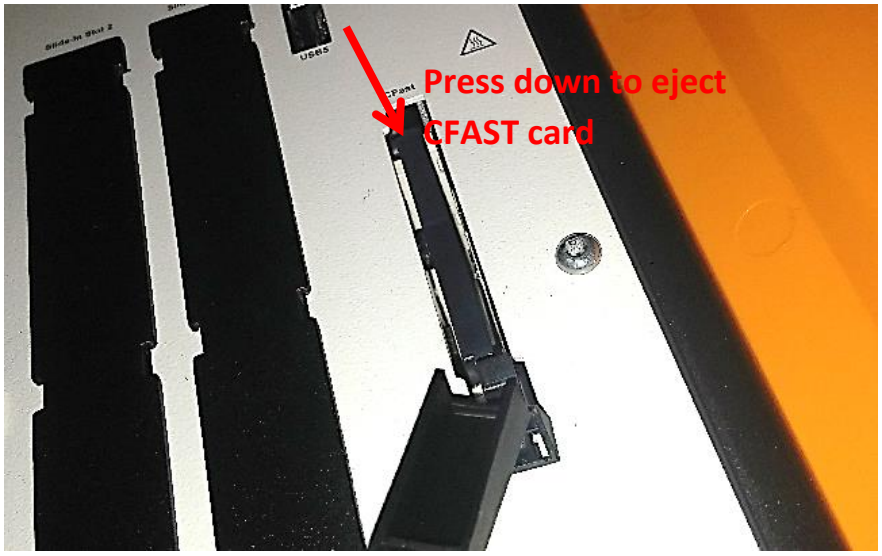


2. Power down the machine using the main power switch to ensure that no power is being delivered to the APC-910 industrial PC



3. Then eject the current CFAST card by flipping up the larger of the two orange panels to reveal the CFAST card slot then flip up the slot cover **and press the card eject button at the end** of the slot.





4. Install the new CFAST card by inserting it and **pressing it down until it seats fully**. Note that very little pressure is required to seat the card, and also note that the mating end of the card is notched to prevent installing the card backwards
5. Close CFAST's slot cover then flip down the larger of the two orange panels
6. Power up the machine using the main power switch



7. Wait until Arwin turn Green then start up the software

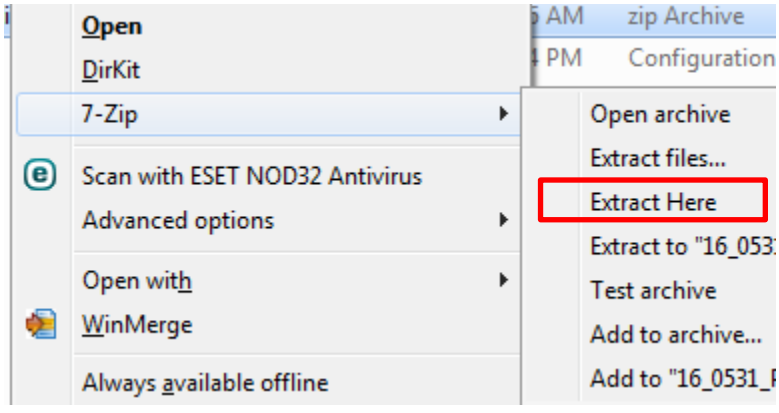


8. Start Pixid software if CFAST card is loaded with machine configuration (Check with Kasalis engineer)

18.3 Restore Software configuration

Software configuration will be restored in C:\Pixid folder.

1. Delete Pixid folder in “C” drive
2. Open D:\Archive and copy zip file “C:\Pixid folder before replacing CFast card” into “C drive “
3. Right click on zip file and select “Extract Here”
   a) Extraction process will create a new Pixid folder with restored configuration

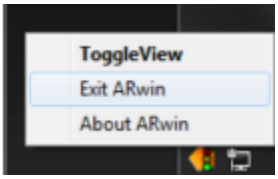


4. Repeat steps 1-3 for RPSHD (E:) and RPSHD (F:)
5. Start Pixid software and reinitialize the machine.

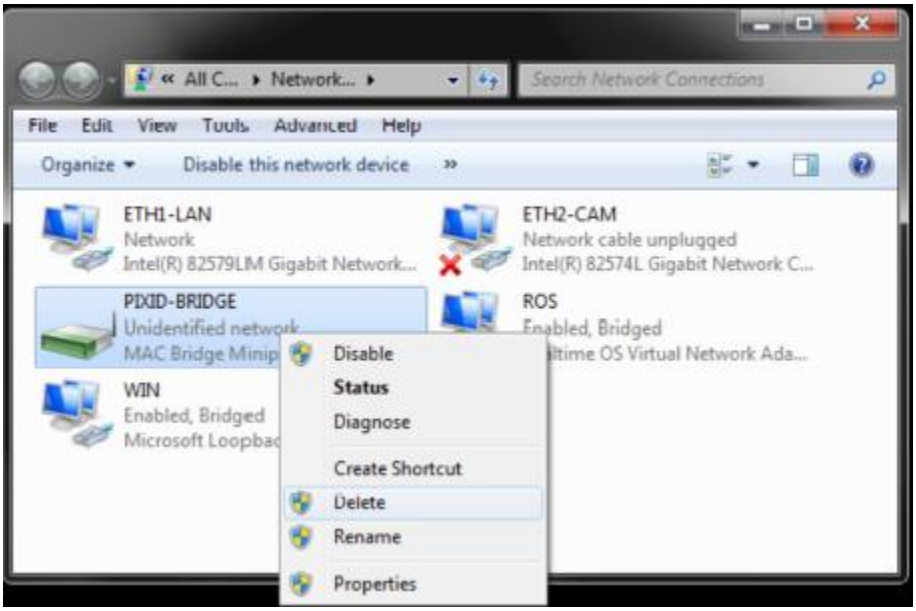
19 Pixid Network Bridge Setup

PIXID-BRIDGE network bridge must be setup to allow communication between the Windows and ARwin (VxWorks) operating systems. The connection between the Windows and ARwin (VxWorks) operating systems could be lost when PIXID system is added to customer internal network or Antivirus has been installed on the system. Steps below must be followed to setup Network Bridge.

1. Shutdown ARwin by right-clicking the traffic light icon in the tray notification area of the Windows taskbar, then selecting “Exit ARwin”



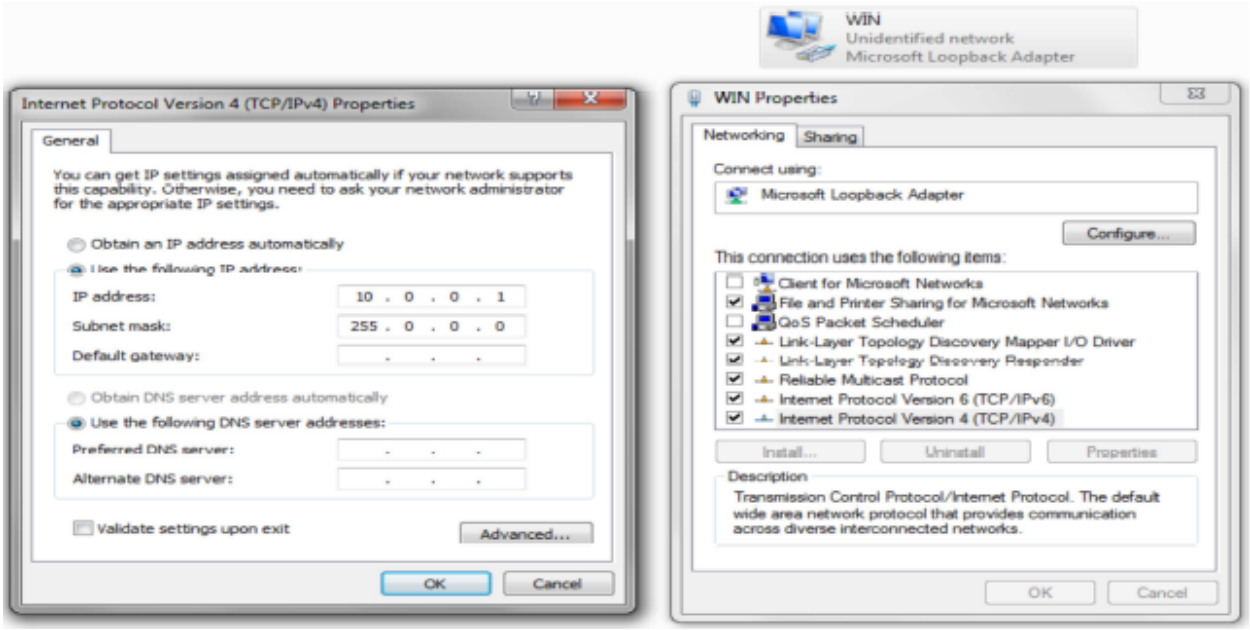
2. Open the network connection list by typing “ncpa.cpl” in the Windows Start Menu search box. In the network connections list right-click the PIXID-BRIDGE and select “Delete”



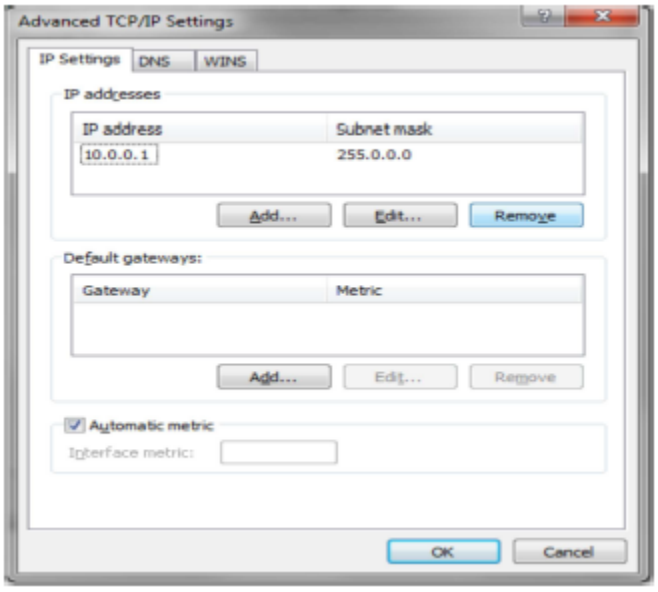
   2.1. NOTE: This section is only applicable if changing the IP addresses used by an existing PIXID-BRIDGE. If setting up the PIXID-BRIDGE for the first time skip to the next section 2.3 Set IP Address for the WIN Network Connection.

3. Open the network connection list by typing “ncpa.cpl” in the Windows Start Menu search box. In the network connections list right-click the WIN connection and select “Properties”. Set the check boxes as shown below in the “This connection uses the following items” list, making sure to clear the box for “Client for Microsoft Networks” and “QoS Packet Scheduler” while leaving all other items checked. Then double-click the “Internet Protocol Version 4 (TCP/IPv4” item to access the dialog to set the connections IP address. Select the option to “Use the following IP address” and enter your

desired IP address and Subnet mask. Note that these are the only two fields in this dialog that need to be set – the other fields should be left blank.

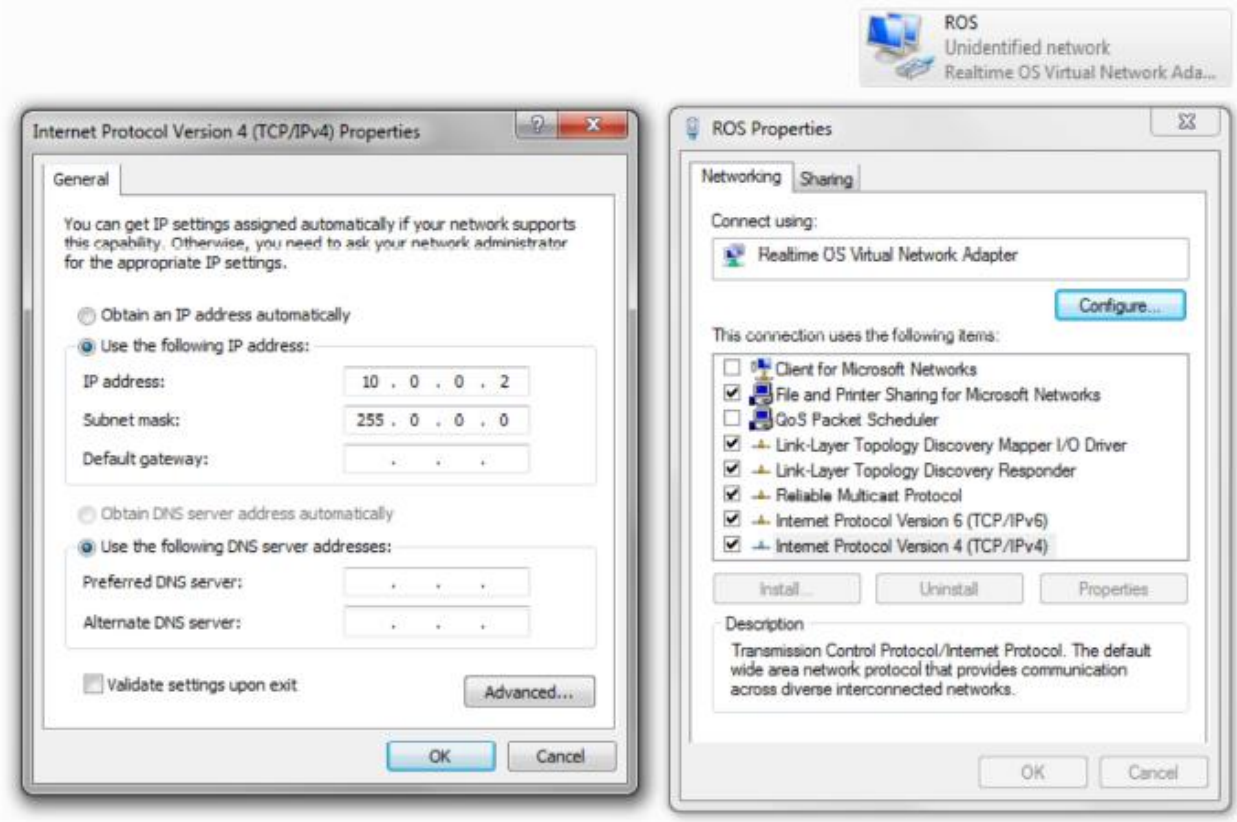


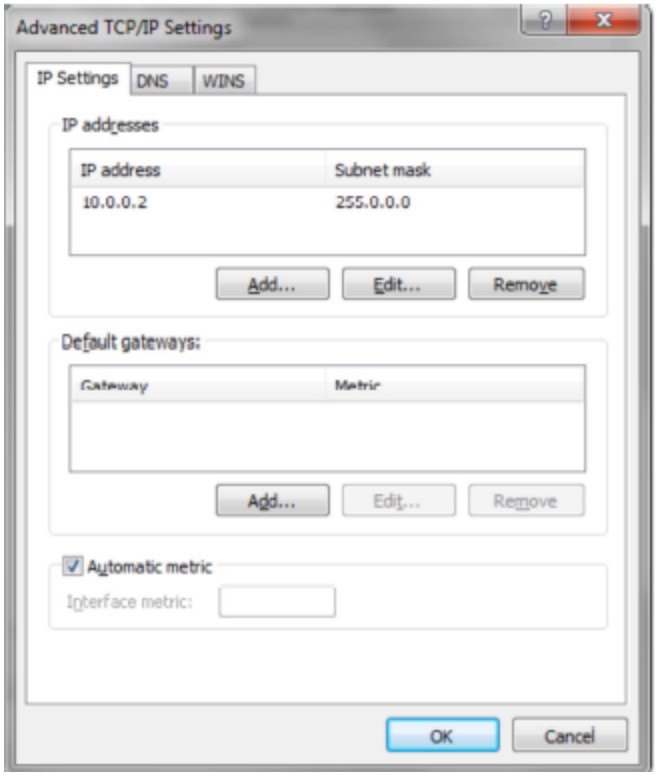
4. Select the Advanced button in the Internet Protocol Version 4 dialog and make sure that the onlyIP address listed in the IP Settings tab is the one that have entered above. Use the Remove button to remove any other entries. The settings in the DNS and WINS tables should not be modified.



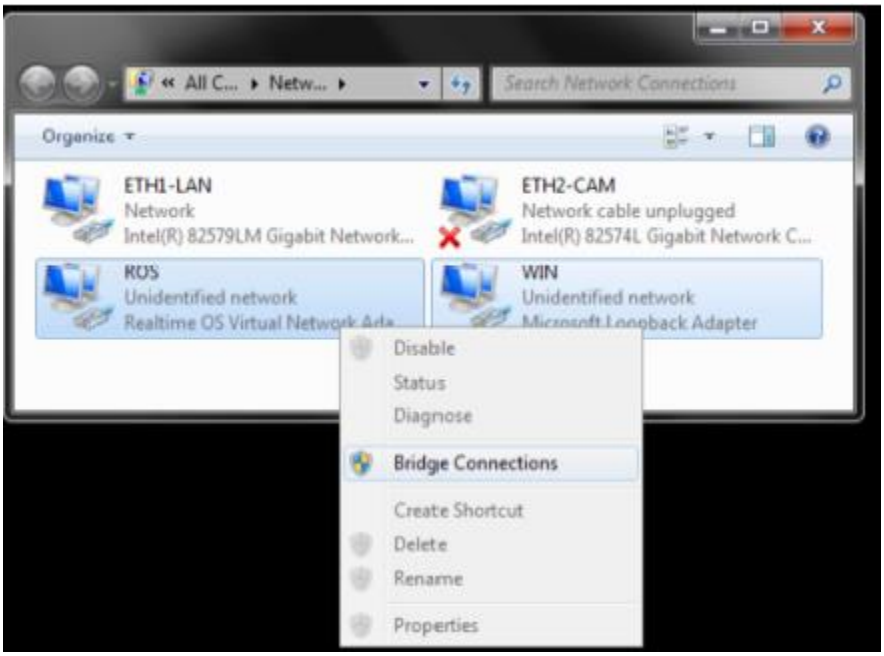
5. Perform the same procedure to set the IP address for the ROS network connection as done above for the WIN network connection with a single exception: The IP address for the ROS connection must have the final octet set to be one greater than the final octet of the WIN connection IP address. For example, if the WIN connection IP address is set to 10.0.0.1, the ROS connection IP address must be set to 10.0.0.2. Both the ROS and WIN connections should specify the same subnet

mask. As with the WIN connection, make sure to verify that only the specified IP address is listed in IP Settings tab of the dialog that is displayed when pressing the Advanced button.

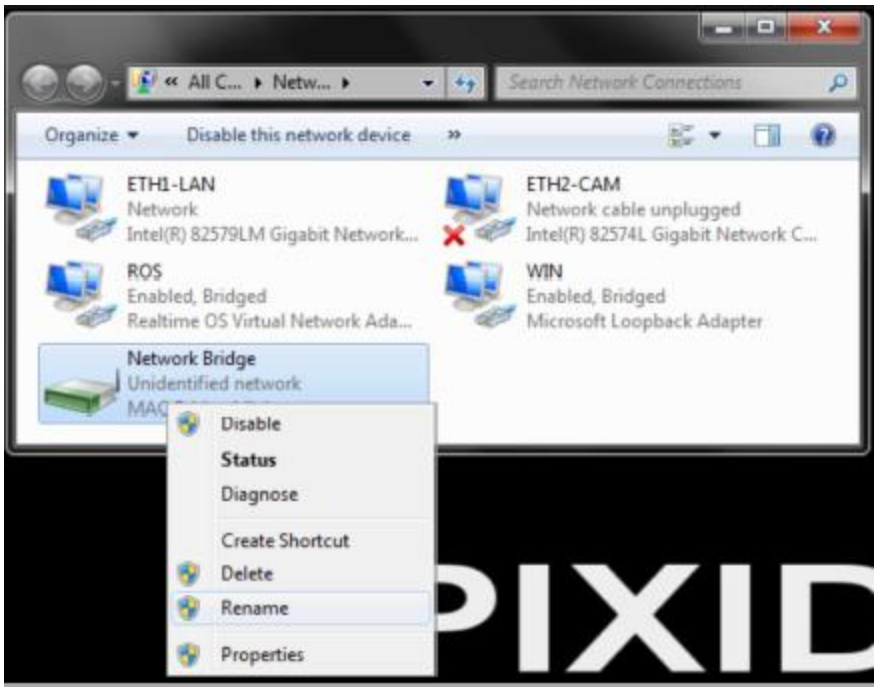




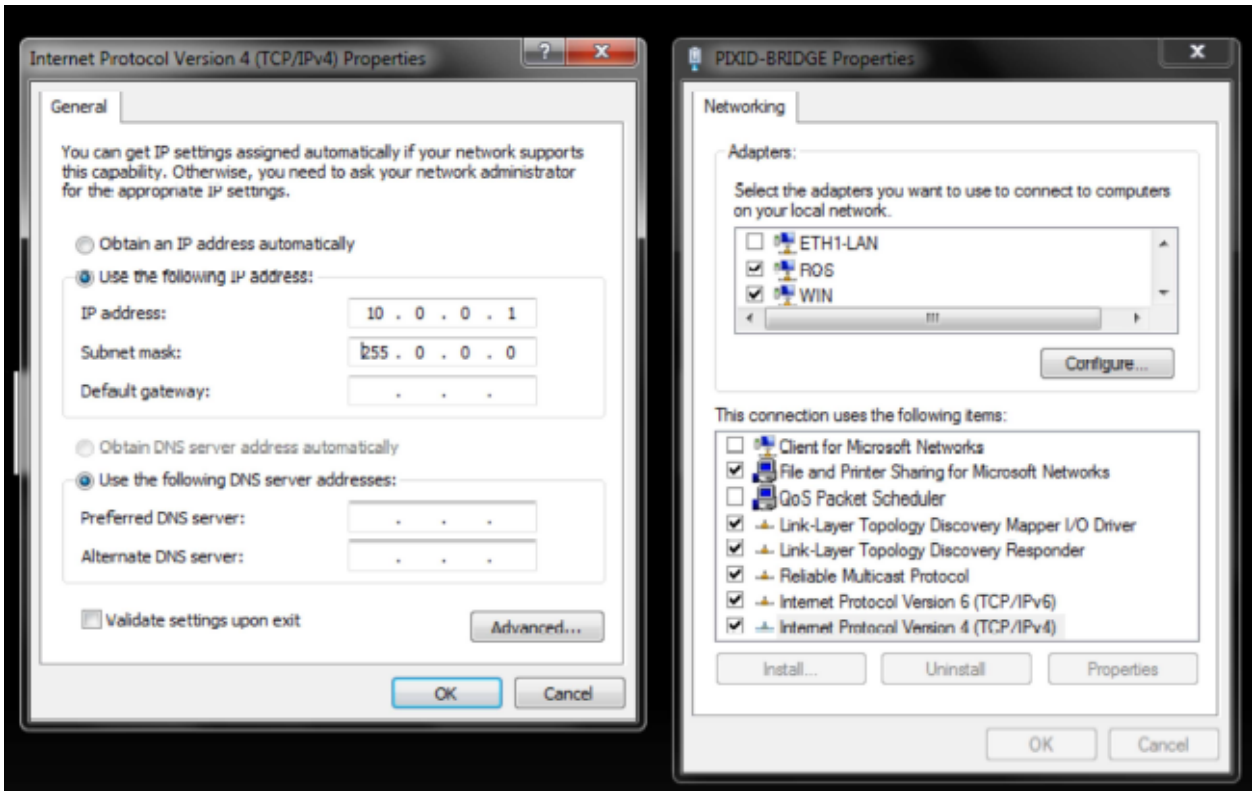
6. Select both the ROS and WIN connections by clicking them while pressing the CTRL key, then right-click either and select “Bridge Connections”.



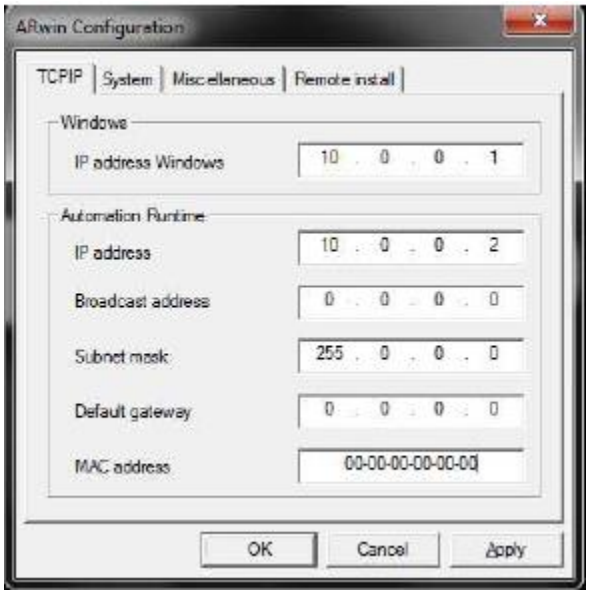
7. Right-click the Network Bridge connection then select Rename and set the name as PIXID-BRIDGE.
8. Right-click the PIXID-BRIDGE connection and select Properties



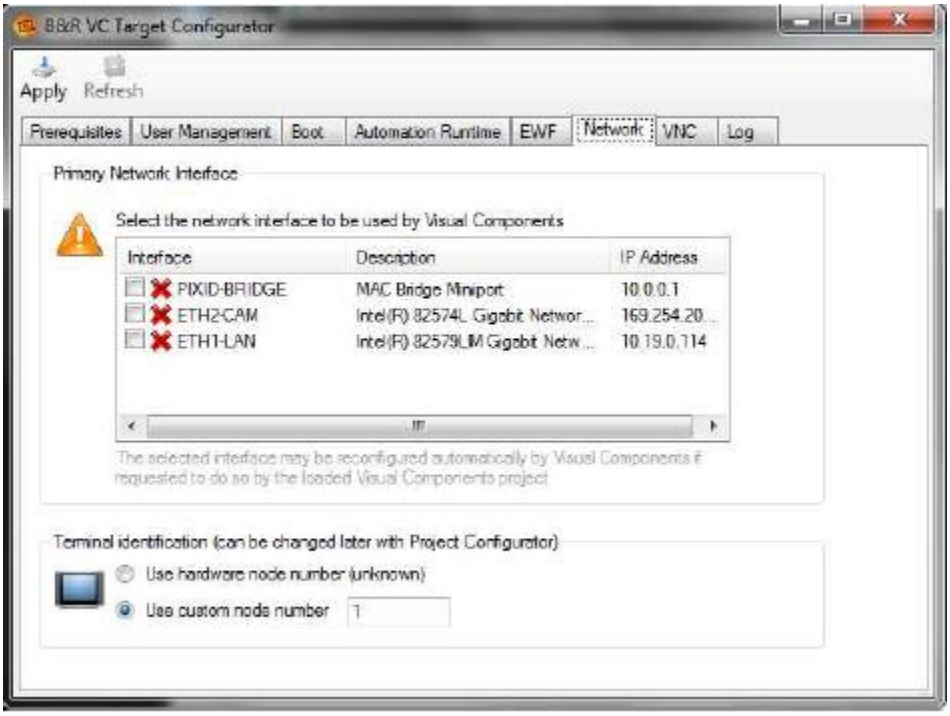
9. Setup the PIXID-BRIDGE in an identical manner as you setup the WIN connection in step 3. The IPv4 IP address and subnet mask of the PIXID-BRIDGE must be identical to the WIN connection. Also again make sure to click the Advanced button to verify that only one IP address is listed in the IP Settings list and that it matches the IP address entered in the Internet Protocol Version 4 dialog.



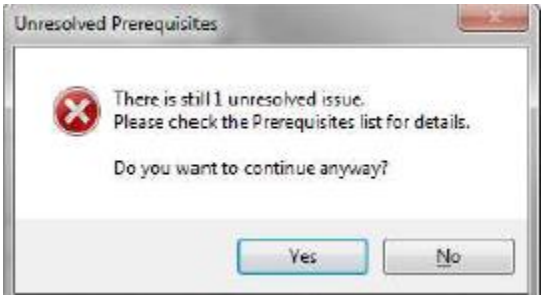
10. Open the ARwin Configuration editor by typing “ARwin Configuration” in the Windows Start Menu search box. In the TCPIP tab set the Windows IP address to the setting used in the WIN connection above, and set the Automation Runtime IP address to the setting used in the ROS connection above (WIN address plus 1). Set the Subnet mask to match the mask used for the WIN and ROS connections above. The settings in the other tabs of the ARwin Configuration should not be modified.



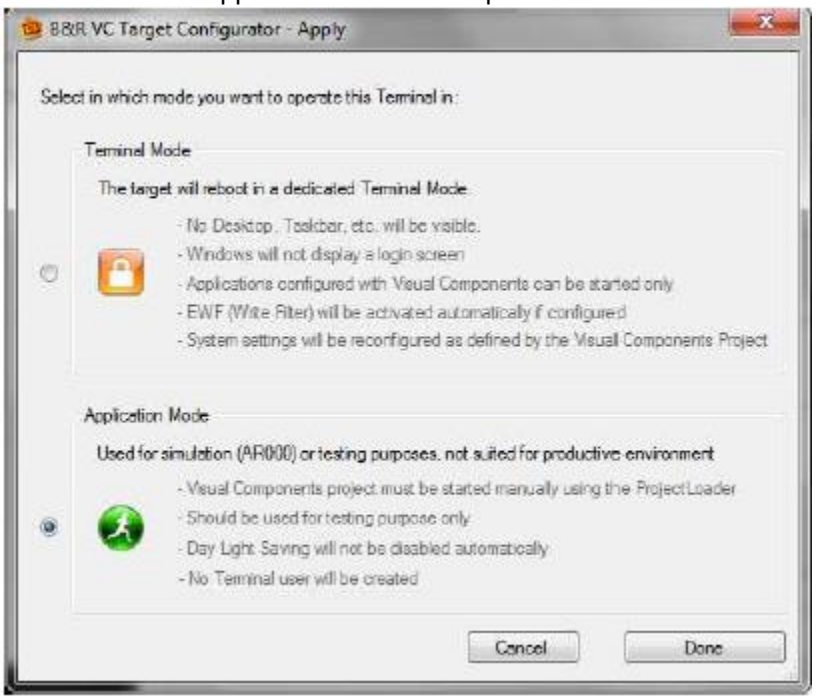
11. Open the VCWT Configuration editor by typing “VCWT Configuration” in the Windows Start Menu search box. In the Network tab ***clear*** the check box for PIXID-BRIDGE as shown below.



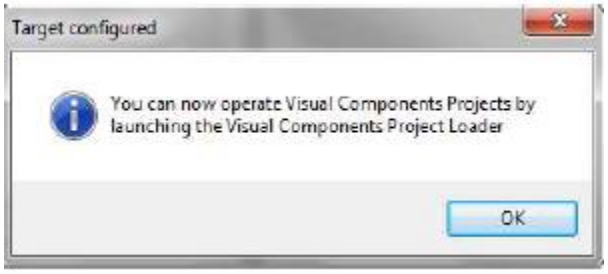
12. Change to the Prerequisites tab and press the Apply button in upper-left area of the dialog. A warning message will be displayed stating that there is an unresolved issue. Click Yes to this message.



13. Another message will be displayed prompting to select the "mode" in which the VC terminal should be operated. Leave this set to "Application Mode" and press the Done button.



14. Next, a message will be displayed confirming that the VCWT target has been configured. Press OK in this message, and then close the main B&R VC Target Configuration window.



15. After closing the VCWT Configuration editor, immediately open it again by typing "VCWT Configuration" in the Windows Start Menu and in the Network tab place a check-mark next to the PIXID-BRIDGE connection that you cleared above. Again, select the Prerequisites tab and press the

Apply button, press Done when prompted to select the operation mode, press OK when prompted that that target has been configured, then close the main configuration window again.

16. Restart ARwin from the Windows Start Menu. When the traffic light icon in tray notification area of the Windows taskbar turns green the system is ready to run the Pixid control software.

20 Maintenance

The Pixid 500 is designed to operate at a high level for years with no maintenance. Within the design, components are not extended to full range, thereby allowing them a longer lifespan and avoiding undue stress that could cause failures.

Clean the Pixid 500 as follows:

20.1 Parker XYZ Stages

20.1.1 Cross Roller Bearing and Screw Lubrication

Materials Required: Replacement bearing and screw lubrication (See below for lubrication type), clean cloth, small brush

1. For positioners with cleanroom preparation **R1**, class 1000 compatible (standard):
   1.1. Use Mobil Vactra Oil No.2 on crossed roller bearings
   1.2. Kyodo Yushi Multemp PS2 Grease on drive ballscrew
   1.3. Drive leadscrews do not require lubrication
2. Lubricant Grease Appearance:
   2.1. Translucent-white, smooth and buttery.
3. For positioners with cleanroom preparation **R2**, class 10 compatible:
   3.1. Use Castrol Braycote 803 Grease on crossed roller bearings and on drive ballscrew
   3.2. Drive leadscrews do not require lubrication
4. Lubricant Grease Appearance:
   4.1. Creamy-white, smooth and buttery

20.1.2 Maintenance Frequency

For both ‘R1’ and ‘R2’ Preparations: Cross roller bearings and drive screws are lubricated at manufacturer’s facility prior to shipment. For lubrication inspection and supply intervals following shipment, apply grease/oil every 1000 hours of usage. The time period may change depending on frequency of use and environment. Inspect for contamination, chips, etc, and replenish according to inspection results.

20.1.3 Lubricant Application

For both ‘R1’ and ‘R2’ Preparations: Wipe the rails or screw down the entire length with a clean cloth. Apply lubrication on the rails or screw, using a small brush.

Notes:

1. Do not use/mix petroleum base grease with synthetic base grease at any time. For lubrication under special conditions consult factory.

2. Shorter lubrication interval may be required in environments with high amounts of dust and other contamination.

20.2 Dymax BlueWave® LED Prime UVA

20.2.1 Lightguide

Clean the lightguide ends monthly or as required. The ends of the lightguide should be kept clean to transmit as much light as possible. Cured adhesive can carefully be removed from liquid-filled lightguide ends with a razor blade. Avoid sharp bends with the light guide since this reduces light output and permanently damages the guide.

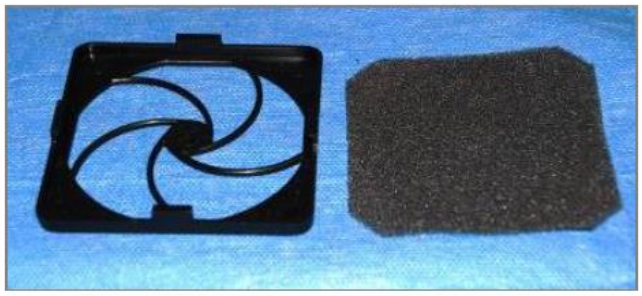
20.2.2 Fan filter

The external fan filter should be inspected and cleaned periodically to prevent dust buildup from affecting airflow through the unit. Spare filters are provided with each unit. The fan filters are washable and may be reused. Remove the fan filter by removing the snap on cover from the rear of the fan.

Cooling Fan



Cooling Fan Cover and Filter



20.2.3 Fuse Replacement

The unit is supllied with two fuses that are installed in the power receptacle. To remove the fuses, unplug the unit and remove the fuse holder with a small screw driver. Remove the fuses from the fuse holder and install new fuses. Replace the fuses with 2.0 Amp fast-blow types (Dymax PN 37236 or equivalent). Reinstall the fuse holder into the power receptacle.

Power Receptacle



Fuse Holder



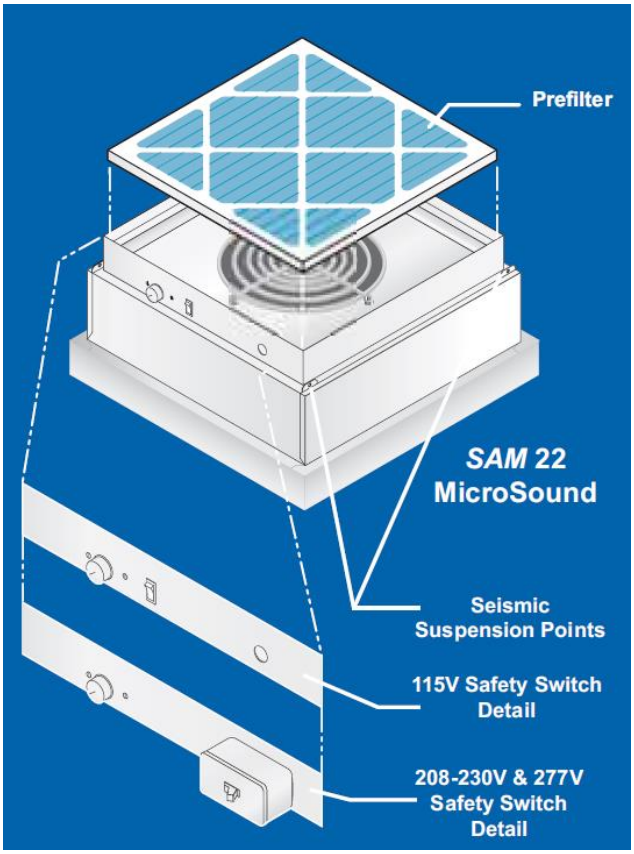
Fuses (PN 37236)



20.3 SAM 22 HEPA Filter and Prefilter

The HEPA filter and Prefilter replacement frequency is based upon the degree of contamination of the environment. Ideally, in a cleanroom, these filters should not need replacing due to the extremely low level of particles in a cleanroom.

Regardless, it is recommended that the filter and pre-filter be inspected every 6-8 months. A diagram of the SAM22 is shown on the right.



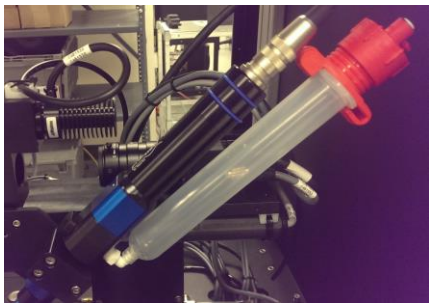
The safety switch should be toggled during service such that the fan is powered off. The prefilter can be simply replaced by removing it and inserting a new one.

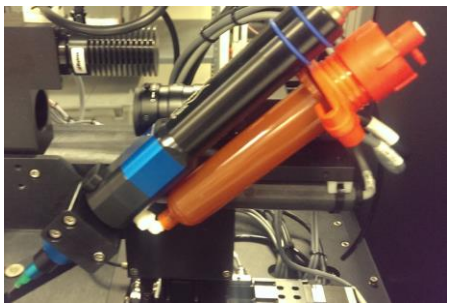
An anodized aluminum frame holds the 53mm media which is protected by a diamond pattern white epoxy coated steel grille. The HEPA filter can be replaced by removing the unit from the machine, placing it on a bench, removing and replacing the filter, then re-installing the unit.

20.4 Installing a new adhesive cartridge

Installing a new adhesive cartridge process is summarized as shown below

1. E-Stop the system, when old dispense cartridge is empty
2. Remove the old dispense cartridge
3. Install an empty cartridge (no glue inside the cartridge “just air”)
4. This will help to push out any glue lift in Eco-pen before i) cleaning purposes ii) installing a new glue cartridge

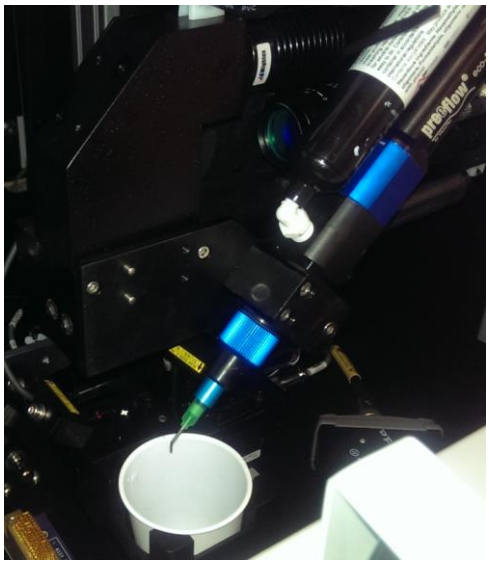






5. Release E-stop, reset the system and initialize the system.
6. Click on prime head1 button under dispense tab and check Box “Do Not Park after Prime”

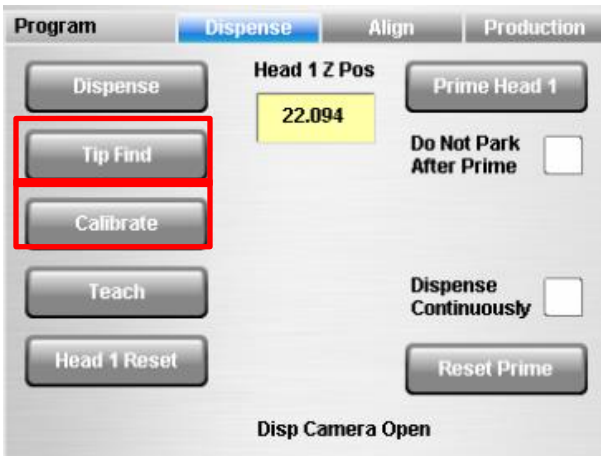




7. In I/O screen click on Head1rate% to setup motor speed to %40-%50 then click on head1start to turn on dispense motor and start purging again into purring cup



8. Stop purging process once there is no glue is flowing out of the dispense tip by clicking on head1start button on I/O screen
9. E-stop the system and remove the empty cartridge
10. Remove the Econ- Pen and clean up the unit
11. Install Eco-pen back in to the system after it has been cleaned properly
12. Install new glue cartridge (with glue)
13. Repeat steps 5, 6, 7 & 8
14. Stop purging process once the glue is starting flow out of the dispense tip
15. Initialize Pixid system
16. Execute system calibration by clicking on calibrate button under dispense tab

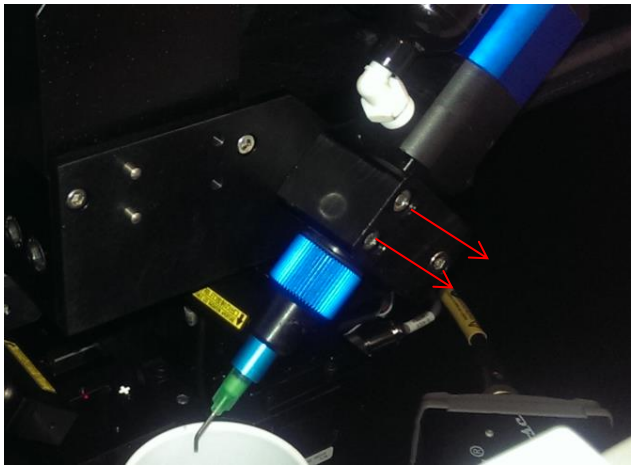


17. Before production, dispense on five parts and verify the glue profile by visual inspection

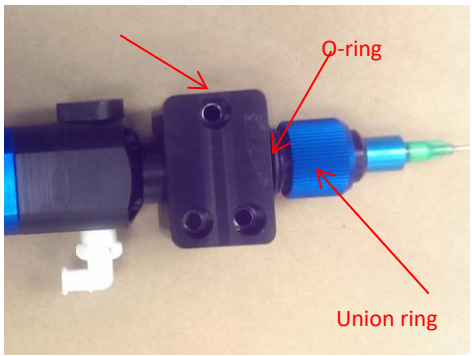
20.5 Dispense unit cleaning procedure

The steps below summarize the procedure of cleaning up the dispense unit

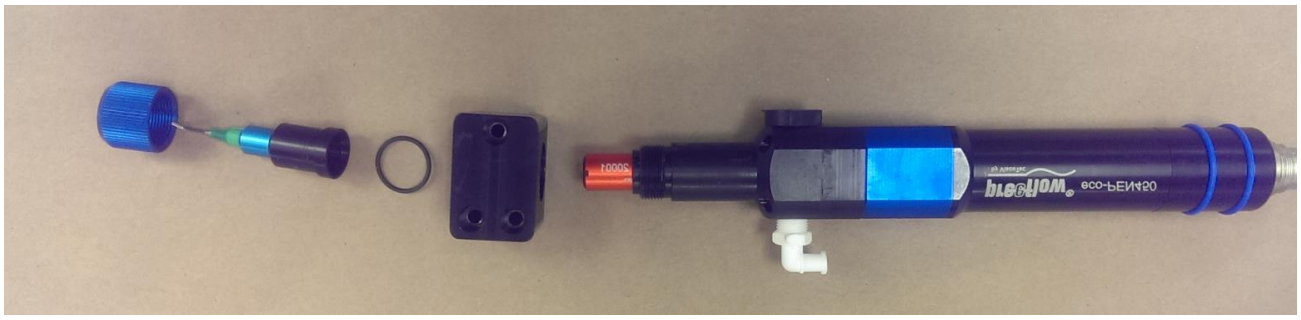
1. E-Stop the system and disconnect the power supply of dispense motor
2. Unmount the dispense unit from the system by loosening the two screws that holds the bracket.



3. Remove the third screw that prevent the dispense unit from rotating



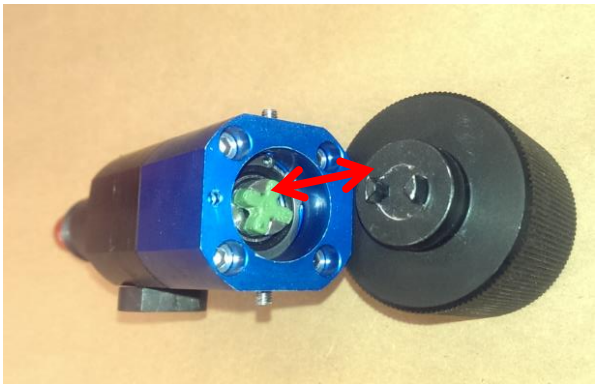
4. Disassemble the unit as shown in the picture below by loosening the union ring and the O-Ring

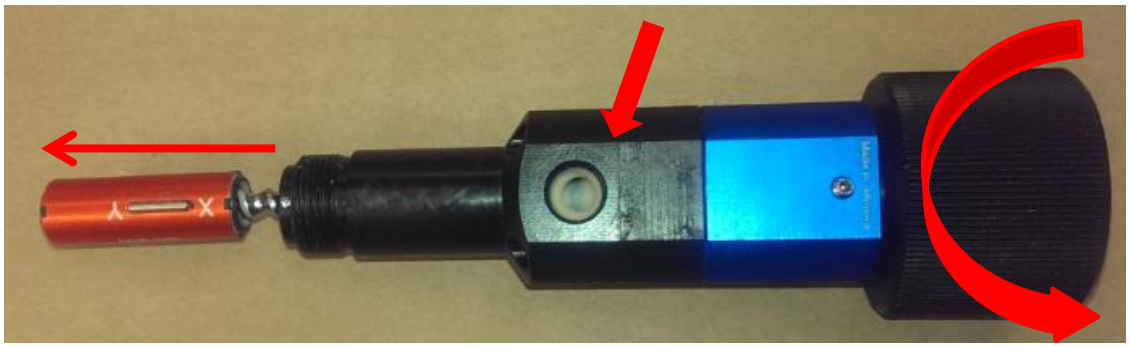


5. Disconnect the motor and uncouple it from the dosing unit
6. Slowly pull back the motor until the motor is disconnect from dosing unit

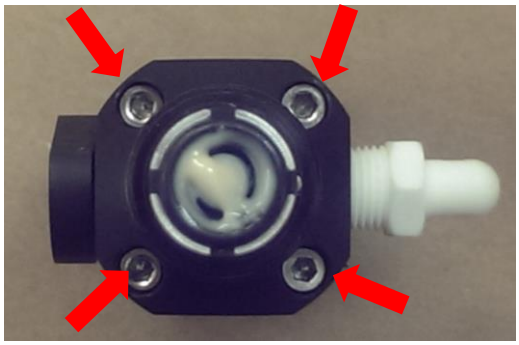


7. Use assembly aid module to lock the rotor housing
8. Pull the stator and rotate (counter clockwise) the rotor housing using assembly aid module
9. Use a wrench to remove the Elbow fitting

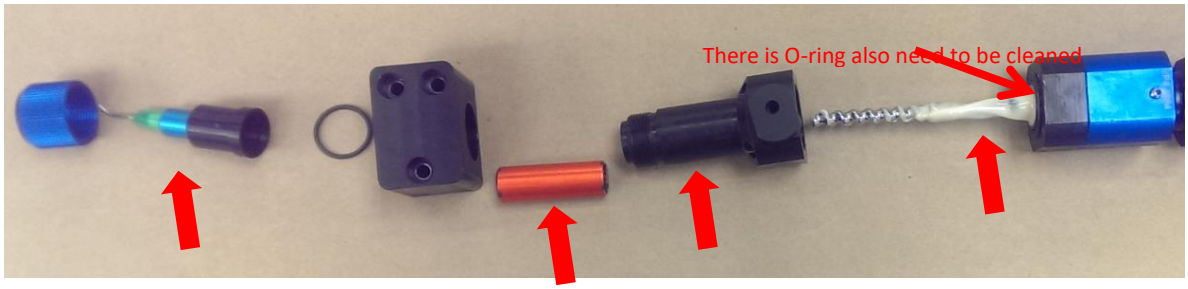




10. Undo the 4 screws to disconnect the rotor housing from dosing unit

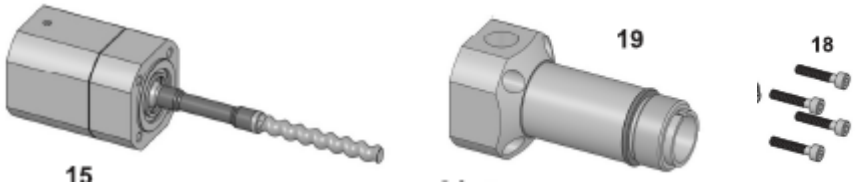


11. Clean all the parts using Alcohol or IPA and Kimwipes or a piece of cloth. Also use compressed air can to dry out any the unit after cleaning.

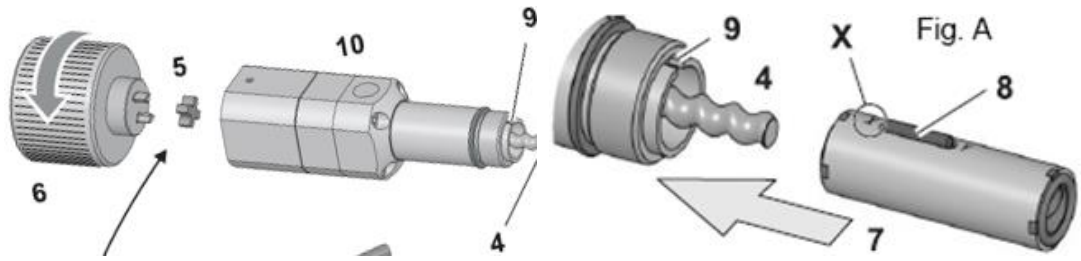


20.6 Reassembly the unit for production purposes

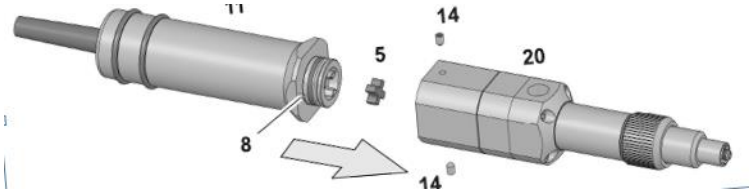
1. Attach the dispenser housing (19) with the O-ring (22) to the module (15) with the screws (18) (do not exceed a tightening torque of 0.35 Nm).



2. Lightly press the stator in the direction of the dispenser housing (10) and turn the assembly aid in the direction of the arrow until the stator has been guided into the dispenser housing.



3. Connecting the motor to dosing unit



4. Turn the set screws (14) in the thread such that they do not protrude into the coupling area. (Danger of damage to the fit)
5. Attach the star-shaped coupling (5) onto the coupling of the drive unit (11).
6. Couple the drive unit (11) to the dosing unit (20) until there is a gap <1 mm between the anti-rotation lock (8) and the dosing unit (20).
7. Set the anti-rotation lock (8) correctly in place by rotating the dosing unit (20).
8. Bring the drive unit (11) and the dosing (20) together completely.
9. Lightly turn the set screws (14), the drive unit (11) has now been centered properly.
10. Mount the bracket to dispense module



20.7 Reinstall dispense unit

1. Insert the dispense unit in to the bracket with no O-ring or union ring
2. Lock the rotation of dispense unit using the third screw
3. Insert the O-ring first
4. Rotate the end piece until the tip is vertical to dispense surface
5. Lock the rotation of end piece by inserting the union ring

